A first-principles graduate self-study textbook

Understanding the Aberration Budget Paper

From waves and diffraction to EUV lithography,
Zernike modes, neural networks, and inverse design

A detailed companion to

Zhishu Chen, Lisong Dong, Huwen Ding, and Yayi Wei,

Aberration budget analysis of EUV lithography from the imaging performance of a contact layer in a 5 nm technology node,
Applied Optics **62**, 7270–7279 (2023).

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Built around the supplied Born–Wolf and Saleh–Teich textbooks

Original explanations, worked derivations, scientific diagrams, solved exercises, a source audit, and reproducible teaching calculations.
Prepared 24 September 2026.

How to study this book

This book assumes no previous optics, lithography, aberration theory, or machine-learning knowledge. Basic algebra is useful. Calculus, complex numbers, Fourier analysis, and matrix ideas are introduced as they become necessary. The aim is to make every technical step in Chen and colleagues' paper [1] understandable, including the steps compressed into phrases such as rigorous simulation, Zernike coefficient, process variation band, and inverse sensitivity analysis.

The research paper contains only three numbered equations. Those equations are not the whole theory behind the work. The missing chain is

$$\text{field} \longrightarrow \text{mask diffraction} \longrightarrow \text{pupil phase} \longrightarrow \text{image} \longrightarrow \text{edges} \longrightarrow \text{constraints}.$$

The neural network is an approximate way to travel backward along part of this chain. Learning the forward physics first makes the backward problem much easier to understand.

The explanations are original teaching material. Source equations are rederived in explicit conventions rather than copied without context. Numerical demonstrations labelled *teaching examples* are calculations made for this book. They are not recovered data from the authors' commercial simulations. In particular, no trained version of their neural network, mask-simulator settings file, or proprietary dataset has been supplied.

A practical study sequence

1. Read Chapters 1–3 with a notebook. You should be able to convert nanometres of optical path into radians of phase and explain why field amplitudes interfere.
2. Study Chapters 4–7. Draw a pupil, an image plane, a defocused wavefront, and a Zernike mode. Do the small integrals yourself.
3. Work through Chapters 8–10. These connect optics to the three measured quantities in the paper.
4. Read the data and learning chapters, then open the paper alongside Chapter 15. Every numbered equation, table, and figure has a corresponding explanation.
5. Finish the budget and laboratory chapters and the solved exercises. Try each exercise before looking at its answer.

Do not try to memorize 25 polynomial labels. Understand the pupil function, the unit convention, and the map from phase to edges. Those ideas remain useful when the numbering convention or lithographic pattern changes.

What was actually available in the source files

- The paper is complete: 10 PDF pages, journal pages 7270–7279. Paper p. 3 means the third PDF page, journal page 7272.
- The supplied *Principles of Optics* scan is the fourth edition of Born and Wolf [2], with 859 PDF pages. Its printed pagination and PDF pagination differ, and a central run is reversed. For example, printed pages 464, 465, and 466 are PDF pages 555, 554, and 553. Use the verified entry points in Appendix C, then follow the printed page numbers.
- The file named `Saleh & Teich Foundations of Photonics.pdf` actually contains *Fundamentals of Photonics*, second edition [3]. It has 300 PDF pages and ends at printed page 278. The useful chapters on ray optics, waves, Fourier optics, electromagnetic optics, and polarization are present. Chapter 11, *Statistical Optics*, is listed in the contents but is not included. This book therefore uses Born–Wolf for the missing partial-coherence foundation and supplies its own complete derivations.

How to interpret statements in this book

Reported result means something explicitly stated or displayed in the paper. **Derived result** follows from equations and assumptions written here. **Teaching example** uses a stated simplified model. **Unresolved detail** means the publication does not supply enough information to settle it. These distinctions matter because an attractive graph is not a complete simulation specification.

Several publication issues deserve particular attention: the pattern tables disagree, the output count and stated Zernike range disagree, a Table 4 column header is repeated, and the source-to-pupil conventions are incomplete. The guide preserves the reported information and identifies the uncertainty. It does not quietly choose a correction and present it as an author's fact.

This textbook concerns the supplied 2023 study. Its conclusions are read in the context of its patterns, illumination, simulation conditions, and validation data. It supplies the underlying education without treating a simulation study as a complete production qualification.

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Part I

The physical language of the paper

Chapter 1

What problem is the paper trying to solve?

Paper location: Title, abstract, Introduction, and Conclusion.

1.1 Printing a pattern with light

A semiconductor chip is made by repeatedly creating patterns in thin layers on a wafer. In optical lithography, light illuminates a patterned mask, an optical system projects a smaller image onto a light-sensitive material called photoresist, and chemical processing turns the exposure into a material pattern. Etching or deposition then transfers or uses that pattern. The paper concerns the optical exposure part of this sequence.

A *contact hole* is an opening intended to connect device regions or conducting layers. A *via* is also an interconnection opening, commonly between conducting levels. For the optical argument, both are small two-dimensional features whose position and size must be controlled. A square or rectangle in the mask does not automatically print as an identical square or rectangle: diffraction rounds corners and mixes light from nearby features.

The *critical dimension*, abbreviated CD, is a specified feature width. A rectangle needs at least a horizontal width CD_x and a vertical width CD_y. *Pitch* is the centre-to-centre repetition distance of a periodic pattern. It is not the width of a hole and not the space between neighbouring edges. For a 14 nm square hole on 38 nm pitch, the nominal edge-to-edge gap along an axis is $38 - 14 = 24\,\mathrm{nm}$.

The name *5 nm technology node* does not mean that every printed line, hole, or transistor dimension is 5 nm. It labels a technology generation. The authors use nominal hole widths of 14 nm and 28 nm. The 2021 IRDS lithography roadmap explicitly separates node labels from physical dimensions [5]. Always identify the layer and feature before assigning a physical length to a node name.

1.2 What makes an image imperfect?

Even a perfectly manufactured finite optical system has diffraction. That unavoidable spreading is different from *aberration*, the departure of an actual optical wavefront from a chosen ideal wavefront. Aberrations change the phases with which light from different pupil locations contributes to an image.

The important objects form a hierarchy:

1. **Physical hardware:** mirror shapes, positions, coatings, the mask stack, and illumination optics.
2. **Optical response:** amplitude, phase, polarization, and the diffraction orders admitted by the pupil.
3. **Image:** a continuous distribution of optical intensity.
4. **Printed geometry:** edges selected by the resist process.
5. **Performance indicators:** widths, shifts, and changes under focus and exposure variations.

A neural network operating on the final indicators does not directly specify a set of manufacturable mirror surfaces. It proposes a wavefront description that still needs an optical realization.

1.3 Why a budget is useful

Suppose a contact must remain within 0.4 nm of its intended vertical position. Many optical errors contribute to that position. A budget asks how much of each error can be tolerated while satisfying the performance requirement. A conventional budget might restrict the total RMS wavefront error. This paper instead explores combinations of wavefront coefficients that are acceptable for a particular contact-layer test set.

Write a vector of aberration coefficients as $\boldsymbol{c} = (c_1, \ldots, c_K)^T$ and the computed imaging indicators as $\boldsymbol{y} = F(\boldsymbol{c})$. The symbol F stands for an entire optical and image-measurement calculation. The forward task calculates $\boldsymbol{y}$ from known $\boldsymbol{c}$. The inverse task finds some $\boldsymbol{c}$ giving acceptable $\boldsymbol{y}$.

The paper trains an approximate inverse map G:

$$\boldsymbol{c}_{\text{candidate}} = G(\boldsymbol{y}_{\text{desired}}), \qquad F(\boldsymbol{c}_{\text{candidate}}) \approx \boldsymbol{y}_{\text{desired}}.$$

It then sends the proposed coefficients back through the commercial forward simulator. That final comparison is essential: a network output is a candidate, and its actual simulated image determines whether it works.

> **Physical meaning.** The central proposal is to choose allowable wavefronts using the imaging needs of a particular layer. The contribution is the layer-specific inverse modelling and candidate-screening workflow, not a new law of diffraction.

1.4 Vocabulary you will repeatedly meet

Term	Meaning in this study
EUV	Extreme ultraviolet light; here the wavelength is 13.5 nm.
DUV	Deep ultraviolet; the Introduction compares EUV with 193 nm light.
Projector / objective	The optical system forming the mask image on the wafer; in EUV it uses reflective optics.
Pupil	A representation of the aperture selecting ray directions or spatial-frequency components.
Wavefront	A surface of equal optical phase at a specified time.
OPD / WFE	Optical path difference / wavefront error relative to a specified ideal reference, usually expressed as a length.

Term	Meaning in this study
Zernike coefficient	The amplitude of one specified polynomial mode in a pupil-wavefront expansion.
PS	Pattern shift: displacement of a specified printed feature centre.
PVB	Process variation band: the spread of printed contours when specified process conditions change.
NILS	Normalized image log slope: relative intensity steepness at an edge.
BPNN	Backpropagation neural network, a trainable nonlinear function.
OPC	Optical proximity correction: modification of the mask layout to improve the eventual printed pattern.

Check your understanding. If two patterns have the same CD, must they have the same pitch, centre position, or process robustness? No. These are separate properties; this is why the paper uses several indicators for many patterns.

Chapter 2

The mathematics we actually need

2.1 Units and dimensionless ratios

One nanometre is 10^{-9} metres; one micrometre is 10^{-6} metres. An angle in radians is the ratio of arc length to radius and is dimensionless. An exponential must have a dimensionless argument. For example, $2\pi W/\lambda$ is legitimate because W and λ are both lengths.

A *wave* of optical path error means one wavelength of OPD. A milliwave, written here as mλ, is 10^{-3} waves:

$$W_{\mathrm{nm}} = \lambda_{\mathrm{nm}} \frac{c_{\mathrm{m}\lambda}}{1000}. \tag{2.1}$$

The paper's MMW is used in this milliwave sense. It is not a millimetre. At 13.5 nm, $1\,\mathrm{m}\lambda = 0.0135\,\mathrm{nm}$ and $20\,\mathrm{m}\lambda = 0.270\,\mathrm{nm}$.

2.2 Complex numbers are a language for phase

The imaginary unit obeys $\mathrm{i}^2 = -1$. A complex number $z = a + \mathrm{i}b$ has conjugate $z^* = a - \mathrm{i}b$, modulus $|z| = \sqrt{a^2 + b^2}$, and squared modulus $zz^* = a^2 + b^2$. Euler's formula is

$$\mathrm{e}^{\mathrm{i}\phi} = \cos\phi + \mathrm{i}\sin\phi. \tag{2.2}$$

Multiplication by $\mathrm{e}^{\mathrm{i}\phi}$ rotates a complex number by ϕ. It changes phase without changing magnitude.

Represent a real harmonic electric field by

$$E(\boldsymbol{r}, t) = \mathrm{Re}\{U(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\omega t}\}, \qquad U = A\mathrm{e}^{\mathrm{i}\phi}. \tag{2.3}$$

The observable field is real. The complex quantity U stores its amplitude and phase compactly. Time differentiation becomes multiplication by $-\mathrm{i}\omega$, a major simplification.

For two scalar fields U_1 and U_2, expand the intensity:

$$\begin{aligned} |U_1 + U_2|^2 &= (U_1 + U_2)(U_1^* + U_2^*) && (2.4)\\ &= |U_1|^2 + |U_2|^2 + U_1U_2^* + U_1^*U_2 && (2.5)\\ &= A_1^2 + A_2^2 + 2A_1A_2\cos(\phi_1 - \phi_2). && (2.6) \end{aligned}$$

The last term is interference. Equal amplitudes in phase give intensity $4A^2$; equal amplitudes out of phase by π give zero. The separate intensities were A^2 in both cases. This is why changing phase can change an image without changing the amount of light admitted by the pupil.

2.3 Derivatives, gradients, and Taylor expansion

A derivative measures a local change. For small displacement δx,

$$I(x+\delta x) = I(x) + I'(x)\delta x + O(\delta x^2).$$

Here $O(\delta x^2)$ denotes terms whose leading size is proportional to the square of the small displacement. A partial derivative varies one argument while holding the others fixed:

$$\nabla_\perp W = \left(\frac{\partial W}{\partial x}, \frac{\partial W}{\partial y}\right).$$

For $W = A(x^2+y^2)^2$, $\nabla_\perp W = 4A(x^2+y^2)(x,y)$. If W is a length, this gradient is dimensionless; it will be related to an angular error.

We frequently use

$$\sqrt{1+\epsilon} = 1 + \frac{\epsilon}{2} - \frac{\epsilon^2}{8} + O(\epsilon^3), \tag{2.7}$$

$$\mathrm{e}^{\mathrm{i}\phi} = 1 + \mathrm{i}\phi - \frac{\phi^2}{2} + O(\phi^3), \tag{2.8}$$

$$\ln(1+\epsilon) = \epsilon + O(\epsilon^2). \tag{2.9}$$

These follow by evaluating successive derivatives at zero. They are approximations for small dimensionless arguments. Calling a formula paraxial or small-aberration specifies the expansion that has been made; it does not remove the need to check the argument's size.

Worked example: Changing to normalized pupil coordinates

Let $x_p = au$, $y_p = av$, where a is the physical pupil radius, and $\widetilde{W}(u,v) = W(au, av)$. The chain rule gives

$$\frac{\partial W}{\partial x_p} = \frac{1}{a}\frac{\partial \widetilde{W}}{\partial u}. \tag{2.10}$$

For $\widetilde{W} = Bu^2$, the physical slope is $2Bu/a$, not $2Bu$. Omitting $1/a$ changes both dimensions and physics.

2.4 Integrals over a disk

Use normalized polar coordinates $u = \rho\cos\theta$, $v = \rho\sin\theta$, $0 \le \rho \le 1$. The small area is $\mathrm{d}u\,\mathrm{d}v = \rho\,\mathrm{d}\rho\,\mathrm{d}\theta$. The uniformly weighted disk average is

$$\langle f \rangle = \frac{1}{\pi}\int_0^{2\pi}\int_0^1 f(\rho,\theta)\rho\,\mathrm{d}\rho\,\mathrm{d}\theta. \tag{2.11}$$

The factor ρ accounts for the growing circumference of rings. Sampling equally spaced radii with equal weights is not uniform area sampling. For $q > -2$,

$$\langle \rho^q \rangle = 2\int_0^1 \rho^{q+1}\mathrm{d}\rho = \frac{2}{q+2}. \tag{2.12}$$

Thus $\langle\rho^2\rangle = 1/2$, $\langle\rho^4\rangle = 1/3$, and $\langle\rho^8\rangle = 1/5$. Also $\langle\rho^2\cos^2\theta\rangle = 1/4$ because the angular average of $\cos^2\theta$ is $1/2$.

The mean-subtracted RMS of a real wavefront is

$$\sigma_W = \sqrt{\langle (W - \langle W\rangle)^2\rangle} = \sqrt{\langle W^2\rangle - \langle W\rangle^2}. \tag{2.13}$$

It is a length. The mean $\langle W\rangle$ is called piston. For $W = B\rho^2$, $\sigma_W = |B|/\sqrt{12}$.

2.5 Vectors, matrices, and least squares

A vector is an ordered list. A matrix applies weighted sums to that list. If A has M rows and K columns, $A\boldsymbol{c}$ has M entries:

$$(A\boldsymbol{c})_i = \sum_{j=1}^{K} A_{ij} c_j.$$

The transpose A^T interchanges rows and columns. The Euclidean norm obeys $\|\boldsymbol{c}\|^2 = \boldsymbol{c}^T\boldsymbol{c} = \sum_j c_j^2$. To fit data $\boldsymbol{b}$ with $A\boldsymbol{c}$, minimize $L = \frac{1}{2}\|A\boldsymbol{c} - \boldsymbol{b}\|^2$. Expansion and differentiation give $\nabla_{\boldsymbol{c}} L = A^T(A\boldsymbol{c} - \boldsymbol{b})$. At a stationary point,

$$A^T A\,\boldsymbol{c} = A^T\boldsymbol{b}. \tag{2.14}$$

These are the normal equations. A unique solution requires independent columns. If a nonzero $\boldsymbol{v}$ satisfies $A\boldsymbol{v} = 0$, then $\boldsymbol{c}$ and $\boldsymbol{c} + \boldsymbol{v}$ predict the same data. Such a vector is in the *null space*. This idea becomes central to the paper's claim that different aberration combinations can have similar imaging performance.

2.6 Averages, random errors, and correlations

For samples $x_1, \ldots, x_N$, the mean is $\bar{x} = N^{-1}\sum_i x_i$. An expectation $\mathbb{E}[x]$ is the corresponding probability-weighted average. Variance is $\mathbb{E}[(x - \mathbb{E}[x])^2]$ and

$$\mathrm{Cov}(x, y) = \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])].$$

Expanding the square proves

$$\mathrm{Var}(x + y) = \mathrm{Var}(x) + \mathrm{Var}(y) + 2\,\mathrm{Cov}(x, y). \tag{2.15}$$

The root-sum-square rule follows only when covariance vanishes. Two fabrication errors that always move together do not behave like independent errors. Fixed errors do not become random simply because their signs are unknown.

2.7 Fourier analysis: a shape as a sum of ripples

A periodic function of period p can be represented as

$$t(x) = \sum_{m=-\infty}^{\infty} a_m \mathrm{e}^{\mathrm{i}2\pi m x/p}, \qquad a_m = \frac{1}{p}\int_{-p/2}^{p/2} t(x)\mathrm{e}^{-\mathrm{i}2\pi m x/p}\mathrm{d}x. \tag{2.16}$$

Substitute the series into the integral. The integral of $\mathrm{e}^{\mathrm{i}2\pi(m-n)x/p}$ over one period is zero unless $m = n$, when it equals p. The integral selects one ripple.

For nonperiodic two-dimensional fields we use

$$\widetilde{U}(\boldsymbol{f}) = \int_{\mathbb{R}^2} U(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}2\pi\boldsymbol{f}\cdot\boldsymbol{x}}\mathrm{d}^2x, \tag{2.17}$$

$$U(\boldsymbol{x}) = \int_{\mathbb{R}^2} \widetilde{U}(\boldsymbol{f})\mathrm{e}^{\mathrm{i}2\pi\boldsymbol{f}\cdot\boldsymbol{x}}\mathrm{d}^2f. \tag{2.18}$$

Spatial frequency $\boldsymbol{f} = (f_x, f_y)$ has units of inverse length. Fine detail needs higher spatial frequencies. A finite pupil removes some frequencies, so an image cannot reproduce arbitrary fine detail.

Convolution, $(h * t)(x) = \int h(x - x')t(x')\mathrm{d}x'$, means summing shifted responses to all object points. Substitution of the Fourier pair shows $\mathcal{F}[h * t] = \widetilde{h}\,\widetilde{t}$: convolution in position becomes multiplication in frequency. This is the algebraic foundation of Fourier optics, not a separate empirical imaging rule.

Chapter 3

Light, phase, rays, and optical path

Textbook connection. Saleh–Teich §§2.1–2.2, printed pp. 40–47 (PDF pp. 62–69), and §2.5, p. 58 (PDF p. 80). Born–Wolf §1.3, p. 14 (PDF p. 43), and §3.1, p. 109 (PDF p. 138).

3.1 From Maxwell's equations to a wave

In a homogeneous, source-free, linear, isotropic, nonmagnetic medium, Maxwell's equations include

$$\nabla \times \boldsymbol{E} = -\mu_0 \frac{\partial \boldsymbol{H}}{\partial t}, \quad \nabla \times \boldsymbol{H} = \epsilon \frac{\partial \boldsymbol{E}}{\partial t}, \quad \nabla \cdot \boldsymbol{E} = 0.$$

Take the curl of the first equation and substitute the second. Using $\nabla \times (\nabla \times \boldsymbol{E}) = \nabla(\nabla \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E}$ gives

$$\nabla^2 \boldsymbol{E} - \mu_0 \epsilon \frac{\partial^2 \boldsymbol{E}}{\partial t^2} = 0. \tag{3.1}$$

The wave speed is $v = 1/\sqrt{\mu_0 \epsilon} = c/n$. For one frequency and a scalar component under a justified approximation,

$$\nabla^2 U + n^2 k_0^2 U = 0, \qquad k_0 = \frac{2\pi}{\lambda_0} = \frac{\omega}{c}. \tag{3.2}$$

This is the Helmholtz equation. A solution $U = A e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ is a plane wave if $|\boldsymbol{k}| = nk_0$. Constant-phase surfaces are planes perpendicular to $\boldsymbol{k}$. given that $|\boldsymbol{k}| = nk_0$, $\lambda_{medium} = \frac{2\pi}{|\boldsymbol{k}|} = \frac{\lambda}{n}$

From the Poynting vector to optical intensity

We now derive the intensity formula rather than introduce it as a separate optical rule. The result for a travelling plane wave in a lossless, nonmagnetic dielectric is

$$I = \frac{n\epsilon_0 c}{2} |\boldsymbol{E}_0|^2.$$

Here $\boldsymbol{E}_0$ is an electric-field phasor, measured in volts per metre, and its components are defined using peak rather than RMS amplitudes. The factor $1/2$ comes from averaging the real oscillating fields. The factor $n\epsilon_0 c$ comes from the relation between the electric and magnetic fields. We will obtain these factors separately.

1. The Poynting vector measures energy flow

For real, instantaneous fields in SI units, the Poynting vector is

$$\boldsymbol{S}(\boldsymbol{r},t) = \boldsymbol{E}(\boldsymbol{r},t) \times \boldsymbol{H}(\boldsymbol{r},t) = \frac{1}{\mu_0}\boldsymbol{E}(\boldsymbol{r},t) \times \boldsymbol{B}(\boldsymbol{r},t), \tag{3.3}$$

where $\boldsymbol{B} = \mu_0 \boldsymbol{H}$ in the nonmagnetic medium considered here. The cross product gives a direction of energy flow. Its units are

$$[\boldsymbol{S}] = \frac{\mathrm{V}}{\mathrm{m}}\frac{\mathrm{A}}{\mathrm{m}} = \frac{\mathrm{W}}{\mathrm{m}^2},$$

so it is power per area. For a surface element $\mathrm{d}A$ with unit normal $\widehat{\boldsymbol{n}}_A$, the signed power crossing it is $\boldsymbol{S} \cdot \widehat{\boldsymbol{n}}_A \, \mathrm{d}A$.

Why does this particular cross product describe energy? For a lossless, nondispersive dielectric with constant ϵ, use the vector identity

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = \boldsymbol{H} \cdot (\nabla \times \boldsymbol{E}) - \boldsymbol{E} \cdot (\nabla \times \boldsymbol{H}).$$

Substituting the source-free Maxwell equations above gives

$$\begin{aligned} \nabla \cdot \boldsymbol{S} &= -\mu_0 \boldsymbol{H} \cdot \frac{\partial \boldsymbol{H}}{\partial t} - \epsilon \boldsymbol{E} \cdot \frac{\partial \boldsymbol{E}}{\partial t} \\ &= -\frac{\partial}{\partial t}\left(\frac{\epsilon |\boldsymbol{E}|^2}{2} + \frac{\mu_0 |\boldsymbol{H}|^2}{2}\right). \end{aligned} \tag{3.4}$$

The expression in parentheses is the electromagnetic energy density u_{em}, in joules per cubic metre. Thus

$$\frac{\partial u_{\mathrm{em}}}{\partial t} + \nabla \cdot \boldsymbol{S} = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\int_V u_{\mathrm{em}} \, \mathrm{d}V = -\oint_{\partial V} \boldsymbol{S} \cdot \widehat{\boldsymbol{n}}_A \, \mathrm{d}A. \tag{3.5}$$

The second equality follows from the divergence theorem: outward energy flow through a boundary reduces the energy stored inside it. This identifies $\boldsymbol{S}$ as an energy-flux density. The simple instantaneous formula for u_{em} assumes no material dispersion; the monochromatic flux calculation below only needs the real, lossless material parameters at the frequency of interest.

2. Average the real fields, not just their phasors

An optical detector or resist exposure averages over many optical cycles. Let $T_{\mathrm{opt}} = 2\pi/\omega$ and define a one-cycle average at a fixed point:

$$\langle \boldsymbol{S} \rangle_t = \frac{1}{T_{\mathrm{opt}}}\int_{t_0}^{t_0+T_{\mathrm{opt}}} \boldsymbol{S}(t) \, \mathrm{d}t.$$

For an exactly monochromatic field this equals an average over any integer number of cycles. Write the real fields in terms of their complex spatial phasors:

$$\begin{aligned} \boldsymbol{E}(\boldsymbol{r},t) &= \mathrm{Re}\{\widetilde{\boldsymbol{E}}(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\omega t}\} = \frac{1}{2}\left(\widetilde{\boldsymbol{E}}\mathrm{e}^{-\mathrm{i}\omega t} + \widetilde{\boldsymbol{E}}^{*}\mathrm{e}^{+\mathrm{i}\omega t}\right), \\ \boldsymbol{H}(\boldsymbol{r},t) &= \mathrm{Re}\{\widetilde{\boldsymbol{H}}(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\omega t}\} = \frac{1}{2}\left(\widetilde{\boldsymbol{H}}\mathrm{e}^{-\mathrm{i}\omega t} + \widetilde{\boldsymbol{H}}^{*}\mathrm{e}^{+\mathrm{i}\omega t}\right). \end{aligned}$$

Suppressing the common position argument, their cross product is

$$\begin{aligned} \boldsymbol{E} \times \boldsymbol{H} = \frac{1}{4}\Big[&(\widetilde{\boldsymbol{E}} \times \widetilde{\boldsymbol{H}})\mathrm{e}^{-2\mathrm{i}\omega t} + \widetilde{\boldsymbol{E}} \times \widetilde{\boldsymbol{H}}^{*} \\ &+ \widetilde{\boldsymbol{E}}^{*} \times \widetilde{\boldsymbol{H}} + (\widetilde{\boldsymbol{E}}^{*} \times \widetilde{\boldsymbol{H}}^{*})\mathrm{e}^{+2\mathrm{i}\omega t}\Big]. \end{aligned}$$

The first and last terms average to zero because $\langle \mathrm{e}^{\pm 2\mathrm{i}\omega t}\rangle_t = 0$. The two remaining terms are complex conjugates of one another. Since $z + z^* = 2\,\mathrm{Re}\,z$, applied component by component,

$$\boxed{\langle \boldsymbol{S}\rangle_t = \frac{1}{2}\,\mathrm{Re}\Big(\widetilde{\boldsymbol{E}} \times \widetilde{\boldsymbol{H}}^{\,*}\Big)\,.} \tag{3.6}$$

The conjugation and the real part are essential: an unmodified cross product of complex phasors is not the measured average power flow. This expression also applies to monochromatic fields that are not plane waves; the following reduction to electric-field magnitude alone uses the plane-wave assumptions.

3. Maxwell's equations relate the two field amplitudes

For a travelling plane wave, write

$$\widetilde{\boldsymbol{E}}(\boldsymbol{r}) = \boldsymbol{E}_0 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}, \qquad \widetilde{\boldsymbol{H}}(\boldsymbol{r}) = \boldsymbol{H}_0 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}, \qquad \boldsymbol{k} = k\widehat{\boldsymbol{s}},$$

where $\widehat{\boldsymbol{s}}$ is a real unit vector in the propagation direction and $k = n\omega/c$. For the time convention $\mathrm{e}^{-\mathrm{i}\omega t}$, Faraday's law becomes

$$\nabla \times \widetilde{\boldsymbol{E}} = \mathrm{i}\omega\mu_0 \widetilde{\boldsymbol{H}}.$$

Taking the curl of the plane-wave exponential gives $\nabla \times \widetilde{\boldsymbol{E}} = \mathrm{i}\boldsymbol{k} \times \widetilde{\boldsymbol{E}}$. Cancel the common exponential and i to obtain

$$\boldsymbol{H}_0 = \frac{\boldsymbol{k} \times \boldsymbol{E}_0}{\omega\mu_0} = \frac{1}{\eta}\widehat{\boldsymbol{s}} \times \boldsymbol{E}_0, \qquad \eta = \sqrt{\frac{\mu_0}{\epsilon}} = \frac{\eta_0}{n}, \qquad \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}. \tag{3.7}$$

We used $\epsilon = n^2\epsilon_0$ and $c = 1/\sqrt{\mu_0\epsilon_0}$. The quantity η is the wave impedance: it sets the ratio of electric to magnetic field amplitude and has units of ohms. Also, $\nabla \cdot \widetilde{\boldsymbol{E}} = 0$ gives $\widehat{\boldsymbol{s}} \cdot \boldsymbol{E}_0 = 0$, so the electric field is transverse. The magnetic field is transverse as well, and its direction follows from the cross product in Eq. (3.7).

4. Obtain the intensity and its direction

Because $\boldsymbol{k}$ is real, the spatial factors $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ and $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$ cancel in Eq. (3.6). Because η is real,

$$\begin{aligned}
\langle \boldsymbol{S}\rangle_t &= \frac{1}{2\eta}\,\mathrm{Re}[\boldsymbol{E}_0 \times (\widehat{\boldsymbol{s}} \times \boldsymbol{E}_0^*)] \\
&= \frac{1}{2\eta}\,\mathrm{Re}[\widehat{\boldsymbol{s}}(\boldsymbol{E}_0 \cdot \boldsymbol{E}_0^*) - \boldsymbol{E}_0^*(\boldsymbol{E}_0 \cdot \widehat{\boldsymbol{s}})] \\
&= \frac{|\boldsymbol{E}_0|^2}{2\eta}\widehat{\boldsymbol{s}}.
\end{aligned}$$

The second line is the vector triple-product identity. Transversality removes its second term, while $|\boldsymbol{E}_0|^2 = \boldsymbol{E}_0 \cdot \boldsymbol{E}_0^* = |E_{0x}|^2 + |E_{0y}|^2 + |E_{0z}|^2$ is real and nonnegative. Define the plane-wave intensity as the average power per area perpendicular to propagation:

$$\boxed{I = \langle \boldsymbol{S}\rangle_t \cdot \widehat{\boldsymbol{s}} = \frac{|\boldsymbol{E}_0|^2}{2\eta} = \frac{n\epsilon_0 c}{2}|\boldsymbol{E}_0|^2.} \tag{3.8}$$

For the last equality, $1/\eta = n/\eta_0 = n\sqrt{\epsilon_0/\mu_0} = n\epsilon_0 c$. The result holds for linear, elliptical, and circular polarization in this medium; it uses the sum of the squared component phasor magnitudes.

If a detector surface normal makes angle α with $\widehat{\boldsymbol{s}}$, its incident power per detector area is $\langle \boldsymbol{S}\rangle_t \cdot \widehat{\boldsymbol{n}}_A = I\cos\alpha$, taking the normal toward the direction of energy flow. Thus one must specify the area orientation when converting flux into detected power.

5. Check the factor of two with a real cosine wave

Take a linearly polarized wave travelling along $+z$:

$$\boldsymbol{E} = \hat{\boldsymbol{x}} E_{\mathrm{pk}} \cos(kz - \omega t), \qquad \boldsymbol{H} = \hat{\boldsymbol{y}} \frac{E_{\mathrm{pk}}}{\eta} \cos(kz - \omega t).$$

Then

$$\boldsymbol{S} = \hat{\boldsymbol{z}} \frac{E_{\mathrm{pk}}^2}{\eta} \cos^2(kz - \omega t), \qquad \left\langle \cos^2(kz - \omega t) \right\rangle_t = \frac{1}{2},$$

which gives $I = E_{\mathrm{pk}}^2/(2\eta)$ directly. If the field is instead specified by $E_{\mathrm{rms}} = E_{\mathrm{pk}}/\sqrt{2}$, the same physical result is $I = E_{\mathrm{rms}}^2/\eta$. The two formulas use different amplitude conventions, so the factor $1/2$ must not be inserted twice. As a units-and-scale check, a vacuum plane wave with $E_{\mathrm{pk}} = 1\,\mathrm{V/m}$ has $I = 1/(2\eta_0) \simeq 1.33 \times 10^{-3}\,\mathrm{W/m^2}$, using $\eta_0 \simeq 376.73\,\Omega$.

6. Why scalar optics can write $I = |U|^2$

For a fixed unit polarization vector $\hat{\boldsymbol{e}}$, write $\widetilde{\boldsymbol{E}} = \hat{\boldsymbol{e}}\,\mathcal{E}$ with $\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{e}}^* = 1$. Here $\mathcal{E}$ is the scalar electric-field phasor in volts per metre. Under the travelling-wave relation just derived,

$$I = \frac{n\epsilon_0 c}{2} |\mathcal{E}|^2.$$

Define a rescaled scalar amplitude

$$U = \sqrt{\frac{n\epsilon_0 c}{2}}\,\mathcal{E} \quad \Longrightarrow \quad I = |U|^2. \tag{3.9}$$

The constant is absorbed into the *amplitude through its square root*. This U has units $\mathrm{W}^{1/2}/\mathrm{m}$ if I is in $\mathrm{W/m^2}$; it is no longer the numerical electric field in V/m. Multiplying by a spatially constant factor leaves the homogeneous Helmholtz equation unchanged. In a scalar, paraxial imaging model with a common polarization and nearly common propagation direction, this normalization is why we may add the scalar fields and then take $|U|^2$ to obtain intensity.

One may also divide intensity by a fixed reference intensity and use dimensionless fields. Neither rescaling changes the interference or edge-position argument, provided the threshold and dose use the same normalization. For absorbing media, standing waves, or fields with substantially different propagation directions, return to the full flux expression in Eq. (3.6); the simple real-n plane-wave prefactor is not a universal replacement for it.

3.2 Optical path is not just distance

Along a geometrical trajectory of arc length ℓ,

$$\mathcal{L} = \int n(\boldsymbol{r})\,\mathrm{d}\ell. \tag{3.10}$$

A uniform slab of thickness t and index n has optical path nt. Phase accumulated along a ray is $k_0\mathcal{L}$ in the $\mathrm{e}^{-\mathrm{i}\omega t}$ convention. Comparing actual and reference optical paths gives OPD W and phase error

$$\phi_W = k_0 W = \frac{2\pi W}{\lambda_0}. \tag{3.11}$$

A reference must be specified: usually an ideal spherical wave converging to a chosen image point. Moving that reference point changes the reported tilt and defocus, even when the physical system is unchanged.

Worked example: The wavelength comparison in the Introduction

For fixed physical RMS OPD of 1.5 nm,

$$\frac{1.5}{193} = 0.0077720 \text{ waves} = 7.7720\,\mathrm{m}\lambda, \qquad \frac{1.5}{13.5} = 0.111111 \text{ waves} = 111.111\,\mathrm{m}\lambda.$$

The ratio is $193/13.5 = 14.2963$. The length error has not grown. Its size measured in wavelengths, and therefore its phase error, has grown. This is the precise meaning of the paper's inverse-wavelength statement.

3.3 How rays emerge from wave theory

Write $U = A(\boldsymbol{r})\mathrm{e}^{\mathrm{i}k_0 S(\boldsymbol{r})}$, where S has length units. Substitute into the Helmholtz equation:

$$\nabla^2 A + 2\mathrm{i}k_0 \nabla A \cdot \nabla S + \mathrm{i}k_0 A \nabla^2 S - k_0^2 A|\nabla S|^2 + n^2 k_0^2 A = 0.$$

When amplitude and index vary slowly on the wavelength scale, the leading real terms require

$$|\nabla S|^2 = n^2. \tag{3.12}$$

This is the eikonal equation. The unit ray direction is $\boldsymbol{d} = \nabla S/n$. Rays are normal to wavefronts. The next terms give amplitude transport; ray geometry alone does not determine interference.

For nearby actual and reference eikonals, $S = S_0 + W$, the first-order transverse direction-cosine difference is

$$\delta \boldsymbol{d}_\perp \simeq \frac{1}{n}\nabla_\perp W. \tag{3.13}$$

Over a paraxial propagation distance L, the corresponding displacement is approximately $L\nabla_\perp W/n$. This is a local geometrical statement. A finite-aperture lithographic image still requires field summation and source averaging.

> **Read carefully.** A coefficient's sign depends on the OPD definition, time convention, pupil axes, and image inversion. This book uses pupil phase $\exp(+\mathrm{i}2\pi W/\lambda)$ and Eq. (2.18). In transfer-function formulas the normalized pupil labels spatial frequency, $\boldsymbol{u} = \boldsymbol{f}/f_c$; it need not have the orientation of the physical exit-pupil coordinates in a ray drawing. Image-shift signs below follow the declared transfer-function convention.

3.4 What a surface error does at a mirror

For a mirror displaced a small normal distance h, the incident and reflected segments each change optical path by approximately $h\cos\alpha$, with α measured from the surface normal. In vacuum and for fixed local ray geometry,

$$|\delta W| \simeq 2|h|\cos\alpha. \tag{3.14}$$

The sign depends on positive surface displacement. At normal incidence, a 0.1 nm height error gives about 0.2 nm OPD. Wavefront tolerance and surface-height tolerance must not be interchanged.

This formula excludes coating phase, changed intersections, multiple-mirror coordinate mappings, and polarization. Those belong in a full sensitivity model. A lens-thickness perturbation in air often gives approximately $(n-1)\delta t$ OPD rather than the mirror factor of two.

3.5 Polarization and the limits of a scalar field

The electric field is a vector perpendicular to the propagation direction for a plane wave in an isotropic medium. The interference cross term is $2\,\mathrm{Re}(\boldsymbol{U}_1 \cdot \boldsymbol{U}_2^*)$. Orthogonal polarizations therefore have no such cross term in total intensity. Reflection gives different amplitudes and phases to fields perpendicular and parallel to the incidence plane.

Scalar optics is useful when these vector effects can be separated or neglected. At an EUV mask with finite-height absorbing features, wavelength-scale layers, oblique illumination, and polarization-dependent scattering, a full electromagnetic calculation may be necessary. Rigorous simulation refers to a more complete numerical treatment of the specified model; it does not eliminate missing inputs or manufacturing and resist uncertainty.

Chapter 4

Pupils, diffraction, resolution, and focus

Textbook connection. Saleh–Teich §§4.1–4.4, printed pp. 105–137. Start at PDF pp. 127, 138, 143, and 149. Born–Wolf §8.3, printed p. 375 (PDF p. 413), and §8.5.2, printed p. 395; the Airy formula is on p. 396 (PDF p. 434).

4.1 Aperture stops, pupils, and numerical aperture

An aperture stop limits the cone of rays that an optical system accepts from an object point. The entrance pupil is the image of that stop seen through the optics before it; the exit pupil is its image through the optics after it. A pupil can therefore be a virtual image rather than a physical hole at the plane where it is drawn.

For a rotationally symmetric image-side cone of half-angle $\alpha_{\max}$,

$$\mathrm{NA} = n \sin \alpha_{\max}. \tag{4.1}$$

In vacuum $n = 1$. Thus $\mathrm{NA} = 0.33$ corresponds to $\alpha_{\max} = \arcsin(0.33) \simeq 19.27^\circ$. The chief ray goes through the centre of the aperture stop; marginal rays reach its edge. These are different rays. The mask chief-ray angle of 6° in Table 1 is not the same angle as the wafer-side numerical-aperture cone.

In a paraxial system with pupil radius a and effective image distance f, $\mathrm{NA} \simeq a/f$ in air. The exact sine definition is the basic one. The optical invariant gives approximately $\mathrm{NA}_{\mathrm{object}} \simeq |M| \mathrm{NA}_{\mathrm{image}}$ for equal media and lateral magnification M. A 4:1 reduction has $|M| = 1/4$; object and image NAs must not be silently interchanged.

4.2 Propagation as a sum of angular components

Insert a transverse Fourier component $\exp[\mathrm{i}2\pi(f_x x + f_y y)] \exp(\mathrm{i}k_z z)$ into the Helmholtz equation in vacuum. The result is

$$k_z^2 = k_0^2 - (2\pi)^2 (f_x^2 + f_y^2).$$

Each propagating component therefore acquires

$$H_z(\boldsymbol{f}) = \exp\left[\mathrm{i}k_0 z \sqrt{1 - \lambda^2 |\boldsymbol{f}|^2}\right]. \tag{4.2}$$

For $|\boldsymbol{f}| > 1/\lambda$, the longitudinal wave number is imaginary and the component is evanescent. The far-field projector does not collect arbitrarily fine near-field detail.

When $\lambda|\boldsymbol{f}| \ll 1$,

$$H_z(\boldsymbol{f}) \simeq \mathrm{e}^{\mathrm{i}k_0 z}\mathrm{e}^{-\mathrm{i}\pi\lambda z|\boldsymbol{f}|^2}. \tag{4.3}$$

Inverse transformation yields the Fresnel propagation integral

$$U(x,y;z) = \frac{\mathrm{e}^{\mathrm{i}k_0 z}}{\mathrm{i}\lambda z}\iint U(x',y';0)\exp\left[\frac{\mathrm{i}\pi}{\lambda z}\{(x-x')^2+(y-y')^2\}\right]\mathrm{d}x'\mathrm{d}y'. \tag{4.4}$$

One way to obtain the same phase is to expand the distance $R = \sqrt{z^2+(x-x')^2+(y-y')^2}$ as $z + \{(x-x')^2+(y-y')^2\}/(2z)$. The amplitude $1/R$ is approximated by $1/z$. These are two approximations, and the phase approximation is often the stricter one because it is multiplied by the large wave number.

If the quadratic source-coordinate phase is negligible, $a^2/(\lambda z) \ll 1$ for characteristic aperture radius a, Eq. (4.4) becomes a Fourier transform:

$$U(x,y;z) \propto \widetilde{U}\left(\frac{x}{\lambda z}, \frac{y}{\lambda z}\right)$$

apart from a known output quadratic phase. This is Fraunhofer diffraction. The dimensionless ratio $N_F = a^2/(\lambda z)$ is the Fresnel number. A focusing optic can cancel the required quadratic phase, giving a Fourier-related focal plane without requiring a physically enormous propagation distance.

4.3 The pupil function and the point-spread function

Define a scalar coherent transfer function

$$P(\boldsymbol{f}) = A(\boldsymbol{f})\,\mathbf{1}_{|\boldsymbol{f}|\le f_c}\exp\left[\frac{\mathrm{i}2\pi}{\lambda}W(\boldsymbol{f}/f_c)\right], \qquad f_c = \frac{\mathrm{NA}}{\lambda}. \tag{4.5}$$

Here $\mathbf{1}$ equals one inside the admitted disk and zero outside, A describes amplitude transmission or reflection, and W is the phase-equivalent OPD. The normalized pupil coordinate is $\boldsymbol{u} = \boldsymbol{f}/f_c = (\rho\cos\theta, \rho\sin\theta)$.

For coherent illumination of a thin scalar object with complex transmittance or reflectance t,

$$U_{\mathrm{image}} = \mathcal{F}^{-1}[\widetilde{t}\,P] = t * h, \qquad h = \mathcal{F}^{-1}[P]. \tag{4.6}$$

h is the amplitude point-spread function (PSF). The intensity PSF is $|h|^2$. A point object contains all transverse spatial frequencies; the pupil selects the part used to reconstruct it. This is why a finite aperture produces a finite spot.

For a clear circular pupil with no aberration, let $r = \sqrt{x^2+y^2}$. Angular integration gives a Bessel function:

$$h(r) = 2\pi\int_0^{f_c} f J_0(2\pi f r)\mathrm{d}f = \frac{f_c J_1(2\pi f_c r)}{r}.$$

Since $h(0) = \pi f_c^2$, the normalized intensity is

$$\frac{I(r)}{I(0)} = \left[\frac{2J_1(v)}{v}\right]^2, \qquad v = 2\pi f_c r. \tag{4.7}$$

The first zero of J_1 is $v \simeq 3.8317$, so

$$r_{\text{first zero}} = 0.6098\,\frac{\lambda}{\mathrm{NA}}. \tag{4.8}$$

This is the Airy pattern. Its rings redistribute energy rather than indicating geometric rays bouncing around the image.

Figure 4.1 shows the intensity given by Eq. (4.7). The central bright region, bounded by the first dark ring, is the *Airy disk*; the full Airy pattern also includes the surrounding rings. The value $0.6098\lambda/\mathrm{NA}$ is the disk's *radius*, not its diameter. At $\lambda = 13.5\,\mathrm{nm}$ and $\mathrm{NA} = 0.33$, this radius is about $24.95\,\mathrm{nm}$.

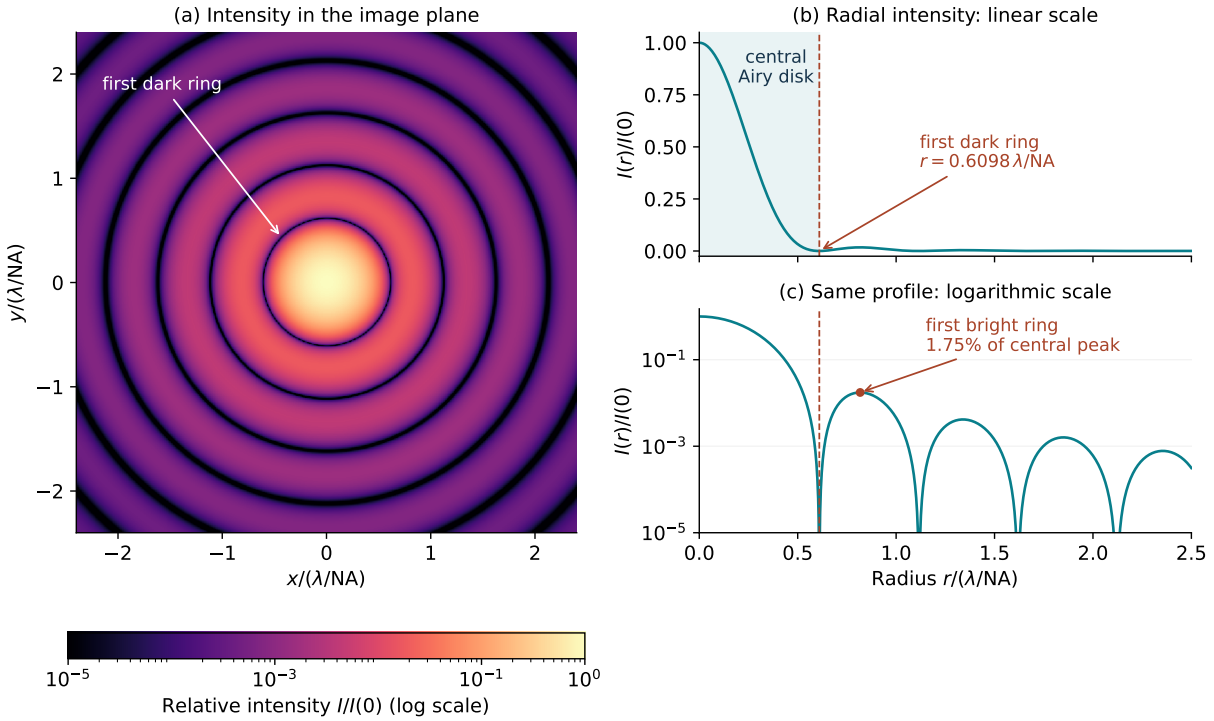


Figure 4.1: Airy intensity pattern for a uniformly illuminated, unobscured circular pupil with no aberration. Coordinates are measured in units of λ/NA, and intensity is normalized to its on-axis value. (a) The image-plane pattern uses a logarithmic colour scale to reveal the faint rings; colour does not represent wavelength. (b) The radial intensity on a linear scale shows the dominant central spot. The dashed line marks the first dark ring at $r = 0.6098\lambda/\mathrm{NA}$. (c) The same radial profile on a logarithmic scale reveals the outer rings. The first bright ring reaches only about 1.75 percent of the central peak. The logarithmic displays stop at $I/I(0) = 10^{-5}$; the ideal dark-ring minima are zero.

4.4 Coherent and incoherent resolution are different questions

For coherent illumination, amplitudes from object points interfere before we square the field. For mutually incoherent object points, cross terms average out and object *intensity* is convolved with $|h|^2$. The incoherent optical transfer function is consequently a pupil autocorrelation:

$$\mathrm{OTF}(\boldsymbol{q}) = \frac{\int P(\boldsymbol{f} + \boldsymbol{q}/2)P^*(\boldsymbol{f} - \boldsymbol{q}/2)\mathrm{d}^2 f}{\int |P(\boldsymbol{f})|^2 \mathrm{d}^2 f}. \tag{4.9}$$

Two disks of radius f_c cease to overlap when their separation exceeds $2f_c$. This explains the incoherent intensity cutoff $2\mathrm{NA}/\lambda$, whereas the on-axis coherent field cutoff is NA/λ. Partial coherence, used in lithography, requires the source-dependent formulation developed later.

The Rayleigh two-point separation $0.61\lambda/\mathrm{NA}$ is a criterion for distinguishing point-source images. It is not a universal minimum thresholded hole diameter. Periodic patterns, oblique illumination,

resist thresholds, and proximity correction pose a different problem. Do not conclude that the paper's 14 nm contacts are impossible merely because $0.61(13.5)/0.33 \simeq 24.95\,\text{nm}$.

Worked example: A sinusoidal mask between the two cutoffs

Figure 4.2 illustrates a case in which spatially incoherent imaging preserves modulation that normally incident coherent illumination loses. We specify both the optical system and the source: changing coherence alone is not a complete experimental description.

Optical system and mask, common to both cases

Use a scalar, paraxial, linear, shift-invariant imaging system in air ($n \simeq 1$), at best focus, with unit lateral magnification $|M| = 1$. The objective is free of aberration and has a clear, uniformly weighted, unobscured circular pupil. Its image-side NA is 0.05, and the wavelength is 500 nm. Both sources are sufficiently narrowband that we may use this single wavelength; the polarization is fixed and treated as scalar. Thus the coherent field transfer is

$$P(f_x, f_y) = \mathbf{1}_{\sqrt{f_x^2+f_y^2}\le f_c}, \qquad f_c = \frac{\text{NA}}{\lambda} = 0.100\ \mu\text{m}^{-1}.$$

Here μm^{-1} denotes cycles per micrometre. The detector measures time-averaged optical intensity, with sufficient spatial sampling; noise, resist processing, and thresholding are outside this example.

Let the thin mask have real, positive amplitude transmittance

$$t(x) = t_0 + \epsilon\cos(2\pi q x), \qquad t_0 = 0.50, \quad \epsilon = 0.05, \quad q = 1.5 f_c. \tag{4.10}$$

It is uniform in y and periodically extended in x, so there are no finite-mask edge effects in the model. Its amplitude varies from 0.45 to 0.55; its intensity transmission is $|t|^2$, not t. The modulation is weak because $\epsilon/t_0 = 0.10$. The spatial frequency is $q = 0.150\ \mu\text{m}^{-1}$, and its period is $p = 1/q = 6.67\ \mu\text{m}$. In particular, $f_c < q < 2f_c$. All positions are expressed in the common object/image length scale.

Source conditions

	Case A: coherent	Case B: incoherent imaging response
Illumination	One collimated, normally incident plane wave, with uniform amplitude over the mask.	A centred, extended angular source: mutually incoherent plane-wave components fill a circular illumination pupil.
Angular support	A single source point at transverse frequency $\boldsymbol{s} = 0$.	Uniform weight per area in transverse-frequency space for $\lvert\boldsymbol{s}\rvert \le 3f_c$; illumination NA = 0.15.
Phase relation	The field across the mask has a stable spatial phase relation.	Different source directions have no time-averaged mutual interference; their image intensities are added.

In both cases the central wavelength is 500 nm. In Case B, an extended narrowband source in a condenser's illumination pupil provides a model for the angular distribution; exposure is long enough to average the relative phase fluctuations. This is not a single normally incident wave with an arbitrary label of incoherence.

The finite source in Case B reproduces the spatially incoherent imaging formula for this particular mask. Its field spectrum contains only $0, +q, -q$. Every source direction that can send one of

those orders through the objective lies within $|\boldsymbol{s}| \le q + f_c = 2.5 f_c$. The uniform source of radius $3f_c$ therefore covers all the relevant pupil-overlap regions. After the calibration below, averaging over these source directions gives the full pupil autocorrelation in Eq. (4.9). A finite source need not have zero spatial coherence at every separation to give this response for the specified spectrum. Chapter 8 develops the source-point calculation.

Intensity calibration. We normalize each case to its response to a uniform mask $t = t_0$, giving the reference level $t_0^2 = 0.25$ in both cases. With equal incident irradiance, the uniform angular source of radius $3f_c$ sends only 1/9 of its power through the objective for a uniform mask in this model. Case B is therefore divided by that reference throughput, or equivalently its source power could be increased by nine. The figure compares pattern contrast after this stated calibration; it does not compare photon efficiency or signal-to-noise ratio.

Case A: the pupil removes the field sidebands

Writing the cosine as two exponentials,

$$t(x) = t_0 + \frac{\epsilon}{2} \mathrm{e}^{+\mathrm{i}2\pi q x} + \frac{\epsilon}{2} \mathrm{e}^{-\mathrm{i}2\pi q x}.$$

Normal incidence places the three field orders at $0, +q, -q$ in the pupil-frequency plane. Since $q > f_c$, both sidebands are blocked:

$$U_A(x) = t_0, \qquad I_A(x) = |U_A(x)|^2 = t_0^2 = 0.25. \tag{4.11}$$

Only a spatially constant field survives, so the image has no sinusoidal modulation.

Case B: some intensity modulation survives

For a clear circular pupil, the normalized overlap area in Eq. (4.9) gives the real, nonnegative intensity transfer

$$T(f) = \begin{cases} \dfrac{2}{\pi}\left[\cos^{-1}\nu - \nu\sqrt{1-\nu^2}\right], & 0 \le \nu \le 1, \\ 0, & \nu > 1, \end{cases} \qquad \nu = \frac{|f|}{2f_c}. \tag{4.12}$$

Here $T(f) = \mathrm{OTF}(f, 0)$ and $T(0) = 1$. Its support extends to $2f_c$, but its value falls toward zero: being below the cutoff does not mean being transmitted at full contrast. At our chosen frequency, $\nu = 0.75$ and $T(q) = 0.144294$.

Expand the object intensity exactly:

$$|t(x)|^2 = t_0^2 + \frac{\epsilon^2}{2} + 2t_0\epsilon\cos(2\pi q x) + \frac{\epsilon^2}{2}\cos(4\pi q x).$$

In incoherent imaging, each *intensity* Fourier component is multiplied by T. The DC term passes unchanged, the term at q is attenuated, and the term at $2q = 3f_c$ is removed. Thus

$$\begin{aligned} I_B(x) &= t_0^2 + \frac{\epsilon^2}{2} + 2t_0\epsilon T(q)\cos(2\pi q x) \\ &= 0.25125 + 0.00721468\cos(2\pi q x). \end{aligned} \tag{4.13}$$

Dropping terms of order ϵ^2 gives the weak-modulation expression $I_B \simeq t_0^2 + 2t_0\epsilon T(q)\cos(2\pi q x)$. The figure retains the exact $\epsilon^2/2$ DC term; this explains why the mean of the incoherent curve sits slightly above the coherent curve.

For a quantitative comparison, define Michelson contrast

$$C = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}.$$

The mask's intensity transmission has $C = 19.80$ percent, the coherent image has $C = 0$, and the incoherent image has $C = 2.87$ percent. The original period survives in Case B with reduced contrast.

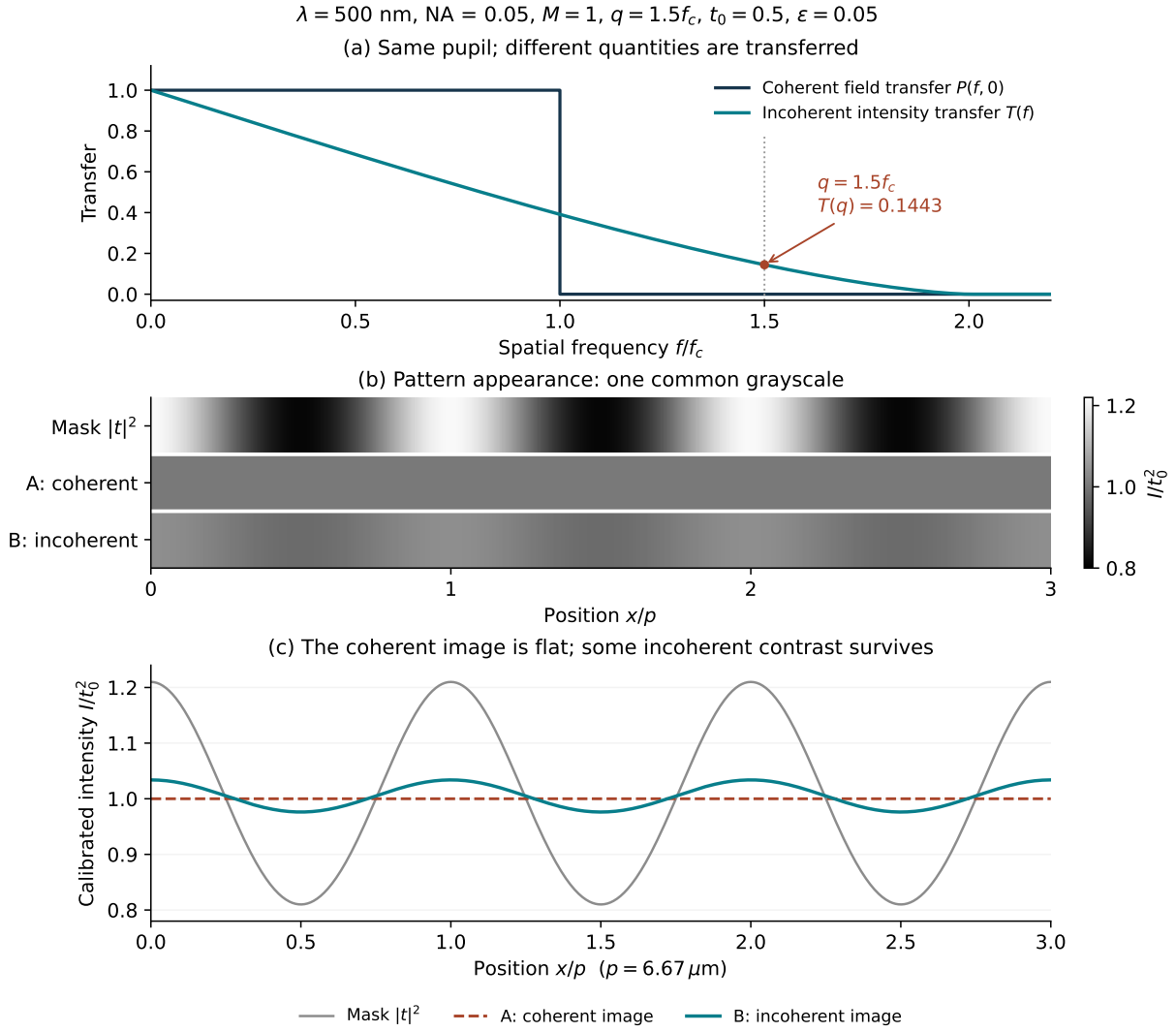


Figure 4.2: Sinusoidal-mask comparison under the optical and source conditions stated in Section 4.4. (a) The coherent curve transfers field amplitude; the incoherent curve transfers intensity modulation. Their common pupil has field cutoff f_c, while $q = 1.5f_c$ lies inside the intensity cutoff $2f_c$. (b) Mask intensity transmission and the two images use one common grayscale, with no separate contrast stretching. (c) The corresponding profiles are divided by t_0^2 after the stated flat-field calibration. Case A uses one normally incident coherent wave; Case B averages a uniform disk of mutually incoherent source directions with illumination NA 0.15. The objective NA is 0.05 in both cases. The incoherent image retains 2.87 percent contrast; the coherent image is uniform.

No objective aperture has become larger in Case B. For example, a source direction $s_x = -q/2$ places the zero and $+q$ orders at $-0.75f_c$ and $+0.75f_c$, both inside the same pupil. Their interference can produce intensity variation at their difference frequency q. The source average retains some of that modulation. This advantage is specific to the stated comparison: oblique

coherent illumination can also transmit an appropriate pair of orders. The paper's quasar source is a separate, partially coherent arrangement, studied in Chapter 8; the present visible-light example isolates the coherence and cutoff mechanism.

4.5 Resolution and depth-of-focus scaling

Lithography often summarizes a process-dependent minimum feature scale as

$$\mathrm{CD} \sim k_1 \frac{\lambda}{\mathrm{NA}}, \tag{4.14}$$

where k_1 absorbs pattern, illumination, mask, resist, and process choices. For 14 nm, 13.5 nm wavelength, and 0.33 NA, the numerical ratio is $k_1 = 14(0.33)/13.5 \simeq 0.3422$. This calculation characterizes the geometry; it does not prove a process window or assign the same k_1 to every pitch.

Defocus adds a relative phase from Eq. (4.2). Subtract the on-axis piston:

$$\phi_z(\rho) = \frac{2\pi z}{\lambda}\left[\sqrt{1-\mathrm{NA}^2\rho^2}-1\right] \tag{4.15}$$

$$\simeq -\frac{\pi z \mathrm{NA}^2}{\lambda}\rho^2. \tag{4.16}$$

When the pupil-edge phase becomes a fixed tolerable number, the corresponding defocus scales as λ/NA^2:

$$\mathrm{DOF} \sim k_2 \frac{\lambda}{\mathrm{NA}^2}. \tag{4.17}$$

Whether DOF denotes a half-range or a full range changes the convention for k_2. The paper specifies sampled focus values, not a universal k_2.

At its conditions,

$$\lambda/\mathrm{NA} = 40.91\,\mathrm{nm}, \qquad \lambda/\mathrm{NA}^2 = 123.97\,\mathrm{nm}.$$

Increasing NA improves lateral resolution but generally makes focus more sensitive. Reducing wavelength improves resolution but also reduces this focus scale at fixed NA. Reducing NA has the opposite resolution effect. The Introduction's combined discussion of smaller wavelength and NA should be read with these two distinct scalings in mind.

4.6 A quantitative focus check

From Eq. (4.16),

$$W_z(\rho) \simeq -\frac{z\mathrm{NA}^2}{2}\rho^2.$$

Remove its mean and use $\mathrm{Var}(\rho^2) = 1/12$:

$$\sigma_{W,z} = \frac{|z|\mathrm{NA}^2}{4\sqrt{3}}. \tag{4.18}$$

At $z = 15\,\mathrm{nm}$, $\sigma_{W,z} = 0.235775\,\mathrm{nm} = 17.4648\,\mathrm{m}\lambda$. The pupil-edge-to-centre OPD is $0.81675\,\mathrm{nm}$. These two values are different because RMS, peak-to-valley, and edge-to-centre error are different statistics.

The exact angular formula gives an edge OPD $15[1-\sqrt{1-0.33^2}] \simeq 0.8403\,\mathrm{nm}$. The paraxial value is a useful approximation, not an exact high-angle propagation result.

Chapter 5

Why EUV needs mirrors and a three-dimensional mask

Paper location: Fig. 1, Table 1, Fig. 3, and the mask discussion in Section 2A.

> **Textbook connection.** Born–Wolf §1.5, printed p. 36 (PDF p. 65), and §1.6, p. 51 (PDF p. 80); its characteristic-matrix discussion begins at p. 55 (PDF p. 84). Saleh–Teich §6.2, p. 209 (PDF p. 231).

5.1 The optical path through an EUV exposure system

The useful sequence is source, collector and illumination optics, reflective patterned mask, projection mirrors, and resist-coated wafer. Illumination optics control the angular distribution incident on the mask. Projection optics determine which mask-diffracted components reach the wafer and with what amplitude and phase.

At 13.5 nm, strong absorption makes ordinary transmissive lens trains and propagation through air unsuitable. EUV scanners use vacuum and multilayer mirrors. ASML's NXE:3400C documentation specifies reflective 4:1 reduction optics and 0.33 NA [4]. The paper's Fig. 1 caption says NXT3400; the relevant EUV family is NXE:3400. This naming issue does not alter the optical calculations.

A geometrical mirror's reflection direction is largely independent of wavelength, so an all-reflective design avoids ordinary refractive chromatic aberration. Its coatings still have wavelength-dependent reflectance and phase. Mirrors also absorb energy, heat, deform, and move. Thus reflective does not mean perfectly achromatic in every optical property or immune to thermal aberration.

5.2 Complex refractive index and absorption

With time convention $\mathrm{e}^{-\mathrm{i}\omega t}$, write a passive EUV material's index as $\widetilde{n} = 1 - \delta + \mathrm{i}\beta$, where $\beta > 0$. A normally propagating field behaves as

$$U(z) = U(0)\mathrm{e}^{\mathrm{i}k_0(1-\delta)z}\mathrm{e}^{-k_0\beta z}.$$

Hence

$$I(z) = I(0)\mathrm{e}^{-\mu_{\mathrm{abs}}z}, \qquad \mu_{\mathrm{abs}} = \frac{4\pi\beta}{\lambda}. \tag{5.1}$$

The real part affects phase; the imaginary part attenuates amplitude. Material optical constants therefore influence both image contrast and phase. Treating an absorber as merely a perfectly black two-dimensional painted region omits this physics.

5.3 Reflection at an interface

Continuity of tangential electric and magnetic fields at an interface gives, for a chosen polarization, equations of the form

$$1 + r = t, \qquad Y_0(1 - r) = Y_1 t.$$

Here Y_j is the appropriate optical admittance for that polarization. Solving,

$$r_{01} = \frac{Y_0 - Y_1}{Y_0 + Y_1}, \qquad t_{01} = \frac{2Y_0}{Y_0 + Y_1}. \tag{5.2}$$

For electric field perpendicular to the incidence plane, Y_j is proportional to $\widetilde{n}_j \cos\theta_j$. For the tangential electric-field convention of parallel polarization it is proportional to $\widetilde{n}_j / \cos\theta_j$. Different polarization amplitude conventions can reverse a reflected amplitude sign; physical intensities and phase comparisons must use one convention consistently.

Snell's relation is $\widetilde{n}_0 \sin\theta_0 = \widetilde{n}_1 \sin\theta_1$. For absorbing layers the transmitted angle and longitudinal wave number are generally complex. The physical branch decays into the medium.

5.4 A thin film: summing all internal reflections

Consider a layer 1 of thickness d between media 0 and 2. The phase for one transit is $\delta_1 = k_0 \widetilde{n}_1 d \cos\theta_1$. The first reflection is r_{01}. The first wave returning from the lower interface contributes $t_{01} t_{10} r_{12} \mathrm{e}^{2\mathrm{i}\delta_1}$. Each additional round trip multiplies this by $r_{10} r_{12} \mathrm{e}^{2\mathrm{i}\delta_1}$. Summing the geometric series,

$$r_{\mathrm{film}} = r_{01} + \frac{t_{01} t_{10} r_{12} \mathrm{e}^{2\mathrm{i}\delta_1}}{1 - r_{10} r_{12} \mathrm{e}^{2\mathrm{i}\delta_1}}.$$

Using $r_{10} = -r_{01}$ and $t_{01} t_{10} = 1 - r_{01}^2$ gives

$$r_{\mathrm{film}} = \frac{r_{01} + r_{12} \mathrm{e}^{2\mathrm{i}\delta_1}}{1 + r_{01} r_{12} \mathrm{e}^{2\mathrm{i}\delta_1}}. \tag{5.3}$$

Replace the lower-interface reflection by the effective reflection of all deeper layers and repeat recursively. This constructs a multilayer response. The equivalent transfer-matrix method propagates tangential fields through each layer and multiplies the matrices in physical order. The reflected intensity is $|r_{\mathrm{stack}}|^2$ in the incident medium, and the reflected phase is $\arg r_{\mathrm{stack}}$.

This derivation explains why a small thickness error can change reflectance and phase simultaneously. It also explains why a multilayer reflector cannot be represented by surface height alone.

5.5 The Mo/Si period and the Bragg estimate

Reflections from successive layers can reinforce when their round-trip phase difference is approximately 2π. Neglecting refractive corrections for an estimate, a repeated period d at angle α from the normal obeys

$$2d \cos\alpha \simeq m\lambda. \tag{5.4}$$

This is a design estimate; the full result follows from the complex multilayer calculation above, including interfaces, absorption, and termination.

The paper gives Mo thickness 0.725 nm, Si thickness 1 nm, and absorber height 15 nm, explicitly in *wafer scale*. Under a 4:1 scale conversion these correspond to physical mask values 2.9 nm, 4 nm, and 60 nm. The inferred period is 6.9 nm, and

$$2(6.9) \cos 6^\circ = 13.7244\,\mathrm{nm},$$

which has the expected scale for 13.5 nm multilayer reflection. This is a consistency check based on a 4:1 interpretation, not a recovered complete mask prescription.

> **Read carefully.** Do not put the wafer-scaled 1.725 nm period directly into a physical 13.5 nm mask-stack solver. Lateral pattern dimensions can be reported at wafer scale for convenience, but a Maxwell calculation must use a consistent physical mask geometry, wavelength, and reduction mapping. The paper does not provide all stack details needed for exact reconstruction.

5.6 Finite absorber height breaks the ideal thin-mask picture

An obliquely incident ray traversing absorber height h has a lateral geometrical displacement $h \tan \alpha$. With the inferred $h = 60\,\text{nm}$ and $\alpha = 6°$, this is about 6.306 nm on the mask, or 1.577 nm after 4:1 reduction. This shows why an apparently small angle matters on nanometre scales. It is not a prediction of the paper's PS: coherent diffraction, material phase, double-pass geometry, and contour formation also contribute.

Different diffraction orders interact with the finite absorber and multilayer differently. Their complex amplitudes can depend on source angle, polarization, hole orientation, and pitch. This produces asymmetric images, orientation-dependent CDs, and focus-dependent placement. A square mask opening can therefore have unequal printed horizontal and vertical dimensions.

The mask chief-ray angle describes the central illumination geometry. The quasar distribution describes angular offsets around that geometry. Off-axis source shaping and a nonzero chief-ray angle are distinct ideas; one should not infer that the chief ray itself supplies every resolution benefit attributed to source shaping.

5.7 What a rigorous mask calculation has to do

A representative electromagnetic workflow is:

1. Specify each material's complex optical constants, layer thickness, feature shape, sidewall angle, period, source angle, and polarization.
2. Solve the frequency-domain Maxwell boundary-value problem. Methods include Fourier modal techniques and finite-element discretizations.
3. Obtain the complex vector amplitudes of reflected diffraction orders.
4. Propagate those orders through the projection system, including pupil phase, amplitude, and defocus.
5. Sum coherent orders for one source point, average intensities over mutually incoherent source points, and extract image or resist contours.

For a periodic permittivity $\epsilon(x, z) = \sum_g \epsilon_g(z) \mathrm{e}^{\mathrm{i}gx}$, expanding the field in the same harmonics turns multiplication by ϵ into convolution of Fourier coefficients. The diffraction orders become coupled equations, not independent rays. Their boundary conditions enforce the material interfaces. This is the basic reason rigorous diffraction is more computationally expensive than a binary thin-mask Fourier transform.

The paper identifies a commercial Synopsys simulator but does not give enough settings to determine its exact electromagnetic solver, numerical convergence, material database, or resist

extraction procedure. Those items remain unknown rather than being supplied by the word rigorous.

Part II

From wavefront errors to printed patterns

Chapter 6

Wavefront aberrations and what RMS does not tell you

Paper location: Introduction, Section 2, and the motivation for an imaging-based budget.

> **Textbook connection.** Born–Wolf §5.1, printed pp. 203–206 (begin at PDF p. 232); §9.1.2, p. 462 (PDF p. 557); §9.1.3, pp. 463–464 (PDF pp. 556–555); §9.3, pp. 468–469 (PDF pp. 551–550). The backward PDF order is real.

6.1 The reference sphere and the aberration function

An ideal converging wave reaches its intended image point with the same phase contribution from every admitted pupil location. On a suitable reference sphere centered at that image point, its phase is constant. The actual wave has residual phase variation $2\pi W/\lambda$. The real-valued function $W(\rho, \theta)$ is the wave aberration.

The coordinate (ρ, θ) identifies a point in the pupil, not a point on the wafer. A pupil map is therefore not a picture of an aberrated contact hole. To get that picture, multiply the pupil field by the phase factor and perform the diffraction calculation.

At a different field point on the wafer, the same optical system may have a different pupil wavefront:

$$W = W(\rho, \theta; \boldsymbol{H}),$$

where $\boldsymbol{H}$ denotes image-field position. A set of coefficients for one field point is not automatically valid across an entire exposure field. Likewise, scan time and thermal state may add dependencies.

6.2 The familiar named aberrations

For a centred rotationally symmetric system, a low-order expansion at field height H contains terms schematically of the form

$$W = A\rho^4 + BH\rho^3 \cos\theta + CH^2\rho^2 \cos^2\theta \tag{6.1}$$

$$+ DH^2\rho^2 + EH^3\rho \cos\theta + \cdots . \tag{6.2}$$

The coefficients and axes depend on convention, but the shapes reveal the standard families:

- **Spherical aberration:** different pupil radii have different focal behavior; the wavefront has a quartic radial contribution.

- **Coma:** an off-axis asymmetric wavefront component, often producing a lopsided point image.
- **Astigmatism:** different meridional directions focus differently. At a suitable intermediate focus, its wavefront has a $\rho^2 \cos 2\theta$ or $\rho^2 \sin 2\theta$ component.
- **Field curvature:** best focus varies across the image field.
- **Distortion:** field-dependent placement error; a point may remain locally sharp while its position is incorrect.

Piston, tilt, and defocus specify a constant phase, a linear pupil phase, and a quadratic radial phase. Whether they are called aberrations or alignment freedoms depends on the measurement purpose. The paper includes low-order coefficients because image placement and focus behavior matter.

Equation (6.2) is an illustrative classical expansion, not the prescription of the paper's off-axis multi-mirror projector. A real EUV system need not have the centred-system symmetries used to derive this form.

6.3 Piston does not change intensity

If W is replaced by $W + c_0$, then every field contribution receives the same factor:

$$U_{\text{image}} \mapsto \mathrm{e}^{\mathrm{i}2\pi c_0/\lambda} U_{\text{image}}.$$

The intensity is unchanged because $|\mathrm{e}^{\mathrm{i}2\pi c_0/\lambda}|^2 = 1$. Consequently intensity-derived CD, PS, and PVB cannot recover global piston. This is an exact invariance within the model, not a failure of neural-network training.

6.4 Tilt moves the image

Consider a linear OPD in the normalized frequency pupil, $W = t_x u + t_y v$. Then

$$P(\boldsymbol{f}) = P_0(\boldsymbol{f}) \exp\left[\mathrm{i}2\pi \left(\frac{t_x}{\mathrm{NA}} f_x + \frac{t_y}{\mathrm{NA}} f_y\right)\right].$$

Substituting in the inverse Fourier integral shows

$$U(\boldsymbol{x}) = U_0\left(\boldsymbol{x} + \frac{\boldsymbol{t}}{\mathrm{NA}}\right), \qquad \Delta\boldsymbol{x} = -\frac{\boldsymbol{t}}{\mathrm{NA}}. \tag{6.3}$$

Thus ideal pure tilt shifts a complete image without changing its shape. An image registered back to its own centre has the same CD and contrast. An unregistered image has a nonzero placement error.

The sign follows our transfer-function convention. The measurable magnitude and the distinction between shape and placement are the important points. When metrology removes tilt before quoting RMS, that RMS cannot subsequently bound the removed placement error.

6.5 Defocus changes the reference image plane

Equation (4.16) gives a quadratic pupil phase when the observation plane moves. Equivalently, a quadratic aberration can be partly compensated by changing focus. A report of best-focus RMS may therefore exclude a defocus term that still matters at a fixed wafer plane.

The paper varies the focus setting by ±15 nm while also changing aberration coefficients. To reconstruct this experiment one must know whether the Zernike coefficients are defined at a fixed reference focus, whether refocusing is permitted, and whether focus is reapplied separately for every candidate. The publication does not fully specify this convention.

6.6 Deriving the small-aberration Strehl relation

For a clear, uniformly illuminated pupil and an on-axis point source, the normalized field at the chosen reference image point is

$$\frac{U(0)}{U_{\text{ideal}}(0)} = \left\langle \mathrm{e}^{\mathrm{i}\phi} \right\rangle, \qquad \phi = \frac{2\pi W}{\lambda}.$$

Expand the exponential:

$$\left\langle \mathrm{e}^{\mathrm{i}\phi} \right\rangle \simeq 1 + \mathrm{i}\,\langle\phi\rangle - \frac{1}{2}\left\langle \phi^2 \right\rangle.$$

Multiplying by its conjugate and retaining terms through second order,

$$S_{\text{ref}} \simeq 1 - \left\langle \phi^2 \right\rangle + \langle\phi\rangle^2 = 1 - \left(\frac{2\pi\sigma_W}{\lambda}\right)^2. \tag{6.4}$$

The exponential form often used for small wavefront errors is

$$S \simeq \exp[-(2\pi\sigma_W/\lambda)^2]. \tag{6.5}$$

It shares the leading expansion. It is also the exact squared modulus of the characteristic function for a zero-mean Gaussian phase distribution under the corresponding average. An arbitrary deterministic aberration does not make that exponential exact.

The usual Strehl ratio compares peak intensities under consistent total power and aperture conditions. Equation (6.4) initially evaluates a chosen reference point. To discuss the optimized peak, remove or optimize the appropriate tilt and focus and specify that choice. With a nonuniform pupil, the amplitude weighting in the coherent field average must also be treated explicitly.

Worked example: Interpreting a $\lambda/50$ RMS requirement

At 13.5 nm, $\lambda/50 = 0.27\,\text{nm} = 20\,\text{m}\lambda$. If this is piston-removed RMS in a suitable small-aberration setting,

$$\sigma_\phi = 2\pi/50 = 0.125664, \qquad S \simeq \mathrm{e}^{-(2\pi/50)^2} = 0.98433.$$

A high Strehl ratio describes concentration of a point image. It does not guarantee a subnanometre PS for every periodic contact image. Pure tilt illustrates the distinction particularly clearly.

6.7 Equal RMS is not equal lithographic performance

RMS measures a mean-square phase error over the pupil. A periodic mask uses particular sets of pupil locations and their phase differences. Two maps with identical RMS can give different phase differences at those locations. Their CD, PS, and process window can therefore differ.

This is the physical motivation for the paper. It does not imply that RMS is useless. RMS remains a compact measure of optical quality and a useful control variable. It means that RMS alone is not a complete description of a pattern-specific imaging requirement.

Chapter 7

Zernike polynomials from the beginning

Paper location: Equation (1), Table 1, and the coefficient axes in Figs. 5, 9, 11, and 12.

> **Textbook connection.** Born–Wolf §9.2.1, printed pp. 464–466 (PDF pp. 555, 554, 553), including its radial-polynomial table. Its normalization differs from the RMS-normalized real basis used here.

7.1 Why use orthogonal functions?

Suppose we tried to describe W with $1, \rho^2, \rho^4, \ldots$. These functions overlap under disk averaging: $\langle \rho^2 \rangle = 1/2$ and $\langle \rho^2 \rho^4 \rangle = 1/4$, not zero. Changing one fitted coefficient can compensate another. Orthogonal functions separate those overlapping contributions.

Define the inner product

$$(f, g) = \langle fg \rangle = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 fg \, \rho \mathrm{d}\rho \mathrm{d}\theta.$$

Orthogonal means $(f, g) = 0$ for different modes. Orthonormal means each mode also has $(f, f) = 1$. The analogy is perpendicular unit axes in ordinary Euclidean geometry, but the vectors here are functions over a disk.

7.2 Constructing the first modes explicitly

The constant mode is $Z_0^0 = 1$. Its RMS is one. For a tilt, $\rho \cos\theta = u$ and $\langle u^2 \rangle = 1/4$, so the unit-RMS tilt is

$$Z_1^1 = 2\rho \cos\theta, \qquad Z_1^{-1} = 2\rho \sin\theta.$$

To create defocus orthogonal to piston, subtract the mean from ρ^2: $\rho^2 - 1/2$. Its mean square is $1/3 - 1/4 = 1/12$, giving

$$Z_2^0 = \sqrt{3}(2\rho^2 - 1). \tag{7.1}$$

For astigmatism, angular averaging already removes piston:

$$\left\langle (\rho^2 \cos 2\theta)^2 \right\rangle = \frac{1}{2} \left\langle \rho^4 \right\rangle = \frac{1}{6}.$$

Hence $Z_2^2 = \sqrt{6}\rho^2 \cos 2\theta$ and $Z_2^{-2} = \sqrt{6}\rho^2 \sin 2\theta$.

For a coma-shaped mode, start from $(\rho^3 - a\rho)\cos\theta$. Remove its overlap with tilt:

$$0 = \left\langle (\rho^3 - a\rho)\rho\cos^2\theta \right\rangle = \frac{1}{2}\left(\frac{1}{3} - \frac{a}{2}\right),$$

which gives $a = 2/3$. Rescale the radial polynomial to equal one at the rim, giving $3\rho^3 - 2\rho$. Its normalized real mode is

$$Z_3^1 = \sqrt{8}(3\rho^3 - 2\rho)\cos\theta. \tag{7.2}$$

The sine partner is Z_3^{-1}. Notice the subtracted tilt. A balanced Zernike coma mode is not just the monomial $\rho^3\cos\theta$.

For spherical aberration, start with $\rho^4 + a\rho^2 + b$ and require orthogonality to 1 and ρ^2:

$$\frac{1}{3} + \frac{a}{2} + b = 0, \qquad \frac{1}{4} + \frac{a}{3} + \frac{b}{2} = 0.$$

Solving gives $a = -1$, $b = 1/6$. Rim normalization and unit RMS produce

$$Z_4^0 = \sqrt{5}(6\rho^4 - 6\rho^2 + 1). \tag{7.3}$$

This explains why the spherical Zernike mode contains defocus and piston terms: they make it independent of the lower radial modes.

7.3 The general definition

Let $n \geq 0$ be radial degree and m be an integer with $|m| \leq n$ and $n - |m|$ even. Define

$$R_n^{|m|}(\rho) = \sum_{s=0}^{(n-|m|)/2} (-1)^s \frac{(n-s)!}{s!\,[(n+|m|)/2 - s]!\,[(n-|m|)/2 - s]!}\rho^{n-2s}. \tag{7.4}$$

The allowed parity ensures all factorials have nonnegative integer arguments and all powers have the correct angular symmetry. The rim value is $R_n^{|m|}(1) = 1$.

Our real RMS-normalized convention is

$$Z_n^m = \begin{cases} \sqrt{n+1}\,R_n^0, & m = 0, \\ \sqrt{2(n+1)}\,R_n^m\cos(m\theta), & m > 0, \\ \sqrt{2(n+1)}\,R_n^{|m|}\sin(|m|\theta), & m < 0. \end{cases} \tag{7.5}$$

The radial orthogonality identity is

$$\int_0^1 R_n^m R_{n'}^m\,\rho\mathrm{d}\rho = \frac{\delta_{nn'}}{2(n+1)}.$$

Angular integration gives 2π for constant modes and π for the square of each nonconstant sine or cosine. Combining those factors with Eq. (7.5) gives

$$\left\langle Z_n^m Z_{n'}^{m'} \right\rangle = \delta_{nn'}\delta_{mm'}. \tag{7.6}$$

The Kronecker delta δ_{ab} equals one if $a = b$ and zero otherwise. This is the formal statement that the modes form independent unit axes under this pupil-area metric.

7.4 A radial table sufficient to understand low-order budgets

Every $m > 0$ entry below has both a cosine and sine partner. Apply the normalization in Eq. (7.5); the table lists only radial factors.

n	$\lvert m\rvert$	$R_n^{\lvert m\rvert}(\rho)$
0	0	1
1	1	ρ
2	0	$2\rho^2 - 1$
2	2	ρ^2
3	1	$3\rho^3 - 2\rho$
3	3	ρ^3
4	0	$6\rho^4 - 6\rho^2 + 1$
4	2	$4\rho^4 - 3\rho^2$
4	4	ρ^4
5	1	$10\rho^5 - 12\rho^3 + 3\rho$
5	3	$5\rho^5 - 4\rho^3$
5	5	ρ^5
6	0	$20\rho^6 - 30\rho^4 + 12\rho^2 - 1$
6	2	$15\rho^6 - 20\rho^4 + 6\rho^2$
6	4	$6\rho^6 - 5\rho^4$
6	6	ρ^6

There are $n + 1$ real modes at degree n. Through degree six there are $1 + 2 + \cdots + 7 = 28$ modes, including piston. This is a complete sequence by radial degree. It is deliberately not presented as the paper's unspecified single-index list of 25 terms.

7.5 Recovering coefficients and calculating RMS

Expand a square-integrable wavefront as

$$W(\rho, \theta) = \sum_{n,m} c_n^m Z_n^m(\rho, \theta). \tag{7.7}$$

Multiply by $Z_{n'}^{m'}$ and average. Orthogonality removes all other terms:

$$c_{n'}^{m'} = \left\langle W Z_{n'}^{m'} \right\rangle. \tag{7.8}$$

Each coefficient has the units of W. If W is in nanometres, so is c. If W/λ is expanded, the coefficients are in waves.

Because $\langle Z_0^0 \rangle = 1$ and all other mode means are zero, $\langle W \rangle = c_0^0$. Expanding the square and using orthogonality gives

$$\sigma_W^2 = \sum_{(n,m)\neq(0,0)} (c_n^m)^2. \tag{7.9}$$

This identity depends on the unit-RMS normalization, complete disk, uniform area weighting, and explicit removal of piston. Other removed modes must also be excluded. With an unnormalized basis, the corresponding modal norm factors remain in the sum.

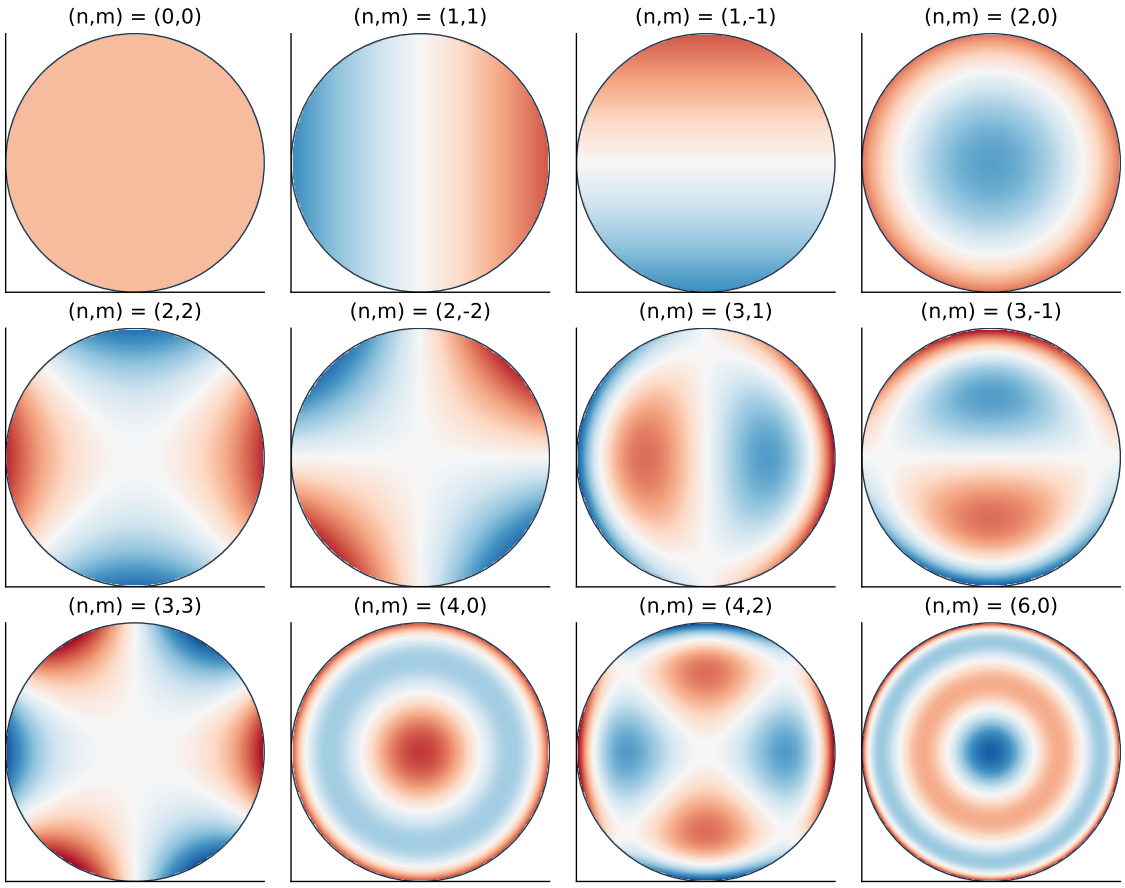


Figure 7.1: Original atlas of selected real, RMS-normalized modes. Red and blue indicate opposite OPD signs; the same colour scale is used throughout. The labels are (n, m), not the paper's single-index coefficient numbers.

Worked example: A coefficient is not a peak-to-valley value

For defocus $W = cZ_2^0$, the centre value is $-\sqrt{3}c$ and the rim value is $+\sqrt{3}c$. Thus PV $= 2\sqrt{3}|c|$, while $\sigma_W = |c|$. For $c = 0.27\,\text{nm}$, RMS is 0.27 nm but PV is about 0.9353 nm. For the normalized tilt $W = 2cu$, PV is $4|c|$. Equal coefficients mean equal RMS for these modes, not equal PV.

Worked example: Why per-mode limits are not a total RMS limit

If 24 independent orthonormal, nonpiston modes each have magnitude $20\,\text{m}\lambda$, then

$$\sigma_W = 20\sqrt{24}\,\text{m}\lambda = 97.9796\,\text{m}\lambda = 1.32272\,\text{nm}.$$

The input box $|c_j| \le 20\,\text{m}\lambda$ therefore does not mean total RMS is at most $\lambda/50$. If all coefficients are independent uniform variables on that interval, then $\mathbb{E}[c_j^2] = (20\,\text{m}\lambda)^2/3$, and

$$\sqrt{\mathbb{E}[\sigma_W^2]} = 20\sqrt{24/3} = 56.5685\,\text{m}\lambda.$$

This is the square root of the expected squared RMS, not the expectation of RMS itself. Both calculations are conditional teaching results: the paper does not state the normalization needed to apply them directly to its coefficient arrays.

7.6 Single-index numbering is a convention, not a law

Many programs replace (n, m) by a single integer j. Noll, Fringe, OSA/ANSI, and software-specific sequences differ. Some start counting at zero, some at one; some normalize each mode to unit RMS and some do not. Axes, sine/cosine assignment, and sign can differ.

The paper writes

$$W(\rho, \theta) = \sum_i Z_i F_i(\rho, \theta).$$

Here its Z_i are *coefficients*, while its F_i are basis functions. In this book Z_n^m names a basis function and c_n^m its coefficient. These are two notation systems for the same expansion idea. Confusing them changes the meaning of an equation.

The paper does not supply the explicit F_i table or normalization. We therefore cannot safely identify its Z_{20}, Z_{21}, or Z_{22} with particular named modes. Its report that these coefficients have narrow ranges is meaningful in its own simulator coordinates, but a physical mode-name interpretation requires that missing dictionary.

7.7 Mode orientation and rotating axes

For a sine/cosine pair of the same n and $m > 0$,

$$c_c \cos m\theta + c_s \sin m\theta = C \cos[m(\theta - \theta_0)],$$

where $C = \sqrt{c_c^2 + c_s^2}$ and $\theta_0 = \text{atan2}(c_s, c_c)/m$. The coefficient pair rotates when axes rotate, but C stays unchanged. Astigmatism has $m = 2$, so a 45° rotation in physical axes interchanges the character of its cosine and sine partners. This matters when comparing horizontal and vertical pattern sensitivities.

7.8 Balanced spherical aberration as a minimization

Suppose a surface produces raw quartic OPD $B\rho^4$. Permit a compensating defocus $d\rho^2$ and remove piston afterward. Its variance is

$$V(d) = B^2 \operatorname{Var}(\rho^4) + d^2 \operatorname{Var}(\rho^2) + 2Bd \operatorname{Cov}(\rho^4, \rho^2).$$

Using disk averages,

$$\operatorname{Var}(\rho^4) = \frac{4}{45}, \quad \operatorname{Var}(\rho^2) = \frac{1}{12}, \quad \operatorname{Cov}(\rho^4, \rho^2) = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Differentiate: $V'(d) = d/6 + B/6 = 0$, giving $d = -B$. After piston removal the balanced residual is

$$W_{\text{bal}} = B(\rho^4 - \rho^2 + 1/6) = \frac{B}{6\sqrt{5}} Z_4^0,$$

so

$$\sigma_{\text{bal}} = \frac{|B|}{\sqrt{180}}. \tag{7.10}$$

The original piston-removed quartic had $\sigma = 2|B|/\sqrt{45}$, exactly four times larger. Allowing refocus changes the budget. That change is meaningful only if refocus is an available adjustment under the actual imaging requirement.

7.9 Sampling, obscurations, and amplitude weighting

Real measured pupils contain discrete points and may have missing regions. For samples $\boldsymbol{w}$, let $B_{\ell j} = Z_j(\rho_\ell, \theta_\ell)$ and let Q be a diagonal matrix of area or uncertainty weights. Weighted least squares minimizes $(B\boldsymbol{c} - \boldsymbol{w})^T Q (B\boldsymbol{c} - \boldsymbol{w})$ and gives

$$(B^T Q B)\boldsymbol{c} = B^T Q \boldsymbol{w}. \tag{7.11}$$

For a masked or nonuniformly weighted pupil, $B^T Q B$ is generally not the identity. Modes orthogonal on a full disk need not remain orthogonal under that new metric.

Pupil *intensity* weighting in wavefront metrology and pupil *amplitude* weighting in coherent on-axis interference are also different. State which quantity is weighted before interpreting RMS, Strehl, or fitted coefficients.

> **Check your understanding.** Why can you not square and sum an arbitrary exported list of coefficients to get RMS? You must first know basis normalization, pupil support and weight, units, and which modes were removed.

Chapter 8

Partially coherent images of periodic contact holes

Paper location: Illumination, test patterns, and the commercial imaging calculations.

> **Textbook connection.** Saleh–Teich §4.4, especially printed pp. 129–134 (PDF pp. 151–156), for coherent transfer functions. Born–Wolf §9.5 for coherent/incoherent imaging; §10.4 for mutual intensity (printed p. 507 is PDF p. 510); Hopkins' propagation formula begins at printed p. 512 (PDF p. 505).

8.1 Why one coherent calculation is not the full image

An extended illumination source contains many emitting regions. The field arriving from one effective source point is coherent across the mask for the idealized source-point calculation. Fields from distinct source points may have rapidly varying relative phases. A resist exposure averages over times long compared with those phase fluctuations.

Let source point s produce U_s with random phase $\phi_s(t)$. Then

$$\left\langle \left| \sum_s U_s \mathrm{e}^{\mathrm{i}\phi_s(t)} \right|^2 \right\rangle_t = \sum_s |U_s|^2 + \sum_{s \neq s'} U_s U_{s'}^* \left\langle \mathrm{e}^{\mathrm{i}(\phi_s - \phi_{s'})} \right\rangle_t .$$

For mutually incoherent source points the last average vanishes. We must sum *intensities* across source points, even though we sum *field amplitudes* across diffraction orders from the same point. Confusing these two sums produces a different physical model.

More generally, the mutual intensity is

$$J(\boldsymbol{r}_1, \boldsymbol{r}_2) = \langle U(\boldsymbol{r}_1, t) U^*(\boldsymbol{r}_2, t) \rangle_t , \qquad \mu_{12} = \frac{J_{12}}{\sqrt{I_1 I_2}}. \tag{8.1}$$

$|\mu_{12}| \leq 1$ measures coherence; unity means full coherence at the two points and zero means no average cross term. If $U_{\mathrm{out}}(x) = \int h(x, r) U_{\mathrm{in}}(r) \mathrm{d}r$, substitution gives

$$I_{\mathrm{out}}(x) = \iint h(x, r_1) h^*(x, r_2) J_{\mathrm{in}}(r_1, r_2) \mathrm{d}r_1 \mathrm{d}r_2 .$$

The incoherent-source integral below is a convenient special form of this general propagation rule.

8.2 The source-point or Abbe imaging equation

Let $\boldsymbol{s}$ be an illumination spatial frequency, expressed in wafer-equivalent coordinates. The incident field on a thin mask is $\mathrm{e}^{\mathrm{i}2\pi\boldsymbol{s}\cdot\boldsymbol{x}}$. Multiplying the mask function t by this phase shifts its spectrum:

$$\mathcal{F}[t(\boldsymbol{x})\mathrm{e}^{\mathrm{i}2\pi\boldsymbol{s}\cdot\boldsymbol{x}}] = \widetilde{t}(\boldsymbol{f}-\boldsymbol{s}).$$

Propagate through pupil P_z, change the integration variable to the mask frequency, and obtain

$$U_{\boldsymbol{s}}(\boldsymbol{x}) = \mathrm{e}^{\mathrm{i}2\pi\boldsymbol{s}\cdot\boldsymbol{x}} \int \widetilde{t}(\boldsymbol{f})P_z(\boldsymbol{f}+\boldsymbol{s})\mathrm{e}^{\mathrm{i}2\pi\boldsymbol{f}\cdot\boldsymbol{x}}\mathrm{d}^2 f. \tag{8.2}$$

The leading carrier has unit magnitude. For nonnegative source weights S normalized so that $\int S(\boldsymbol{s})\mathrm{d}^2 s = 1$,

$$I(\boldsymbol{x}; W, z) = \int S(\boldsymbol{s}) \left| \int \widetilde{t}(\boldsymbol{f})P_z(\boldsymbol{f}+\boldsymbol{s})\mathrm{e}^{\mathrm{i}2\pi\boldsymbol{f}\cdot\boldsymbol{x}}\mathrm{d}^2 f \right|^2 \mathrm{d}^2 s. \tag{8.3}$$

The source shifts the spectrum relative to the pupil. It does not enlarge the physical projection aperture.

8.3 The Hopkins transmission cross coefficient

Expand the squared modulus in Eq. (8.3) and interchange integrals:

$$I(\boldsymbol{x}) = \iint \widetilde{t}(\boldsymbol{f}_1)\widetilde{t}^*(\boldsymbol{f}_2)T(\boldsymbol{f}_1, \boldsymbol{f}_2) \tag{8.4}$$

$$\times\, \mathrm{e}^{\mathrm{i}2\pi(\boldsymbol{f}_1-\boldsymbol{f}_2)\cdot\boldsymbol{x}}\mathrm{d}^2 f_1 \mathrm{d}^2 f_2, \tag{8.5}$$

$$T(\boldsymbol{f}_1, \boldsymbol{f}_2) = \int S(\boldsymbol{s})P_z(\boldsymbol{f}_1+\boldsymbol{s})P_z^*(\boldsymbol{f}_2+\boldsymbol{s})\mathrm{d}^2 s. \tag{8.6}$$

T is a transmission cross coefficient (TCC). It tells us how strongly each pair of object frequencies interferes after source averaging. Its phase contains the pupil-phase difference between the two sampled locations. The two formulations are equivalent under their shared thin-mask, scalar, shift-invariant assumptions.

A thick EUV mask can have a source-dependent scattering response. Then its order amplitudes are $a_{mn}(\boldsymbol{s}, \text{polarization})$, and one cannot generally factor out a single source-independent $\widetilde{t}$ as above. The source-point sum remains useful with the rigorously computed amplitudes inserted separately.

8.4 Diffraction coefficients of a rectangular opening

Consider an ideal binary opening of width a repeated every p_x and height b repeated every p_y. Within a period, $t(x, y) = 1$ inside the centred rectangle and zero elsewhere. Its Fourier series is

$$t(x, y) = \sum_{m,n} a_{mn}\mathrm{e}^{\mathrm{i}2\pi(mx/p_x + ny/p_y)}.$$

Direct integration gives

$$a_{mn} = \frac{1}{p_x p_y}\int_{-a/2}^{a/2} \mathrm{e}^{-\mathrm{i}2\pi mx/p_x}\mathrm{d}x \int_{-b/2}^{b/2} \mathrm{e}^{-\mathrm{i}2\pi ny/p_y}\mathrm{d}y \tag{8.7}$$

$$= \frac{ab}{p_x p_y}\operatorname{sinc}\left(\frac{ma}{p_x}\right)\operatorname{sinc}\left(\frac{nb}{p_y}\right), \tag{8.8}$$

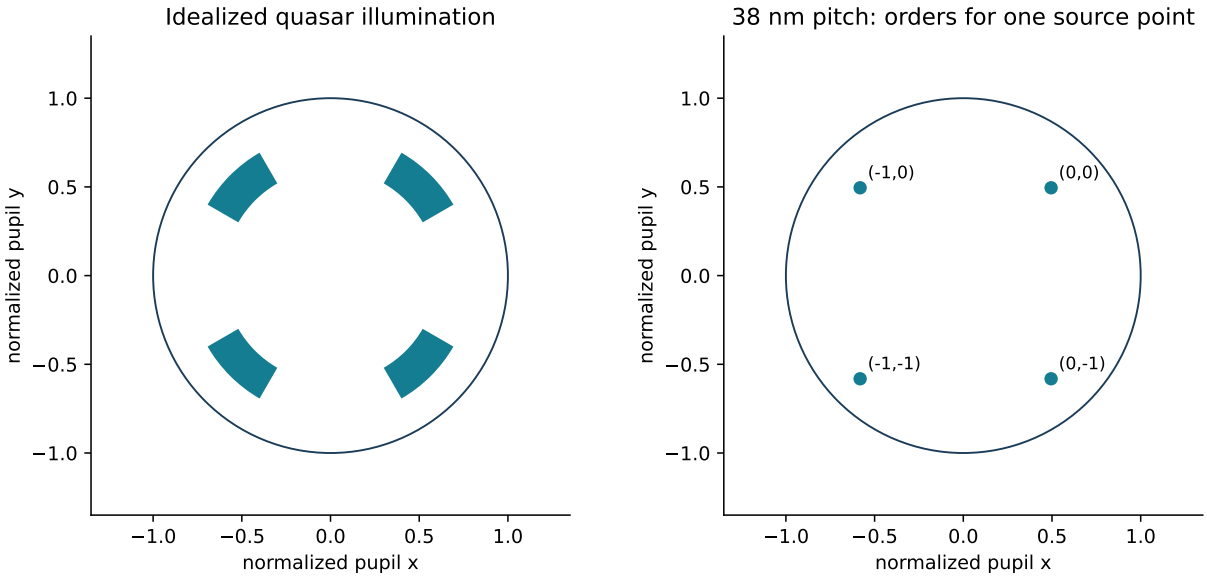


Figure 8.1: Original pupil diagrams. Left: an idealized four-sector quasar source. Right: shifted order locations for one diagonal source point and a 38 nm square pitch. Accepted orders can interfere to create a two-dimensional contact image. This is a thin-mask explanation, not a reproduction of the commercial simulator.

where $\operatorname{sinc} q = \sin(\pi q)/(\pi q)$ and $\operatorname{sinc} 0 = 1$. The prefactor is the open-area fraction. A narrower hole spreads its spectrum more widely, but the pupil still limits the orders that contribute.

For one source point, neglecting the irrelevant carrier,

$$U_{\boldsymbol{s}}(x,y) = \sum_{m,n} a_{mn} P_z\left(\boldsymbol{s} + \left(\frac{m}{p_x}, \frac{n}{p_y}\right)\right) \mathrm{e}^{\mathrm{i}2\pi(mx/p_x + ny/p_y)}. \tag{8.9}$$

The integer pair (m, n) names a diffraction order. Each admitted order is a plane-wave component of the image field. Their relative phases and amplitudes determine the shape.

8.5 Understanding the quasar illumination

Table 1 specifies inner and outer normalized source radii 0.6 and 0.8, angle 30°, and rotation 45°. Figure 2 shows four off-axis source regions around diagonal directions. Their radii are normalized to the relevant projection-pupil scale; 0.8 is not an angle of 0.8 radians. The exact software interpretation of the angle setting is not documented. The drawing in this book uses four annular sectors with full angular width 30° as an explicit teaching convention.

At pitch 38 nm,

$$\Delta u = \frac{1/p}{f_c} = \frac{\lambda}{\mathrm{NA}p} = \frac{13.5}{0.33(38)} \simeq 1.07656.$$

For on-axis illumination the first axial orders lie beyond a unit pupil. Only the zero order survives in the ideal scalar square lattice, so it cannot create the desired periodic modulation. For a source point at radius 0.7 and azimuth 45°, $\boldsymbol{u}_s = (0.49497, 0.49497)$. Now the orders $(0,0), (-1,0), (0,-1), (-1,-1)$ all lie inside the pupil. The oblique source has made several useful orders available simultaneously.

Increasing pitch reduces order spacing in the pupil. Different pitches therefore sample different phases even when CD stays fixed. Changing a rectangle's aspect ratio changes the weights a_{mn} in

the two directions. This is why the authors keep many pitches and both orientations instead of one averaged contact pattern.

8.6 A two-order derivation of pattern shift

Consider

$$U(x) = a + b\mathrm{e}^{\mathrm{i}(qx+\Delta\phi)}, \qquad q = 2\pi/p,$$

with real positive a, b. Its intensity is

$$I(x) = a^2 + b^2 + 2ab\cos(qx + \Delta\phi). \tag{8.10}$$

The maximum previously at $x = 0$ is now at

$$\Delta x = -\frac{\Delta\phi}{q} = -\frac{p}{2\pi}\Delta\phi. \tag{8.11}$$

If the differential phase is due to pupil OPD, $\Delta\phi = 2\pi(W_b - W_a)/\lambda$, then

$$\Delta x = -\frac{p}{\lambda}(W_b - W_a).$$

At $p = 38\,\mathrm{nm}$, a differential OPD of $0.1\,\mathrm{nm}$ gives a displacement magnitude of $0.28148\,\mathrm{nm}$ in this two-order example. The controlling quantity is a *difference at sampled pupil points*, not the pupil's global RMS.

8.7 An even phase perturbation changes shape

With symmetric side orders,

$$U(x) = a + b\mathrm{e}^{\mathrm{i}\phi}\mathrm{e}^{\mathrm{i}qx} + b\mathrm{e}^{\mathrm{i}\phi}\mathrm{e}^{-\mathrm{i}qx} = a + 2b\mathrm{e}^{\mathrm{i}\phi}\cos qx,$$

so

$$I(x) = a^2 + 4b^2\cos^2 qx + 4ab\cos\phi\cos qx. \tag{8.12}$$

The image remains even in x, but its intensity profile changes. A fixed threshold therefore gives a different printed width. Because $\cos\phi = 1 - \phi^2/2 + \cdots$, the leading response at this symmetric operating point is second order. There need not be a nonzero linear sensitivity to every aberration.

8.8 Defocus, symmetry, and source design

An order at normalized radius ρ acquires the defocus phase $-\pi z\mathrm{NA}^2\rho^2/\lambda$ in the paraxial model. Two orders at equal radius receive the same defocus phase and preserve their relative phase. This is one reason suitable off-axis illumination can improve focus robustness for selected patterns. For unequal radii, the difference is

$$\Delta\phi_z = -\frac{\pi z\mathrm{NA}^2}{\lambda}(\rho_2^2 - \rho_1^2).$$

One source design cannot generally equalize every important order pair for all pitches and both rectangular dimensions. Source optimization is a pattern-dependent interference problem.

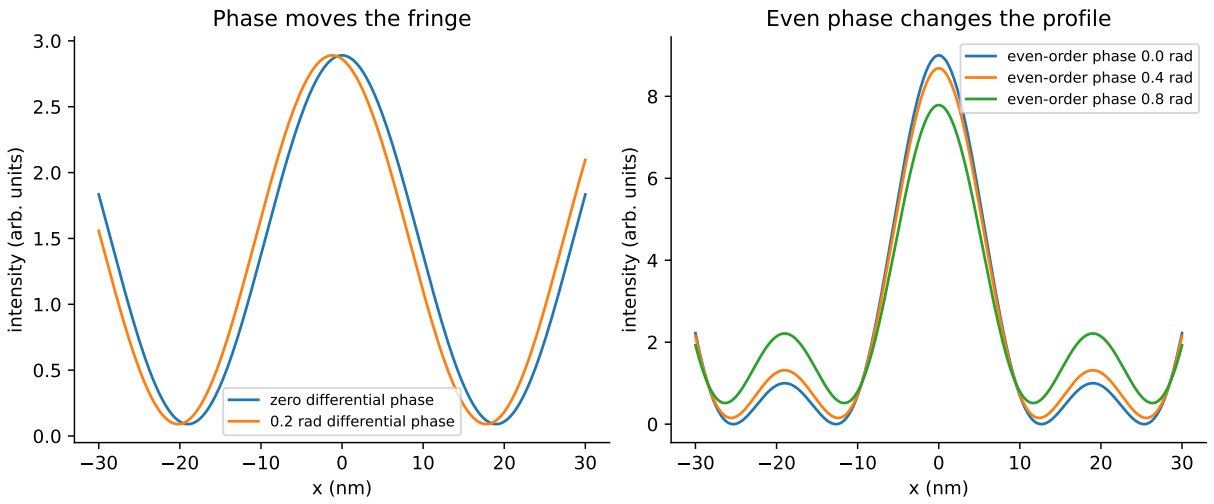


Figure 8.2: Exact teaching examples. A differential phase between two orders shifts fringes. A common phase on symmetric side orders changes an even intensity profile. A real contact combines many such interference terms and an incoherent source average.

Chapter 9

From light intensity to CD, PS, PVB, and NILS

Paper location: The six imaging indicators used throughout the paper.

9.1 An aerial image is not yet a resist contour

The aerial image is the optical intensity in the image plane. Exposure dose is energy per area, obtained by integrating intensity over time. A full resist model may include photon absorption, secondary electrons, chemical reactions, diffusion, and development. The paper does not provide enough resist-model detail to reconstruct that stage exactly.

To derive the basic geometry, use a declared constant-threshold model:

$$D\,I(x, y; z, W) = T. \tag{9.1}$$

D is a relative dose multiplier and T is a fixed exposure threshold in consistent units. This is a teaching approximation. It captures how intensity perturbations move edges without claiming a full EUV resist model. For a bright contact opening, the printed interior is taken to be the region above threshold; resist tone can reverse that interpretation.

9.2 Critical dimension and centre position

On a specified horizontal cut through a feature, let threshold crossings be x_L and x_R. Define

$$\mathrm{CD}_x = x_R - x_L, \qquad x_c = \frac{x_R + x_L}{2}, \qquad \mathrm{PS}_x = x_c - x_{\mathrm{target}}. \tag{9.2}$$

Use analogous definitions in y. For a nonrectangular contour, the measurement cut, bounding box, fitted shape, or centroid definition must be specified. A centroid computed from the whole intensity distribution need not equal the midpoint between two threshold edges.

Small edge changes satisfy

$$\delta\mathrm{CD}_x = \delta x_R - \delta x_L, \qquad \delta\mathrm{PS}_x = \frac{\delta x_R + \delta x_L}{2}. \tag{9.3}$$

Equal edge motion shifts the pattern. Opposite edge motion changes its width. These two combinations explain why CD and PS provide different information about a wavefront.

9.3 Deriving the edge-displacement formula

Let an original edge satisfy $I_0(x_e) = T$ at nominal dose one. Perturb the image by δI and the edge by δx:

$$I_0(x_e + \delta x) + \delta I(x_e + \delta x) = T.$$

Keeping first-order terms, $I_0(x_e) + I_0'(x_e)\delta x + \delta I(x_e) = T$. Cancel the original threshold equation:

$$\delta x = -\frac{\delta I(x_e)}{I_0'(x_e)}. \tag{9.4}$$

The same intensity error causes a larger edge displacement when the edge slope is shallow. For a two-dimensional contour, normal displacement is

$$\delta n = -\frac{\delta I}{|\nabla I_0|},$$

with the signed normal chosen along increasing intensity. At a vanishing slope, the linear formula fails: an opening may appear, disappear, split, or merge. These topology changes also make learned metric maps difficult to approximate smoothly.

9.4 NILS makes the slope dimensionless

For characteristic width w, define the magnitude of normalized image log slope at an edge:

$$\mathrm{NILS} = w \left| \frac{\partial \ln I}{\partial x} \right|_{x_e} = w \frac{|I'(x_e)|}{I(x_e)}. \tag{9.5}$$

Different reporting conventions retain or remove the slope sign. The dimensionless magnitude is what matters for this sensitivity argument.

A relative dose change $\epsilon = \delta D/D$ multiplies the image by approximately $1 + \epsilon$, so $\delta I = \epsilon I$. Equation (9.4) gives

$$|\delta x| \simeq \frac{w}{\mathrm{NILS}} |\epsilon|. \tag{9.6}$$

For a symmetric bright feature with equal slope magnitudes at the two edges, both edges move outward when dose increases:

$$|\delta \mathrm{CD}| \simeq \frac{2w}{\mathrm{NILS}} |\epsilon|.$$

The factor of two comes from two moving edges. It must not be silently carried into a one-edge placement formula.

Worked example: A 5 percent dose perturbation

If $w = 14\,\mathrm{nm}$ and NILS is 2, one edge moves approximately $14(0.05)/2 = 0.35\,\mathrm{nm}$ under a 5 percent dose change. The full CD changes by approximately $0.70\,\mathrm{nm}$ in the symmetric example. These assumed slopes are not measured values from the paper. The example explains why contrast loss can enlarge process variation even if nominal CD is adjusted back to target by changing dose.

9.5 Focus-exposure matrices and process windows

For each focus z and relative dose D, simulate a contour and compute $\mathrm{CD}(z, D)$, $\mathrm{PS}(z, D)$, and other required measures. Plots of CD against focus at several doses are often called Bossung curves. A process window is the set of focus-dose pairs satisfying *all* chosen acceptance conditions:

$$\mathcal{P} = \{(z, D) : \mathrm{CD}_{\min} \le \mathrm{CD}(z, D) \le \mathrm{CD}_{\max},\ |\mathrm{PS}(z, D)| \le b, \ldots\}.$$

An attractive nominal image at $z = 0, D = 1$ can coexist with a very small process window. Aberration-induced sensitivity to focus and dose is therefore physically significant.

The paper gives $z = -15, 0, +15\,\mathrm{nm}$ and dose variation $\pm 5\%$. If one uses doses $0.95, 1.00, 1.05$, a teaching focus-dose matrix has nine conditions. This is a declared sampling choice here, not proof that the authors used exactly that grid or only those dose samples.

9.6 What a process variation band can mean

For contours $C(z, D)$, their envelope is the region swept out by edge positions over the chosen process conditions. On a horizontal cut, define edge-band widths

$$B_L = \max x_L - \min x_L, \qquad B_R = \max x_R - \min x_R. \tag{9.7}$$

A scalar horizontal PVB might be the larger edge-band width, a specified average, or another software-defined contour measure. By contrast, CD range is

$$B_{\mathrm{CD}} = \max(x_R - x_L) - \min(x_R - x_L). \tag{9.8}$$

These quantities are not generally equal. For pure translation, $B_{\mathrm{CD}} = 0$ while both edge bands can be large. For symmetric expansion/contraction, each edge band can be half the CD range.

> **Read carefully.** The paper does not give a mathematical extraction rule for PVB_x and PVB_y. Use its term as a reported process-contour variation indicator. Do not assume without verification that it is simply maximum CD minus minimum CD. The laboratory in this book reports the specific edge-band widths of Eq. (9.7).

9.7 Placement error, overlay, and edge placement error

Pattern shift concerns one feature relative to its intended location. Overlay concerns alignment between layers or exposures. An aberration-induced shift can consume part of an overlay allowance, but stage alignment, mask registration, wafer distortion, and other contributors also matter.

For a one-dimensional target with left and right edges, small edge placement errors combine CD and PS:

$$\mathrm{EPE}_R = \mathrm{PS} + \frac{\delta\mathrm{CD}}{2}, \qquad \mathrm{EPE}_L = \mathrm{PS} - \frac{\delta\mathrm{CD}}{2}. \tag{9.9}$$

This algebra shows why controlling only the feature centre is insufficient to control every edge. A contact can be correctly centred but too large. Conversely, its width can be perfect while its entire position is wrong.

9.8 Why horizontal and vertical errors differ

The mask incidence plane establishes a preferred direction. A rectangular hole has different spectra in x and y. A quasar source has discrete angular structure. A wavefront mode has its own orientation. The interplay can make one direction more sensitive than the other.

The paper's relatively large PS_x and small PS_y should not be interpreted as a universal property of all EUV contacts. The incidence azimuth, coordinate conventions, baseline placement, and definition of the reported shifts would need to be known. A small denominator in a percentage error is a separate statistical issue, addressed in Chapter 14.

Chapter 10

How an aberration coefficient changes a metric

10.1 Differentiating the pupil and the field

Let $W = \sum_j c_j Z_j$ with coefficients in length units. Then

$$\frac{\partial P}{\partial c_j} = \frac{\mathrm{i}2\pi}{\lambda} Z_j P. \tag{10.1}$$

For a fixed thin-mask spectrum and one source point,

$$\frac{\partial U_{\boldsymbol{s}}}{\partial c_j} = \frac{\mathrm{i}2\pi}{\lambda} \int \widetilde{t}(\boldsymbol{f})\, Z_j\!\left(\frac{\boldsymbol{f}+\boldsymbol{s}}{f_c}\right) P(\boldsymbol{f}+\boldsymbol{s}) \mathrm{e}^{\mathrm{i}2\pi \boldsymbol{f}\cdot\boldsymbol{x}} \mathrm{d}^2 f.$$

Differentiate $I = UU^*$:

$$\frac{\partial I}{\partial c_j} = 2\,\mathrm{Re}\left\{ U^* \frac{\partial U}{\partial c_j} \right\}. \tag{10.2}$$

For a partially coherent source, integrate this expression over source weights. In a vector calculation, replace the scalar product with the appropriate polarization sum.

The derivative is physically a measure of how a phase change in one mode perturbs the constructive and destructive interference in the image. It is not determined merely by how large the mode looks in a pupil plot.

10.2 Differentiating a printed edge

Apply implicit differentiation to $I(x_e(\boldsymbol{c}), \boldsymbol{c}) = T$:

$$\frac{\partial I}{\partial x}\frac{\partial x_e}{\partial c_j} + \frac{\partial I}{\partial c_j} = 0.$$

Therefore

$$\frac{\partial x_e}{\partial c_j} = -\frac{\partial I/\partial c_j}{\partial I/\partial x}. \tag{10.3}$$

Combine left and right derivatives using Eq. (9.3) to obtain the sensitivity of CD and PS. This supplies an explicit physics-based path from coefficient to metric.

For M measured indicators, define

$$J_{ij} = \left.\frac{\partial F_i}{\partial c_j}\right|_{\boldsymbol{c}_0}, \qquad \delta \boldsymbol{y} \simeq J\,\delta \boldsymbol{c}. \tag{10.4}$$

J is the imaging Jacobian. Its rows correspond to particular patterns, orientations, and metrics; its columns correspond to modes. If coefficients and metrics are both in nm, its entries are dimensionless. If coefficients are in milliwaves, the entries have units nm per milliwave.

10.3 A numerical derivative when an analytic derivative is unavailable

With a black-box simulator, a centred finite difference gives

$$J_{ij} \simeq \frac{F_i(\boldsymbol{c}_0 + h\boldsymbol{e}_j) - F_i(\boldsymbol{c}_0 - h\boldsymbol{e}_j)}{2h}. \tag{10.5}$$

Here $\boldsymbol{e}_j$ changes only coefficient j. Taylor expansion shows the leading truncation error is $O(h^2)$ for a sufficiently smooth map. Making h extremely small is not always beneficial: numerical contour interpolation, solver convergence error, and cancellation can dominate. Check derivative stability over a sensible range of h and ensure the same physical feature is being measured.

10.4 Interactions and nonlinear response

The second-order expansion of indicator i is

$$\delta y_i \simeq \sum_j J_{ij}\delta c_j + \frac{1}{2}\sum_{j,k} H_{ijk}\delta c_j \delta c_k, \tag{10.6}$$

where H_{ijk} is a second derivative. Terms with $j \neq k$ describe interactions between modes. Single-mode sweeps do not by themselves measure those cross terms.

For example, if $y = c_1 c_2$, varying either coefficient while keeping the other zero produces no response. Yet both nonzero together produce a finite response. This elementary example explains why the paper supplements individual-mode sweeps with combined aberrations.

10.5 Symmetry predicts some zero sensitivities

If a complete imaging configuration is invariant under reflection $x \mapsto -x$, an even phase perturbation cannot by itself generate an odd placement change under conditions preserving that symmetry. Likewise, a quantity may depend on a coefficient through c^2 near zero, giving zero first derivative but nonzero second derivative.

These conclusions require symmetry of the whole experiment: source, mask, pupil amplitude, aberration, focus, and metric definition. Oblique EUV mask scattering can destroy a symmetry present in a simplified scalar calculation. Mode names alone cannot establish which physical direction shifts.

10.6 Why a physical inverse is difficult but not impossible

The paper says required aberrations cannot be deduced using physical laws or equations because of complexity. The careful interpretation is that a convenient closed-form inverse is generally unavailable. The forward map is still governed by wave physics; its derivatives can be computed, and iterative constrained optimization can use them. The neural network offers an approximate, data-driven route to candidate solutions. It does not replace Maxwell's equations or prove that physics-based inverse design is impossible.

> **Check your understanding.** Can a coefficient with a small derivative at zero be given an unlimited budget? No. Higher-order effects, interactions, changes in the active contour, and other patterns can become important away from zero.

Part III

Understanding the authors' inverse model

Chapter 11

Reconstructing the paper's dataset and assumptions

Paper location: Section 2A, Tables 1–3, and the preprocessing in Section 2C.

11.1 The optical conditions that define this particular experiment

An aberration-to-image map is conditional on many fixed inputs. More explicitly, one should write

$$\boldsymbol{y} = F(\boldsymbol{c}; \lambda, \mathrm{NA}, S, \mathcal{M}, \mathcal{R}, \mathcal{Q}),$$

where S is illumination, $\mathcal{M}$ is mask geometry and material response, $\mathcal{R}$ is resist or contour extraction, and $\mathcal{Q}$ is the set of sampled process conditions. Changing any of these changes the map to be learned.

Table 11.1: The paper's Table 1, expressed with explicit meanings. Values are reported from the publication; qualifications below are this guide's interpretation.

Quantity	Reported value
Vacuum wavelength	13.5 nm
Projection NA	0.33
Chief-ray angle at object	6°
Illumination	Quasar: inner radius 0.6, outer radius 0.8, angle 30°, rotation 45°
Absorbing topography	Ta_6N_4, 15 nm in wafer scale
Reflecting multilayer	Mo/Si, respectively 0.725 nm and 1 nm in wafer scale
Focus settings	$-15, 0, +15$ nm
Exposure-dose variation	±5%
Zernike indices	Z_2 through Z_{25}
Coefficient sweep	-20 to $+20$ mλ, step 1 mλ

The optical constant database, complete multilayer termination, number of periods, exact source discretization, polarization, mask sidewalls, pupil apodization, and resist settings are not specified well enough for a numerical reconstruction. A student should understand their role rather than infer them from typical values.

11.2 The intended pattern families in Table 2

Through-pitch testing means holding a nominal feature type or size while varying its neighbourhood spacing. This probes proximity effects and different pupil sampling. It does not mean changing

Table 11.2: Pattern specification reported in the paper's Table 2. Each pitch list contains seven entries; rectangular pitch pairs are read position by position.

Family	$(\mathrm{CD}_x, \mathrm{CD}_y)$ (nm)	p_x (nm)	p_y (nm)
Square vias	$(14, 14)$	38, 40, 42, 46, 50, 80, 120	38, 40, 42, 46, 50, 80, 120
Rectangular vias	$(14, 28)$	56, 60, 64, 70, 80, 100, 150	70, 74, 78, 84, 94, 114, 164

the wavelength or continuously scanning every possible pitch.

The count suggested by two families and seven variants is 14 patterns. Each contributes six scalar indicators:

$$(\mathrm{CD}_x, \mathrm{CD}_y, \mathrm{PS}_x, \mathrm{PS}_y, \mathrm{PVB}_x, \mathrm{PVB}_y).$$

Thus $14 \times 6 = 84$ explains the reported network input width. This is a consistent reconstruction of the feature count, not a published list of the exact vector ordering.

11.3 The conflicting plot-label table

The paper's Table 3 assigns the following parameters to the bars labelled REC and SQ. It is important to preserve this discrepancy: The pitch families appear exchanged relative to the shape

Table 11.3: Paper Table 3 as printed. Every row uses CD $(14, 28)$ nm for REC and $(14, 14)$ nm for SQ. These pitch assignments disagree with Table 2.

	REC pitch (nm)		SQ pitch (nm)	
Pattern	p_x	p_y	p_x	p_y
1	38	38	56	56
2	40	40	60	74
3	42	42	64	78
4	46	46	70	84
5	50	50	80	94
6	80	80	100	114
7	120	120	150	164

assignments in Table 2. The first SQ pair $(56, 56)$ also differs from the rectangular pair $(56, 70)$ in Table 2. The text says REC means rectangle and SQ means square, but does not resolve which table reflects the actual simulation layout.

When reading the graphs, retain the authors' labels and pattern numbers. When performing this book's educational calculation, use explicitly declared geometry. Do not claim that a chosen correction reproduces the authors' dataset.

11.4 Three groups of aberration inputs

The stated simulation sets are:

1. **Individual-mode sweeps:** 41 coefficient values $-20, -19, \ldots, +20\,\mathrm{m}\lambda$ for each of 24 indices, with all others zero. This gives $41 \times 24 = 984$ rows.
2. **Definitive screening design:** 77 combined cases using levels $-20, 0, +20\,\mathrm{m}\lambda$.
3. **Random combinations:** 972 additional cases.

Their nominal total is $984 + 77 + 972 = 2033$ before any filtering. The paper does not give a complete split into training, validation, and testing, nor the exact random distribution, seed, or retained row count.

The individual sweeps include the all-zero vector 24 times. They therefore contain at most $984 - 23 = 961$ distinct vectors. Duplicates between groups may further reduce the distinct count. This matters in a replication: randomly splitting duplicate optical cases between training and test sets would weaken the independence of the test. The paper does not establish that such leakage occurred; it simply leaves the handling undocumented.

11.5 Why a screening design is used

A local quadratic response in K variables has

$$1 + K + K + \frac{K(K-1)}{2}$$

coefficients: intercept, linear terms, squares, and pairwise interactions. For $K = 24$ this is 325. An efficient screening design seeks useful information about dominant effects from far fewer than a full factorial set of runs, usually under sparsity or low-order assumptions.

Three levels allow curvature to be distinguished from a simple straight line. Carefully chosen combinations can separate some effects better than arbitrary combinations. However, 77 runs cannot independently estimate all 325 coefficients of an unrestricted quadratic model. The exact design matrix and its aliasing properties are not supplied. No specific orthogonality guarantee for these 77 rows can be reconstructed.

If every coefficient took all 41 grid values, a full 24-dimensional grid would contain

$$41^{24} = 5.091110945\ldots \times 10^{38}$$

points. This count is a hypothetical grid comparison, not a claim that the random group follows that grid. It illustrates why a few thousand cases can only sample a minute portion of the possible coefficient space.

11.6 What is one machine-learning sample?

The natural forward row is

$$\boldsymbol{c}^{(r)} \xrightarrow{F} \boldsymbol{y}^{(r)} \in \mathbb{R}^{84},$$

where the vector holds all patterns and metrics for aberration case r. The inverse training row reverses the pair: $\boldsymbol{y}^{(r)}$ is input, $\boldsymbol{c}^{(r)}$ is target. The 14 patterns are therefore features within one optical case. They are not automatically 14 independent realizations of the same input.

Keeping them separate is physically useful. For two patterns with responses $\delta y_1 = c$ and $\delta y_2 = -c$, averaging produces zero and loses all sensitivity to c. A feature vector $(c, -c)$ preserves it. This is the paper's motivation for avoiding averages over pattern performance during learning.

11.7 The 24-versus-25 output issue

The inclusive range $Z_2, \ldots, Z_{25}$ contains 24 coefficients. The $984 = 41 \times 24$ count and plotted axes support 24 varied terms. However, the prose repeatedly refers to the first 25 terms and the reported network has 25 output neurons.

One possible explanation is inclusion of a fixed piston coordinate, but the paper does not confirm that explanation. If piston is included as an unconstrained output, intensity indicators cannot determine it. A reconstruction must establish the exact output dictionary rather than fabricate a twenty-fifth observable aberration.

11.8 Filtering and scaling are part of the model

Section 2C describes filtering poorly fitted or low-valued data and mentions a 0.001 threshold, but its wording mixes input data and output Zernike terms. It does not establish which rows or coordinates were removed, the units before filtering, or whether the decision used test information.

This ambiguity affects reproducibility and performance interpretation. For example, removing all near-zero shift examples can improve a percentage-error score while making predictions near the desired zero shift less well supported. Filtering must be defined from the training workflow and recorded; it is not just a harmless plotting choice.

Chapter 12

The inverse problem: existence, ambiguity, and feasible targets

12.1 Three separate questions

Given desired metrics $\boldsymbol{y}_*$, ask:

1. **Existence:** is there any allowed wavefront with $F(\boldsymbol{c}) = \boldsymbol{y}_*$, or sufficiently close?
2. **Uniqueness:** if a solution exists, is it the only one?
3. **Stability:** do small changes in desired metrics lead to small changes in the required coefficients?

A forward simulator may be perfectly deterministic even when the inverse fails one or more of these tests.

12.2 More indicators than coefficients does not settle uniqueness

With about 24 varied coefficients and 84 metrics, there are more reported numbers than unknowns. This does not imply either uniqueness or nonuniqueness by itself. Metrics can be correlated, insensitive, or redundant; nonlinear maps can also have multiple branches.

Locally, J has 84 rows and 24 columns. If it has full column rank, small coefficient changes are locally distinguishable by the measured metrics. If its rank is lower, some first-order changes are invisible. Even full local rank does not rule out two distant coefficient vectors with the same metrics.

Piston is an exact unobservable direction if included. A simple additional teaching example is $F(c) = c^2$: both $c = 1$ and $c = -1$ give the same output. In symmetric optical settings, sign ambiguities can arise from similar even dependence, but they should be established for the actual source and mask rather than assumed from a mode name.

12.3 The singular-value decomposition explains sensitivity

Any real Jacobian can be written

$$J = U\Sigma V^T. \tag{12.1}$$

The columns of V are orthogonal coefficient-space directions. Applying J to direction $\boldsymbol{v}_j$ gives $J\boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$. The singular value σ_j measures how much that coefficient combination changes the measured outputs.

A small σ_j means weak observability. Formally inverting this component divides by σ_j, amplifying measurement and modelling error. A zero value is a linear null direction. This is a more precise concept than calling one single Zernike term unimportant: a weak direction can be a correlated combination of many terms.

Before calculating an SVD, scale coordinates appropriately. Otherwise replacing nm by milliwaves changes numerical singular values simply by changing units. A useful scaled Jacobian uses coefficient allowance scales in its columns and metric tolerance scales in its rows.

12.4 Linear inverse and regularization

For desired change $\delta\boldsymbol{y}$, a weighted least-squares inverse solves

$$\min_{\delta \boldsymbol{c}} \frac{1}{2} \left\| Q_y^{1/2} (J\delta\boldsymbol{c} - \delta\boldsymbol{y}) \right\|^2 + \frac{\alpha}{2} \left\| L\delta\boldsymbol{c} \right\|^2 . \tag{12.2}$$

Q_y weights metric errors; L can penalize coefficient magnitude or difficult-to-manufacture combinations; $\alpha \geq 0$ balances the terms. Differentiation gives

$$(J^T Q_y J + \alpha L^T L)\delta\boldsymbol{c} = J^T Q_y \delta\boldsymbol{y}. \tag{12.3}$$

With $Q_y = I$ and $L = I$, an SVD component is multiplied by $\sigma_j/(\sigma_j^2 + \alpha)$ rather than $1/\sigma_j$. Small, unreliable directions are suppressed. The penalty is a design preference, not additional optical data.

12.5 Feasible outputs occupy a structured subset

If $\boldsymbol{c}$ varies in a K-dimensional region, its smooth image in 84-dimensional metric space has at most K local degrees of freedom. Many metric combinations are therefore unattainable.

Being inside the minimum and maximum of each training feature is a necessary range check, but not a sufficient physical-feasibility check. For example, data satisfying $y_2 = y_1^2$ with $-1 \leq y_1 \leq 1$ have coordinate ranges $y_1 \in [-1, 1]$ and $y_2 \in [0, 1]$. The point $(0, 1)$ lies inside both ranges but not on the curve.

The paper's 118 target vectors come from other simulations under the same lithographic conditions and were not used in training or validation, according to Section 3. That is stronger than choosing 84 coordinates independently within a box. The paper nevertheless does not release the vectors or their generation procedure, so their precise coverage is unknown.

12.6 Why averaging inverse solutions can fail

Suppose training pairs have identical input $y = 1$ and two valid targets $c = +1$ and $c = -1$ under $F(c) = c^2$. A deterministic model trained by squared coefficient error minimizes

$$L(\hat{c}) = \tfrac{1}{2}[(\hat{c} - 1)^2 + (\hat{c} + 1)^2] = \hat{c}^2 + 1.$$

Its minimizer is $\hat{c} = 0$, but $F(0) = 0 \neq 1$. The mean of valid inverse solutions is not necessarily valid.

More generally, squared-error regression favors the conditional mean $\mathbb{E}[\boldsymbol{c} \mid \boldsymbol{y}]$, which can fall between distinct feasible branches. A good test is therefore the forward consistency

$$\boldsymbol{r}_{\text{image}} = F(G(\boldsymbol{y})) - \boldsymbol{y}, \tag{12.4}$$

not only the coefficient residual $G(\boldsymbol{y}) - \boldsymbol{c}_{\text{original}}$. The paper's forward re-simulation is an important part of its logic. It checks whether a proposed alternative actually retains useful imaging.

12.7 One output per input versus a distribution of solutions

A deterministic feedforward network returns one coefficient vector for one specified metric vector. Giving it 118 different target vectors can produce 118 candidates. That does not mean the same trained network samples all allowable wavefronts for a single fixed target.

To obtain diverse alternatives for the same target one could use multiple optimization initializations, explicit latent inputs, or a conditional generative model, followed by forward verification. These are possible extensions, not methods reported in this paper.

12.8 The feasible set is the real object of a budget

Let permitted coefficient conditions be $\boldsymbol{c} \in \mathcal{C}$ and imaging limits be $\boldsymbol{y} \in \mathcal{Y}_{\text{allowed}}$. The acceptable wavefront set is

$$\mathcal{A} = \{\boldsymbol{c} \in \mathcal{C} : F(\boldsymbol{c}) \in \mathcal{Y}_{\text{allowed}}\}. \tag{12.5}$$

The paper identifies a finite collection of candidates in or near this set, as judged by its simulator. The shape, volume, boundary, robustness, and manufacturability of the entire set require further analysis. Chapter 16 develops the distinction between candidate examples and a guaranteed tolerance region.

Chapter 13

The neural network and its training, step by step

Paper location: Sections 2B–2C, Fig. 4, and the reported network architecture.

13.1 A neuron is a weighted sum followed by a function

Given inputs $a_1, \ldots, a_d$, a neuron forms

$$q = \sum_{j=1}^{d} w_j a_j + b, \qquad a_{\text{out}} = g(q).$$

The weights w_j determine which inputs contribute and with what sign; the bias b shifts the response. The activation g introduces nonlinearity. A layer of neurons applies this operation in parallel:

$$\boldsymbol{q}^{(\ell)} = A^{(\ell)}\boldsymbol{a}^{(\ell-1)} + \boldsymbol{b}^{(\ell)}, \qquad \boldsymbol{a}^{(\ell)} = g_\ell(\boldsymbol{q}^{(\ell)}). \tag{13.1}$$

The function acts component by component. Repeating layers produces a composition of nonlinear functions. With only linear activations, all layers collapse into a single affine map, no matter how many are used.

For example, $g(q) = \tanh q$ is smooth, ranges from -1 to $+1$, and has derivative $1 - \tanh^2 q$. A rectified linear unit is $\max(0, q)$ and is piecewise linear. The paper does not report its activation functions, so neither choice can be attributed to the authors. Likewise, an ordinary unbounded linear regression output is a possible choice, but the actual output activation is undocumented.

13.2 What the network is asked to learn

Here the input is the 84-component imaging vector, not a pupil image. The output is a vector of Zernike coefficients, not CD or PVB. The desired relationship is

$$G_\Theta : \boldsymbol{y} \longmapsto \boldsymbol{c},$$

where Θ collects all trainable weights and biases. During training, the known simulator pair $(\boldsymbol{y}^{(r)}, \boldsymbol{c}^{(r)})$ supplies the input and target. During use, a new desired imaging vector supplies the input.

Calling this a backpropagation network describes how derivatives of the training loss are computed. It does not mean light is physically propagated backward through a Maxwell solver. Back prediction refers to the direction of the learned inverse mapping; backpropagation refers to the chain rule used to train it.

13.3 Interpreting the reported layer widths

The paper gives

$$84,\ 84,\ 64,\ 64,\ 50,\ 36,\ 32,\ 25,\ 25.$$

Read literally, the network has 84 inputs and 25 outputs. Its seven hidden layers have widths $84, 64, 64, 50, 36, 32, 25$. Figure 4 is a simplified schematic and does not depict these exact widths or the full hidden-layer count.

A fully connected layer taking d_{in} inputs to d_{out} outputs has $d_{\text{in}}d_{\text{out}}$ weights and d_{out} biases. The literal architecture therefore has

$$\sum_{\ell=1}^{8}(d_{\ell-1}+1)d_{\ell} = 7140 + 5440 + 4160 + 3250 + 1836 + 1184 + 825 + 650 = 24\,485$$

trainable parameters. This calculation assumes the conventional dense architecture; the publication does not provide implementation code to confirm all details.

The parameter count exceeds the number of simulated aberration cases. That does not mathematically prove overfitting: each case has multiple outputs, and training structure matters. It does make independent testing, preprocessing disclosure, and regularization choices important. The 25-output ambiguity discussed in Chapter 11 remains.

13.4 Why feature normalization matters

CD may be of order tens of nanometres while PS_y can be a small fraction of a nanometre. An optimizer acting on raw coordinates may assign very different numerical scales to these quantities. Featurewise min–max normalization to $[-1, 1]$ is

$$\widetilde{x} = 2\frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1. \tag{13.2}$$

Solving for the physical value,

$$x = x_{\min} + \frac{\widetilde{x}+1}{2}(x_{\max} - x_{\min}). \tag{13.3}$$

The paper names MATLAB's `mapminmax`, which performs rowwise range mapping and supports applying stored settings to new data [6].

The minimum and maximum must be fitted on training data and saved. Use those same settings for validation, testing, and deployment. Recomputing them independently for a new batch changes the meaning of every network coordinate. Including held-out data when fitting them lets test information enter preprocessing.

If $x_{\max} = x_{\min}$, the feature has no training variation and the formula divides by zero. It needs explicit handling. If a new value lies outside the training range, its normalized value lies outside $[-1, 1]$; min–max mapping does not by itself clip it. Similarly, undoing output normalization does not impose a coefficient bound unless the output is explicitly bounded or clipped.

13.5 The training loss

Let N be the number of training samples and K the number of output coordinates actually trained. One common mean-square loss is

$$L(\Theta) = \frac{1}{NK}\sum_{r=1}^{N}\sum_{j=1}^{K}\left[\widehat{c}_j^{(r)} - c_j^{(r)}\right]^2. \tag{13.4}$$

Often the coefficients here are normalized coordinates. The paper says training uses MSE and a goal of 10^{-8}, but does not supply enough preprocessing detail to turn that number into a guaranteed physical coefficient error.

If, purely as an example, one output spans $[-20, +20]\,\mathrm{m}\lambda$ and is mapped to $[-1, 1]$, then a normalized RMSE 10^{-4} corresponds to $0.002\,\mathrm{m}\lambda$ for that coordinate. The square root of an average training MSE does not bound each output, does not describe held-out error, and does not bound image error after the nonlinear forward map.

13.6 Backpropagation is repeated application of the chain rule

For clarity use a one-sample loss $L = \frac{1}{2}\,\|\widehat{\boldsymbol{c}} - \boldsymbol{c}\|^2$. For a linear output layer, the derivative with respect to its preactivation is

$$\boldsymbol{\delta}^{(L)} = \widehat{\boldsymbol{c}} - \boldsymbol{c}.$$

For a nonlinear output, multiply componentwise by g_L'. At each earlier layer,

$$\boldsymbol{\delta}^{(\ell)} = \left[(A^{(\ell+1)})^T \boldsymbol{\delta}^{(\ell+1)}\right] \odot g_\ell'(\boldsymbol{q}^{(\ell)}). \tag{13.5}$$

The symbol $\odot$ means componentwise multiplication. To understand the transpose, note that one hidden neuron influences every neuron in the next layer; its derivative sums all those downstream paths, weighted by the connecting weights.

The parameter derivatives are

$$\frac{\partial L}{\partial A^{(\ell)}} = \boldsymbol{\delta}^{(\ell)}(\boldsymbol{a}^{(\ell-1)})^T, \qquad \frac{\partial L}{\partial \boldsymbol{b}^{(\ell)}} = \boldsymbol{\delta}^{(\ell)}. \tag{13.6}$$

Average or sum over samples as required by the loss normalization. Backpropagation computes these derivatives efficiently by reusing intermediate derivatives, rather than perturbing every weight separately.

Worked example: One hidden neuron

Take $a = \tanh(wx + b)$, $\hat{c} = va + d$, and $L = \frac{1}{2}(\hat{c} - c)^2$. With $e = \hat{c} - c$,

$$\frac{\partial L}{\partial v} = ea, \quad \frac{\partial L}{\partial d} = e, \quad \frac{\partial L}{\partial w} = ev(1 - a^2)x, \quad \frac{\partial L}{\partial b} = ev(1 - a^2).$$

Every factor has a role: residual, downstream weight, activation slope, and upstream input. A deep network repeats the same multiplication and summation pattern across more paths.

13.7 Deriving the Levenberg–Marquardt update

Flatten all training residuals into a vector $\boldsymbol{e}(\Theta)$ and let $J_e = \partial \boldsymbol{e}/\partial\Theta$. This *training Jacobian* differentiates residuals with respect to network parameters. It is distinct from the imaging Jacobian $\partial \boldsymbol{y}/\partial \boldsymbol{c}$.

For a small parameter change $\Delta\Theta$, $\boldsymbol{e}(\Theta + \Delta\Theta) \simeq \boldsymbol{e} + J_e\Delta\Theta$. Minimize the locally linearized residual with a damping penalty:

$$\frac{1}{2}\,\|\boldsymbol{e} + J_e\Delta\Theta\|^2 + \frac{\mu}{2}\,\|\Delta\Theta\|^2\,.$$

Differentiate with respect to $\Delta\Theta$, set the derivative to zero, and obtain

$$(J_e^T J_e + \mu I)\Delta\Theta = -J_e^T \boldsymbol{e}. \tag{13.7}$$

Solve this linear system; do not explicitly form a matrix inverse in a numerical implementation.

For small μ, the step approaches Gauss–Newton. For very large μ, $\Delta\Theta \simeq -\mu^{-1} J_e^T \boldsymbol{e}$, which is a small gradient-descent step. The optimizer adjusts damping according to whether a proposed step improves the loss. This is the mathematical reason LM can combine rapid local convergence with caution away from a good fit.

For a squared-residual loss, the exact Hessian contains $J_e^T J_e + \sum_i e_i \nabla^2 e_i$. Gauss–Newton drops the second term, which is often small near a good fit. LM adds the stabilizing diagonal term. It does not guarantee finding a global optimum of a deep network.

The paper identifies MATLAB's LM training function. Its official documentation distinguishes the MSE performance goal from validation-based stopping and documents the damping update [7]. These software facts confirm the interpretation of the algorithm; the derivation above follows directly from the local least-squares problem.

13.8 Training, validation, and test have different jobs

Training data determine the weights. Validation data guide choices such as stopping time or architecture. Test data estimate performance after those choices are fixed. Repeatedly changing the architecture after looking at the same test set makes that set part of model selection.

The paper says training stops below MSE 10^{-8} or after six rounds without an obvious decrease. Standard MATLAB `trainlm` uses a six-failure default for *validation* performance. The paper's wording does not settle whether its implemented condition was precisely this default or a different training criterion. Record that ambiguity rather than equating training stagnation with validation failure.

13.9 What the learning theory does and does not promise

The Introduction cites approximation results for neural networks. Such results motivate using nonlinear networks to approximate continuous functions under suitable conditions. They do not establish that this particular architecture, finite dataset, and optimizer will recover a stable inverse of an ambiguous optical map.

In particular, they do not guarantee extrapolation beyond the training region, compliance with coefficient bounds, a manufacturable optical system, or reliable behavior when the source or mask changes. The paper's useful evidence is its forward-simulation validation, not the general existence of approximation theorems.

Chapter 14

Errors, percentages, and what validation means

Paper location: Equations (2)–(3), Figs. 5–10, and the statements about small PS values.

14.1 Root mean square error

For predictions $\hat{y}_i$ and reference values y_i, define $e_i = \hat{y}_i - y_i$. Then

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2}. \tag{14.1}$$

This is the paper's Eq. (2). Squaring prevents positive and negative errors from cancelling. Taking the square root restores the units. RMSE of a CD in nm is in nm. RMSE of a coefficient in waves is in waves. The figure axis must be checked before comparing magnitudes.

If $\bar{e}$ is the error mean and $s_e^2 = N^{-1}\sum_i (e_i - \bar{e})^2$, direct expansion gives

$$\mathrm{RMSE}^2 = \bar{e}^2 + s_e^2.$$

RMSE combines systematic bias and scatter. It is not the maximum error. For example, 99 exact predictions and one 1 nm error have RMSE 0.1 nm over 100 cases.

14.2 Mean absolute percentage error

The paper's Eq. (3) is

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\left|\frac{\hat{y}_i - y_i}{y_i}\right|. \tag{14.2}$$

The absolute value makes each term nonnegative, including for negative reference values. The denominator makes the statistic dimensionless. It also gives large weight to small reference magnitudes and is undefined when $y_i = 0$.

For a 14 nm CD, an error of 0.014 nm is 0.1 percent. For a 0.01 nm pattern shift, the same 0.014 nm error is 140 percent. The second percentage looks alarming, but the physical displacement error is identical. Conversely, a 1 percent error on a large quantity can exceed a stringent absolute tolerance.

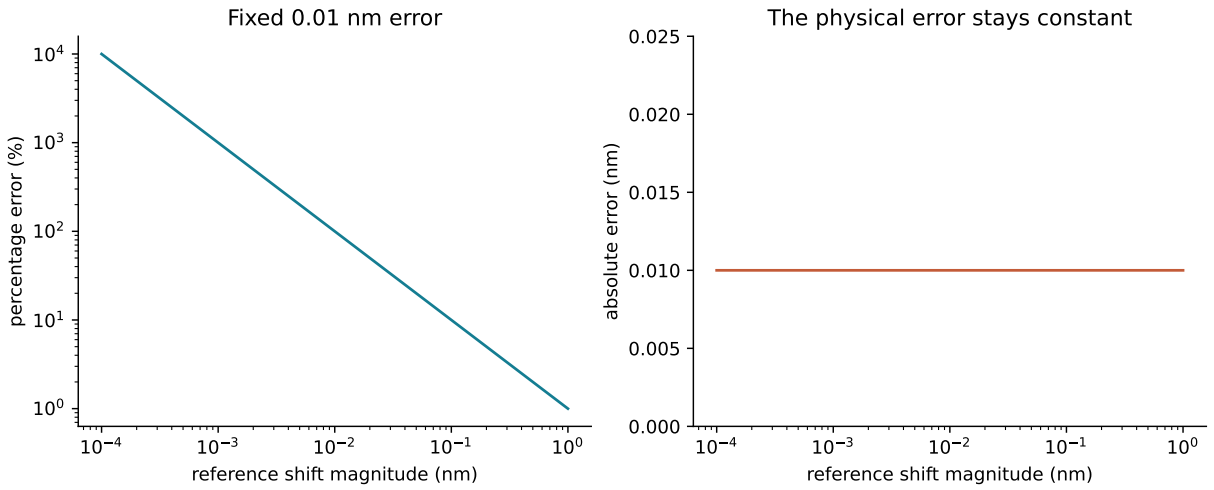


Figure 14.1: A fixed absolute error of 0.01 nm produces an arbitrarily large percentage as the reference shift approaches zero. The right panel shows that the physical error does not diverge.

14.3 Why near-zero shifts are especially awkward

Zero is often the desired shift. A metric that becomes undefined exactly at the target is a poor sole measure of control quality. The paper separates values around 0.14 nm in Fig. 8 and around 0.1 nm in Fig. 10. The 0.14 nm scale is 1% of the nominal 14 nm width; it is not the paper's final ± 0.4 nm acceptance boundary.

The text describes values above and below these thresholds but does not fully document how signed shifts, exact zeros, or removed values were handled. A careful replication would define whether the split uses $|y_i|$, explicitly state any denominator floor, and report the sample count in each group.

A worsening MAPE for smaller $|y|$ can occur with unchanged absolute error. Therefore the graph alone does not demonstrate that a neural network intrinsically cannot predict small numbers. Noise, feature scaling, cancellation between coefficients, weak observability, target density, and preprocessing can all contribute. The paper's explanation should be treated as a suggested account, not a mathematical inevitability of neural networks.

14.4 Better companion measures for a placement budget

Keep RMSE and add interpretable quantities:

$$\mathrm{MAE} = \frac{1}{N}\sum_i |e_i|, \tag{14.3}$$

$$e_{\max} = \max_i |e_i|, \tag{14.4}$$

$$\text{pass fraction} = \frac{\#\{i : |\mathrm{PS}_{y,i}^{\text{verified}}| \le b\}}{N}. \tag{14.5}$$

For a specified tolerance $b > 0$, a useful normalized error is $|e_i|/b$. Its denominator remains meaningful at zero target shift. Quantiles show tail behavior; signed mean error detects systematic bias. None of these removes the need to state which patterns and operating conditions are included.

Worked example: A candidate near the boundary

Suppose desired shift is 0.395 nm and the prediction-to-simulation error is +0.020 nm. The verified shift is 0.415 nm and fails a 0.4 nm limit. A small absolute discrepancy is consequential here. The same discrepancy at a target of zero would still meet the limit. The relevant question is error relative to remaining margin, not just whether a number sounds small.

14.5 Coefficient error versus image error

For a held-out original pair $(\boldsymbol{c}, \boldsymbol{y} = F(\boldsymbol{c}))$, the network outputs $\hat{\boldsymbol{c}} = G(\boldsymbol{y})$. Two separate residuals are

$$\boldsymbol{r}_c = \hat{\boldsymbol{c}} - \boldsymbol{c}, \qquad \boldsymbol{r}_y = F(\hat{\boldsymbol{c}}) - \boldsymbol{y}.$$

Figure 5 concerns coefficient differences. Figures 6 and 7 concern the imaging differences after forward simulation. A sizeable $\boldsymbol{r}_c$ can coexist with a small $\boldsymbol{r}_y$ if the two wavefronts are nearly equivalent for the selected indicators. However, coefficient disagreement by itself does not prove a valid alternative: the forward residual must actually be checked.

Agreement in a few scalar metrics also does not establish identical full contours, NILS, stochastic defect behavior, or performance on unseen pattern classes. The paper validates the indicators it measures.

14.6 What an independent simulation test establishes

The 118 additional target vectors provide a useful test beyond the model's training/validation set, as described by the authors. Their proposed coefficients are sent through the same type of forward simulator under the same lithographic conditions. This tests consistency with the simulated model and conditions. It is not an experimental measurement on a fabricated projector or wafer.

Using the same simulator for training labels and final checking can validate the inverse approximation while leaving a common forward-model bias invisible. Actual engineering deployment also needs calibrated mask, resist, optical, and metrology models. These limitations define the scope of the evidence rather than negate the usefulness of a simulation-based study.

Part IV

Reading the results and constructing a defensible budget

Chapter 15

Read the paper with this chapter beside it

15.1 Title, abstract, and the actual claim

The title joins three ideas: an aberration budget, an EUV projector, and the image quality of a contact layer. The budget is conditional on that layer's selected patterns and imaging indicators. The paper does not claim to characterize every possible mask pattern.

The abstract's phrase back propagating neutral network should be read as *backpropagation neural network*. The learned map takes imaging indicators as inputs and proposes wavefront coefficients as outputs. Forward re-simulation tests the resulting images. The important conclusion is that the authors obtained candidate wavefront distributions with useful simulated imaging performance under their chosen conditions.

The phrase wavefront aberration margin should not immediately be read as a certified independent tolerance on each mirror or each Zernike coefficient. A margin needs a defined region, perturbation model, and worst-case or probabilistic guarantee. The paper primarily demonstrates successful coefficient combinations.

15.2 Introduction: why this work was attempted

15.2.1 Shorter wavelength makes a fixed OPD more serious

The 193 nm versus 13.5 nm comparison is worked through in Chapter 3. In physical length units, the error is unchanged. In waves and radians, it is about 14.3 times larger at EUV wavelength. The consequences listed in the paper have distinct meanings:

- Reduced process window means fewer focus-dose combinations meet all imaging tolerances.
- Lower image fidelity means the printed contour departs more from its intended shape.
- Lower NILS means a shallower relative edge slope and generally greater edge sensitivity.
- Pattern centre offset is a placement error even if width is unchanged.
- Unwanted artifacts can include distorted, bridged, or otherwise undesired image features; the paper does not provide a separate artifact-rate study.

15.2.2 The earlier aberration-control methods

The Introduction discusses several classes of prior work. They are context, not additional algorithms implemented and validated in this paper.

Pattern-specific wavefront adjustment uses available optical controls to improve selected pattern images. The Nikon TAO example illustrates that an optimum for one pattern need not minimize a pattern-independent wavefront norm.

Thermal prediction and control models how absorbed power changes optical shape or index and adjusts compensators before or during exposure. A simple control description is $\boldsymbol{c}(t) = \boldsymbol{c}_{\text{thermal}}(t) + B\boldsymbol{u}(t)$, where $\boldsymbol{u}$ represents actuator commands and B their wavefront influence. Predictive control uses a future estimate of $\boldsymbol{c}_{\text{thermal}}$ when choosing $\boldsymbol{u}$. This equation is explanatory, not the controller in any cited instrument.

Wavefront manipulators provide practical degrees of freedom to alter the pupil phase. The Introduction's FlexWave example concerns DUV immersion equipment. It should not be read as a statement that arbitrary 25-dimensional phase changes are independently available in an EUV scanner.

Optical prescription optimization adjusts mirror shapes, locations, and aspheric coefficients. Grouping methods divide a complicated optical system into tractable subassemblies. Ray tracing calculates actual surface intersections and directions. Finite-aberration methods evaluate finite departures beyond a solely infinitesimal paraxial analysis. These approaches operate on physical design parameters rather than directly on an unconstrained image-to-mode map.

Global search heuristics mentioned in the references include simulated annealing and particle swarm methods. Annealing sometimes accepts a worsening objective by a temperature-dependent rule to escape local minima. Particle swarm search updates candidate locations using their own and the group's best positions. The present paper's stated training algorithm is LM; it does not report applying those design algorithms in its own 118-candidate test.

The quoted $\lambda/50$, $\lambda/70$, and $\lambda/30$ wavefront scales belong to the cited contexts. At 13.5 nm they are 0.270, 0.19286, and 0.450 nm respectively. Comparisons require the same RMS definition, removed modes, aperture weighting, field points, and wavelength.

15.3 Figure 1: the optical system

Paper location: PDF p. 1 / journal p. 7270. Read the drawing as a map of functional stages: illumination creates the source distribution at the mask; the reflective mask diffracts the field; projection optics form the wafer image. The colored boxes identify these components, not separate mathematical wavefront-error budgets.

The EUV system is reflective, unlike an ordinary camera drawn with glass lenses. The naming issue in the caption was noted in Chapter 5. Nothing in the figure gives a mirror prescription, surface sensitivity matrix, or actuator range. Consequently it cannot turn predicted Zernike coefficients into a unique hardware design.

15.4 Equation (1): the entire wavefront represented by coefficients

Paper location: PDF p. 2 / journal p. 7271. The equation

$$W(\rho, \theta) = \sum_{i=1}^{\infty} Z_i F_i(\rho, \theta)$$

is a basis expansion. Its technical content is explained in Chapter 7. A finite implementation replaces the infinite sum by a selected set of terms. Truncation neglects all remaining wavefront structure; it does not prove that those higher terms vanish in a real projector.

The coefficients are not independent physical defects merely because the basis is orthogonal. A single mirror alignment error may excite several coefficients, and a single coefficient may receive contributions from many surfaces. The missing basis normalization and index dictionary are necessary for quantitative comparisons with another optical program.

15.5 Table 1 and Figure 2: illumination and process conditions

Paper location: PDF p. 3 / journal p. 7272. Table 1 fixes wavelength, numerical aperture, incidence geometry, source shape, nominal stack, focus range, and coefficient sweep. Together these define the conditional forward map. Figure 2 shows the angular distribution of illumination in normalized pupil coordinates. Bright regions are active source directions. They are not four holes in the physical mask and not four printed contacts.

The source is separated from the projection pupil: the source chooses incident directions, and the projection pupil selects diffracted outgoing components. Chapter 8 derives how they interact. At the end of the paper the authors state that this source was not well optimized for the contact layer. That limitation applies to the learned aberration preferences as well.

15.6 Table 2 and Figure 3: what the mask represents

Paper location: PDF p. 3 / journal p. 7272. The top view in Fig. 3 depicts the opening and absorber in a periodic mask cell. The side view depicts absorber topography on a multilayer. The square example is 14 nm wide on 38 nm pitch in wafer-referred units. The diagram makes it clear that a hole in the absorber exposes a reflective region; it is not a transparent hole through an ordinary lens mask.

The absorber and multilayer phases affect the complex diffraction orders. Their geometry is one reason horizontal and vertical imaging can differ. Table 2 gives two feature families and through-pitch examples. The conflict with Table 3 must be remembered whenever translating a plot's Pattern_1 through Pattern_7 into an actual mask layout.

15.7 Figure 4: the simplified network

Paper location: PDF p. 3 / journal p. 7272. Each circle is a scalar neuron and each connection represents a trainable weight. The arrows of evaluation run from imaging metrics to aberration coefficients. During training, derivative information moves backward through the computational graph.

The figure is explicitly simplified. The reported architecture in Section 2C has seven hidden layers, while the schematic shows fewer hidden groups. Use the written widths for the literal parameter-count calculation, while recognizing the implementation details remain incomplete. The diagram provides no activation functions, output constraints, or training/test split.

15.8 Equations (2) and (3): comparing errors

Paper location: PDF p. 4 / journal p. 7273. Equation (2) is RMSE; Eq. (3) is MAPE. Their derivations and limitations are in Chapter 14. Every time an error is quoted, identify the target quantity, physical units, aggregation set, and treatment of zeros.

An RMSE across coefficients is not an RMSE across printed CDs. A percentage error in a small signed shift is not interchangeable with a percentage error in a 14 nm width. These distinctions explain why several later plots can look very different without a logical contradiction.

15.9 Table 3: labels for all subsequent bar charts

Paper location: PDF p. 4 / journal p. 7273. REC and SQ are described as rectangular and square patterns, with seven variants each. The pitch assignments conflict with Table 2, as transcribed in Chapter 11. Until this is resolved, a comparison such as REC versus SQ can be reported using the authors' labels but should not support a precise claim about a particular reconstructed pitch family.

The bars retain information by pattern and direction rather than averaging all of them into one number. That preserves, for example, a poor result for one sparse pitch that an overall average might conceal.

15.10 Figure 5: different coefficients for similar requested images

Paper location: PDF p. 4 / journal p. 7273. Each bar summarizes disagreement between a predicted coefficient and the original simulator coefficient associated with the input image metrics. The vertical axis is in wavelengths. Larger bars mean the inverse model has not simply recovered the original wavefront.

The authors interpret this as evidence of alternative aberration combinations. That interpretation is plausible only together with the subsequent image re-simulation. By itself, Fig. 5 is also compatible with inverse prediction error. Nearly equivalent images may come from weakly observable combinations or nonlinear inverse ambiguity, as discussed in Chapter 12.

The plotted coefficient labels run from Z_2 to Z_{25}. They reinforce the 24-varied-coordinate interpretation, but do not resolve the network's 25-output statement. The figure is not a measurement of mirror surface error or a tolerance specification.

15.11 Figure 6: physical errors after forward re-simulation

Paper location: PDF p. 5 / journal p. 7274. **Panel (a), CD:** the simulator is run with predicted coefficients, and the resulting widths are compared with the input target widths. Bars are separated into x/y and REC/SQ. The vertical unit is nm. The graph shows noticeably different errors by pattern family and pitch, so an overall accuracy number would hide structure.

Panel (b), PS: this is the same comparison for feature-centre shift. Direction matters. Several vertical-shift bars for the SQ-labelled family are visibly larger than the corresponding horizontal-shift bars. The physically relevant result is their absolute size in nm and their relation to a specified placement tolerance.

Panel (c), PVB: the comparison concerns process-contour variation, not nominal feature width. A good CD prediction does not imply a good PVB prediction because the latter depends on

several focus-dose conditions.

The figure gives aggregate RMSE values, not maximum per-case residuals. The paper does not release numerical arrays for these bars; fine numerical values should not be inferred from raster-image heights.

15.12 Figure 7: percentages tell a different story

Paper location: PDF p. 6 / journal p. 7275. Panels (a) and (b) show MAPE for CD and PVB. Panel (c) shows horizontal PS, and panel (d) vertical PS. Most plotted errors for the first three categories are relatively small percentages, while vertical PS has substantially larger percentages.

This is the expected danger when PS_y is close to zero: the reference denominator becomes small. The statement that most errors are below 1 percent does not describe every bar, especially not the vertical-shift category. RMSE and MAPE summarize different functions of the residuals, so their relative bar ordering need not match.

15.13 Figure 8: splitting around 0.14 nm

Paper location: PDF p. 6 / journal p. 7275. Panel (a) concerns the larger-shift subset and panel (b) the smaller-shift subset, using the stated 0.14 nm threshold. The smaller-reference subset has much worse percentage errors. The graph supports that descriptive statement.

To decide whether the underlying physical prediction is worse, one also needs absolute errors, counts, and uncertainty within each subset. The mathematical denominator effect alone can generate the observed trend. The threshold is a diagnostic choice motivated by 1 percent of 14 nm; it is not a pass/fail criterion for lithographic placement.

15.14 Section 3: the additional 118 cases

Paper location: PDF pp. 5–7 / journal pp. 7274–7276. The authors refer to a 2021 IRDS white paper and infer a requirement that vertical contact placement be within 1 nm. The paper does not give an explicit allocation derivation from that roadmap to a standalone projector-wavefront PS allowance. Treat 1 nm as the authors' stated motivation rather than a universal independently established per-mode limit.

They then use a tighter target range, -0.4 to $+0.4$ nm, for vertical shift in 118 sets of imaging indicators. The remaining indicators are kept within training-data ranges. The targets originate in other simulations under the same stated conditions, not in arbitrary independent coordinate draws. The model produces 118 coefficient vectors and the forward simulator checks their imaging.

15.15 Table 4: the remaining target ranges

Paper location: PDF p. 6 / journal p. 7275. The publication labels both final columns PVB_X. Using one as PVB_Y is a reasonable hypothesis because six metrics include both directions, but it is not confirmed.

The REC horizontal widths in this table are also near 27–30 nm despite the nominal horizontal CD of 14 nm elsewhere. Printed CD can differ from nominal CD, but the discrepancy and the earlier table conflict prevent us from deciding whether this is a measurement convention, strong image bias, an axis swap, or a publication error. Do not relabel the numbers to force agreement.

Table 15.1: Paper Table 4 transcribed without silently correcting its repeated last header. All entries are nm. The final column is likely intended to represent vertical PVB, but that is an inference.

Family	Bound	CD_x	CD_y	PS_x	PVB column 1	PVB column 2
REC	Maximum	29.636	27.447	4.233	2.888	2.676
REC	Minimum	26.927	25.520	2.319	1.850	1.881
SQ	Maximum	13.488	12.663	4.190	7.238	8.755
SQ	Minimum	12.023	9.7623	2.083	3.079	3.791

These are training-range conditions, not independently justified production acceptance limits for all five indicators. In particular, allowing horizontal shifts of several nm within the training range does not prove they satisfy a complete overlay budget.

15.16 Figure 9: candidate ranges and accuracy

Paper location: PDF p. 7 / journal p. 7276. **Panel (a):** the minimum and maximum predicted values are plotted for each coefficient. The authors identify small ranges for Z_3, Z_{20}, Z_{21}, and Z_{22} and connect them with vertical PS. These are observations within their selected candidate population. They do not by themselves identify independent sensitivities or prove that each coefficient must always be tightly bounded.

The coefficient axis is in *waves*. Some visible extrema are roughly -0.065λ or $+0.05\lambda$, well beyond the stated training sweep of $\pm 0.020\lambda$. Those are approximate visual readings, not digitized data. They flag an important issue: the inverse output may extrapolate beyond the sampled coefficient range, or the publication may contain an unexplained scale/reporting difference. The paper does not establish a hard output-bound enforcement rule.

The caption's reference to the first 25 sets is also unclear in relation to the 118 proposed combinations and 24 plotted coefficient indices. It should not be converted into an undocumented sample count.

Panel (b): absolute RMSE of verified imaging indicators versus target indicators is shown by pattern. **Panel (c):** MAPE is shown for the indicators other than vertical PS. **Panel (d):** vertical-PS MAPE is shown separately and is large for some patterns. The paper text refers to panels (b) and (c) when discussing PS percentage behavior, but the dedicated PS MAPE panel is (d); use the actual axis and legend to interpret it.

The panel (a) intervals are *marginal ranges across candidates*. Taking the largest allowed value from every bar and combining them does not generally produce another valid candidate. Correlations between coefficients may be responsible for successful compensation. Chapter 16 provides an explicit counterexample.

15.17 Figure 10: a second small-value split

Paper location: PDF p. 8 / journal p. 7277. This figure repeats the larger-versus-smaller vertical-shift comparison for the new application cases using 0.1 nm instead of 0.14 nm. The smaller-reference group again has larger MAPE. The two split thresholds belong to different analyses; there is no single fixed 0.14 nm filtering rule governing all later results.

Since the new targets were deliberately selected near zero, strong percentage penalties are unsurprising. To judge a 0.4 nm budget, inspect the signed verified values and worst-case absolute errors as well. The figure does not provide those complete arrays.

15.18 Figures 11 and 12: two individual candidates

Paper location: PDF pp. 8–9 / journal pp. 7277–7278. Each figure has four parts:

1. **(a) Coefficient vector:** a particular proposed wavefront, plotted in waves. The vertical scales differ between the figures. Compare numerical axes rather than visual bar height.
2. **(b) CD:** paired bars compare the relevant target/prediction and simulator-reference widths for both families and directions.
3. **(c) Horizontal PS and PVB:** the same candidate is evaluated for additional imaging requirements and process robustness.
4. **(d) Vertical PS:** the small signed shifts are shown separately so that discrepancies are visible near zero.

The prose defines the suffixes `_pred` and `_ref`. Operationally, however, the network outputs *coefficients*; it does not output the CD bars directly in the described architecture. The meaningful comparison is the requested/input image vector against the forward-simulated image vector for the proposed coefficients. This distinction prevents misreading these graphs as an independently trained forward imaging network.

Figure 11 includes small shifts near or across zero. Figure 12 includes negative vertical shifts. These signed examples reinforce why an absolute-value tolerance must be used and why zero handling matters in MAPE. The authors report that all 118 proposed combinations satisfy the intended 0.4 nm vertical-shift condition after simulation. The two illustrated examples support the qualitative comparison, but are not a substitute for the unreleased full validation table.

15.19 Conclusion and the limits explicitly acknowledged

The authors conclude that the inverse model can propose aberration distributions satisfying selected imaging requirements and can assist budget analysis. They also state that the chosen source is not well optimized for the contact layer and that the patterns were not corrected by OPC. They propose including improved illumination and corrected patterns in future work.

OPC changes $\widetilde{t}$ or the rigorous mask-order amplitudes. Source optimization changes S. Both change the forward map F. A coefficient combination preferred before those changes is not guaranteed to remain preferred afterward.

The data-availability statement says the underlying data are not public and may be obtained from the authors on request. No such request has been made for this guide. The absent data limit exact numerical reproduction; they do not prevent deriving the physics or critically understanding the published experiment.

15.20 The claim you should be able to explain accurately

Under a fixed simulated EUV contact-layer setup, the authors train an inverse neural model from selected image metrics to Zernike coefficients, then verify proposed combinations by forward simulation. They find alternative wavefronts with useful agreement in the selected indicators and report acceptable vertical shifts for their 118 application cases. This supports pattern-dependent aberration selection within the studied setup. It does not establish a unique inverse, independent per-mode tolerances, a manufacturable mirror prescription, or general production performance across arbitrary layers.

Chapter 16

Turning candidate wavefronts into an engineering budget

16.1 State the acceptance condition before assigning tolerances

An engineering budget begins with an explicit list of requirements. For pattern p, examples are

$$|\mathrm{PS}_{y,p}| \le b_{y,p}, \quad |\mathrm{CD}_{x,p} - \mathrm{CD}_{x,p}^{\mathrm{target}}| \le b_{\mathrm{CD},p}, \quad \mathrm{PVB}_{x,p} \le b_{\mathrm{PVB},p}.$$

The conditions should identify the field point, process range, measurement definition, and whether compensation such as refocus or global image registration is allowed.

The paper emphasizes a vertical-shift target near $\pm 0.4\,\mathrm{nm}$ and keeps other requested indicators in training-data ranges. That is a particular candidate-generation experiment. A production budget would require explicit physical limits for all relevant indicators, rather than training coverage alone.

16.2 A local constraint is a slab in coefficient space

Let $\boldsymbol{c}_0$ be a verified candidate. For indicator i, write its residual from the target as r_i and its gradient row as $\boldsymbol{g}_i^T$. Locally,

$$r_i(\boldsymbol{c}_0 + \delta\boldsymbol{c}) \simeq r_i + \boldsymbol{g}_i^T \delta\boldsymbol{c}.$$

The two-sided acceptance condition is

$$|r_i + \boldsymbol{g}_i^T \delta\boldsymbol{c}| \le b_i. \tag{16.1}$$

This is the region between two parallel hyperplanes. Intersecting all such slabs gives a local feasible polytope. An optical or mechanical constraint on coefficients further restricts it. This geometry makes clear why tolerances are coupled.

16.3 The largest local RMS ball around a candidate

Suppose an orthonormal coefficient norm represents the retained RMS. For any $\|\delta\boldsymbol{c}\| \le R$, Cauchy–Schwarz gives

$$|\boldsymbol{g}_i^T \delta\boldsymbol{c}| \le \|\boldsymbol{g}_i\|\, R.$$

A sufficient local guarantee is $|r_i| + \|\boldsymbol{g}_i\| R \le b_i$ for every i. Hence a radius satisfying

$$R \le \min_i \frac{b_i - |r_i|}{\|\boldsymbol{g}_i\|} \tag{16.2}$$

fits inside the linearized constraints, omitting rows with zero gradient unless their residual already violates the limit.

This is a bound on *perturbations around a chosen candidate*. It differs from the candidate's own RMS and from a maximum admissible absolute RMS anywhere in the feasible set. A system can have an appreciable baseline wavefront and still have very little remaining tolerance to perturbation.

16.4 Independent box limits require a joint worst-case test

Suppose each coefficient perturbation is independently allowed within $|\delta c_j| \le a_j$. Then

$$\max_{|\delta c_j| \le a_j} |\boldsymbol{g}_i^T \delta \boldsymbol{c}| = \sum_j |g_{ij}| a_j. \tag{16.3}$$

The upper bound follows from the triangle inequality. It is attained by choosing the sign of each δc_j to align with g_{ij}. A safe linear box therefore requires

$$|r_i| + \sum_j |g_{ij}| a_j \le b_i \quad \text{for every required metric } i. \tag{16.4}$$

Checking one coefficient at a time while fixing the others cannot establish this joint guarantee.

Worked example: The error in treating marginal ranges as a budget

Suppose a required shift is proportional to $c_1 + c_2$ and must obey $|c_1 + c_2| \le 0.4$ in chosen units. The two candidates $(1, -1)$ and $(-1, 1)$ both give zero shift. Their observed coordinate ranges are $[-1, 1]$ for both coefficients. Combining the two maximum values gives $(1, 1)$, whose shift is 2 and fails badly.

16.5 Ellipsoidal budgets and covariance

For an ellipsoidal perturbation region $\delta \boldsymbol{c}^T Q \delta \boldsymbol{c} \le 1$ with positive definite Q, set $\boldsymbol{z} = Q^{1/2} \delta \boldsymbol{c}$. Then

$$\boldsymbol{g}_i^T \delta \boldsymbol{c} = (Q^{-1/2} \boldsymbol{g}_i)^T \boldsymbol{z}.$$

Maximizing over $\|\boldsymbol{z}\| \le 1$ gives

$$\max |\boldsymbol{g}_i^T \delta \boldsymbol{c}| = \sqrt{\boldsymbol{g}_i^T Q^{-1} \boldsymbol{g}_i}. \tag{16.5}$$

This allows tolerances to be wider along insensitive combinations and tighter along sensitive combinations.

For random perturbations with covariance C_c, linear propagation gives

$$C_y = J C_c J^T, \qquad \sigma_{y_i}^2 = \boldsymbol{g}_i^T C_c \boldsymbol{g}_i. \tag{16.6}$$

The covariance formula is a statistical statement, not a deterministic guarantee. If Gaussian assumptions are justified, a multiple of σ_{y_i} can be related to a tail probability. For non-Gaussian tails or systematic errors, that interpretation may fail.

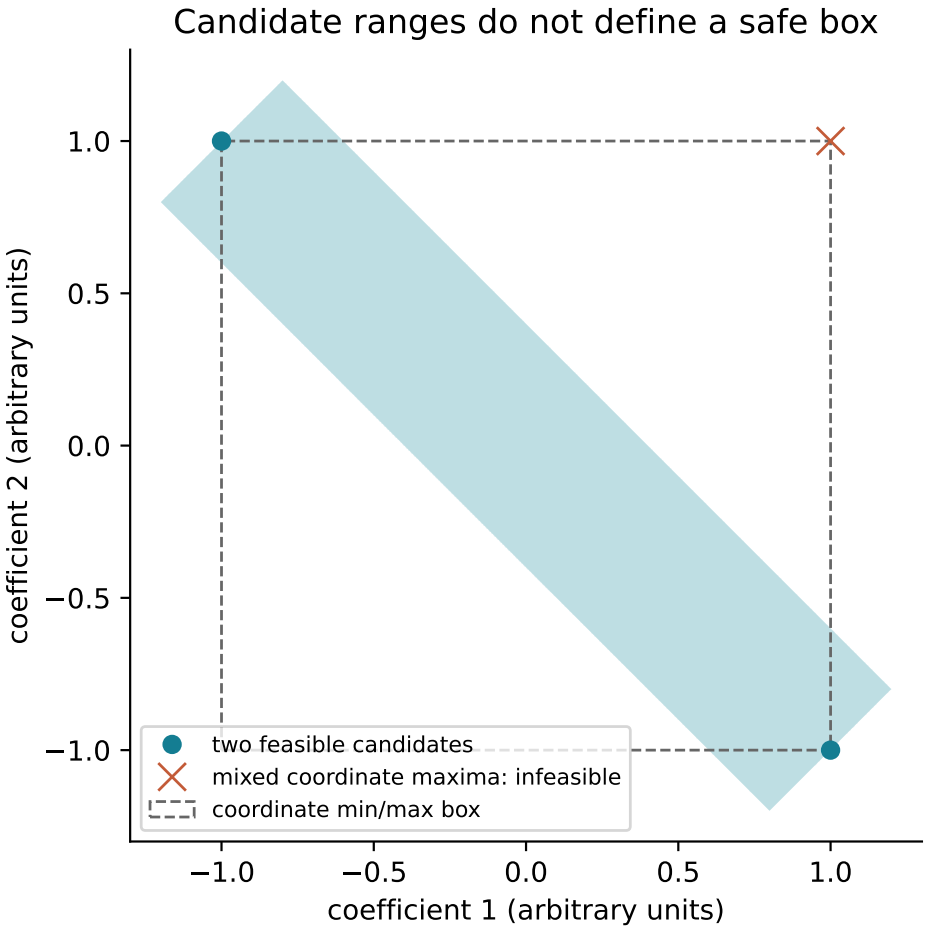


Figure 16.1: A correlated feasible region and two feasible candidates. The dashed box contains all independent mixtures of the candidates' coordinate ranges. Its upper-right corner is not feasible. This is why the min/max bars in the paper's Fig. 9(a) are not independent guaranteed tolerances. The additional diagonal bound in this illustration only makes the plotted region finite.

16.6 A mode-specific placement calculation

For our unit-RMS vertical tilt, $W = c_y Z_1^{-1} = 2c_y v$. Equation (6.3) gives

$$|\mathrm{PS}_y| = \frac{2|c_y|}{\mathrm{NA}}. \tag{16.7}$$

At NA = 0.33, a pure-tilt requirement $|\mathrm{PS}_y| \le 0.4\,\mathrm{nm}$ implies

$$|c_y| \le 0.066\,\mathrm{nm} = 4.88889\,\mathrm{m}\lambda \quad (\lambda = 13.5\,\mathrm{nm}).$$

A 20 milliwave tilt would shift the image by about 1.63636 nm. This is an exact translation result within the stated scalar shift-invariant pupil convention. It is not a numerical bound on the paper's Z_3 because its normalization and axis convention are not given, and compensating modes or registration could change the relevant allowance.

16.7 From wavefront coefficients to physical hardware

Let $\boldsymbol{q}$ collect physical design or adjustment variables: mirror decentres, tilts, surface modes, or actuator commands. Linearizing the optical system gives

$$\delta \boldsymbol{c} = B\,\delta \boldsymbol{q}, \qquad \delta \boldsymbol{y} \simeq JB\,\delta \boldsymbol{q}. \tag{16.8}$$

B is a physical influence matrix. Its column space contains the wavefront changes actually attainable to first order. A candidate coefficient vector outside that attainable set cannot be implemented by the available controls, even if its simulated image is good.

A practical adjustment problem might minimize

$$\left\| Q_y^{1/2}[F(\boldsymbol{c}_0 + B\delta \boldsymbol{q}) - \boldsymbol{y}_*] \right\|^2 + \alpha \left\| L_q \delta \boldsymbol{q} \right\|^2$$

subject to displacement, surface-slope, stress, alignment, or actuator bounds. The paper does not supply B, so it does not solve this hardware-realization step.

Worked example: An illustrative mirror-height allocation

Assume six equal, statistically independent mirror contributions, all with local incidence angle 6°, negligible coating contribution, and total permitted RMS OPD 0.27 nm. Using Eq. (3.14),

$$\sigma_{W,\mathrm{total}}^2 = 6(2\sigma_h \cos 6^\circ)^2.$$

Thus

$$\sigma_h = \frac{0.27}{2\cos 6^\circ \sqrt{6}} = 0.05542\,\mathrm{nm}.$$

This is about 55 pm per surface under the stated assumptions. Six perfectly correlated equal wavefront contributions would add in amplitude and give a different allocation, $0.27/[12\cos 6^\circ] \simeq 0.02262\,\mathrm{nm}$. Neither number is a recovered tolerance of the projector in the paper. They demonstrate how geometry and correlation enter a surface budget.

16.8 Nonlinear robustness and process variation

The Jacobian approximations are local. If a bound M_i on the relevant Hessian norm is known throughout a perturbation region, Taylor's theorem gives a remainder estimate

$$|R_i| \le \frac{1}{2} M_i \|\delta \boldsymbol{c}\|^2 .$$

A conservative ball test can then include $|r_i| + \|\boldsymbol{g}_i\| R + \frac{1}{2} M_i R^2 \le b_i$. Without such a bound, a local linear result needs nonlinear checking over the proposed tolerance region.

Likewise, checking focus at $-15, 0, +15$ nm does not mathematically prove every intermediate focus passes. The extremum of a nonlinear CD or PS curve can lie between samples. PVB envelopes involve maxima and minima, and can become nonsmooth when the process condition defining an extreme changes.

Useful engineering checks include nominal cases, candidate corners, random perturbations from an explicit distribution, and targeted searches for the worst violation. A finite test is evidence about the tested region and sampler; it is not automatically a continuous worst-case proof.

16.9 Finite success counts and probability

As a teaching calculation, suppose N cases are independently drawn from a fixed operational distribution and every case passes. If the true failure probability is p, the probability of seeing no failures is $(1-p)^N$. Setting this to 0.05 gives the usual one-sided 95 percent upper limit

$$p_{\text{upper}} = 1 - 0.05^{1/N}.$$

For $N = 118$, this is about 2.51 percent. Even under those strong sampling assumptions, zero observed failures would not prove zero failure probability. The paper does not establish that its selected 118 targets are independent random samples from a manufacturing distribution, so this calculation is not a statistical confidence claim about its results.

16.10 Optical determinism does not remove photon and resist noise

The paper's reported metrics are deterministic simulation indicators. EUV photon energy is $E_\gamma = hc/\lambda \simeq 91.84$ eV at 13.5 nm. For a deliberately illustrative dose of $20\,\text{mJ/cm}^2 = 200\,\text{J/m}^2$ incident on a $14\,\text{nm} \times 14\,\text{nm}$ area,

$$N_\gamma \approx \frac{200(14 \times 10^{-9})^2}{hc/(13.5 \times 10^{-9})} \simeq 2.66 \times 10^3.$$

This is an incident photon estimate for uniform exposure, not the actual absorbed count in a printed contact image.

For ideal Poisson counting, $\operatorname{Var} N = \mathbb{E}N$ and relative count noise is $1/\sqrt{N}$, about 1.94 percent for this example. Absorption, secondary-electron transport, resist chemistry, and nonuniform intensity add further structure. The IRDS discussion identifies stochastic defects and pattern control as major lithographic concerns [5]. A wavefront that passes deterministic CD and PS criteria does not thereby establish an acceptable missing-contact probability.

16.11 A defensible use of the paper's method

Use the inverse network as a candidate generator. Verify each candidate with the forward model and all required patterns. Check coefficient-domain coverage, physical realizability, and robustness to manufacturing and process uncertainty. Then express the final budget as an explicit joint set, distribution, or perturbation allowance with documented conditions.

This preserves the paper's useful idea: imaging performance can guide wavefront choices more directly than a single RMS number. It also supplies the additional steps needed to turn promising examples into an engineering tolerance statement.

Chapter 17

A reproducible laboratory for the key ideas

17.1 What the supplied calculation does

The script `tools/reproduce_examples.py` generates this book's original figures and the files `numerical_checks.txt` and `teaching_model_results.json`. It verifies disk-mode orthogonality and several analytic identities, then calculates a simple periodic contact image. It contains no trained version of the authors' network and does not claim to reproduce the commercial reflective-mask simulation.

The dependencies are NumPy, SciPy, and Matplotlib. The PDF and figures already included with this book can be read without running Python.

17.2 Declared optical and numerical assumptions

The contact example uses:

- wavelength 13.5 nm, image NA 0.33, and an unobscured unit pupil;
- a scalar binary thin mask containing one centred $14\,\mathrm{nm} \times 14\,\mathrm{nm}$ opening per $38\,\mathrm{nm} \times 38\,\mathrm{nm}$ period;
- four idealized source sectors, radii 0.6–0.8, centred at $45°, 135°, 225°, 315°$;
- 60 equally weighted source samples: three radii uniformly spaced in squared radius and five midpoint angular samples per sector;
- exact analytic rectangle Fourier coefficients from Eq. (8.8), retaining every order that can enter the pupil for the selected source support;
- pupil phase $2\pi W/\lambda$ and paraxial defocus phase $-\pi z \mathrm{NA}^2 \rho^2/\lambda$;
- a fixed threshold chosen so that the no-aberration, zero-focus, unit-dose reference has a 14 nm CD on its central cuts.

Equal spacing in squared radius approximates uniform annular area weighting because $\rho \mathrm{d}\rho = \frac{1}{2}\mathrm{d}(\rho^2)$. These choices make the example reproducible; they are not inferred hidden settings from the paper.

17.3 How the calculation is organized

For each source point:

1. Compute normalized order locations $\boldsymbol{u}_{mn} = \boldsymbol{u}_s + (m/p_x, n/p_y)/f_c$.
2. Reject orders with $|\boldsymbol{u}_{mn}| > 1$.
3. Multiply each retained mask coefficient by its aberration and defocus phase.
4. Sum the complex plane-wave orders along an image cut.
5. Square the field magnitude.

Average those intensities across source points. Find left and right crossings of the fixed threshold by linear interpolation on a dense spatial grid. Then calculate CD and midpoint shift. For pure tilt, orthogonal cuts pass through the analytically translated feature centre; otherwise a fixed off-centre cut could falsely suggest a width change even for a rigid translation.

17.4 Same RMS, different images

The three nonzero cases use one unit-RMS mode with $c = 0.27\,\mathrm{nm} = 20\,\mathrm{m}\lambda$: horizontal tilt, defocus, and cosine astigmatism. All have the same piston-removed wavefront RMS in the declared disk basis.

Table 17.1: Computed teaching-model results, rounded for readability. The exact arrays are saved with the source. These are not paper data.

Case	CD_x (nm)	PS_x (nm)	CD_y (nm)	PS_y (nm)
Reference	14.00000	0	14.00000	0
Horizontal tilt	14.00000	−1.63636	14.00000	0
Defocus	13.94249	0	13.94249	0
Astigmatism	13.97132	0	13.97132	0

Tilt agrees with the analytic displacement $-2c/\mathrm{NA}$ and preserves the registered width. Defocus and astigmatism change the image profile by different amounts. In this particularly symmetric in-focus example, the astigmatism case has equal central-cut CDs in the two axes. That is not a universal statement about astigmatism; different focus, source, or mask conditions can break this equality.

17.5 Focus and dose variation

The demonstration samples focus at $-15, 0, +15\,\mathrm{nm}$ and dose at $0.95, 1.00, 1.05$. It collects left and right edge positions over all nine combinations. Both edge-band widths are approximately

$$B_L = B_R = 0.974586\,\mathrm{nm}$$

in this symmetric teaching example. The equality follows from the model symmetry, not from a universal PVB identity. Because the centre remains fixed here, the full CD range is twice one edge-band width.

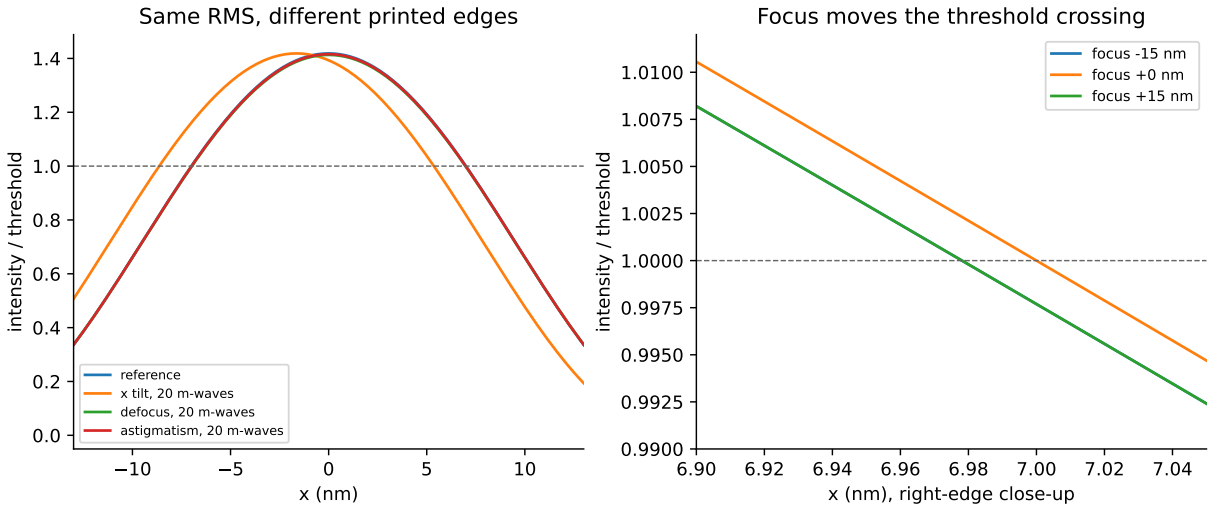


Figure 17.1: Original computed contact profiles. Left: equal-RMS phase modes have different effects on a fixed-threshold image. Right: a close-up of the right edge reveals the small change in threshold crossing through focus. The +15 nm and −15 nm curves coincide by symmetry in this example. All curves use the same reference threshold rather than retuning dose to conceal profile changes.

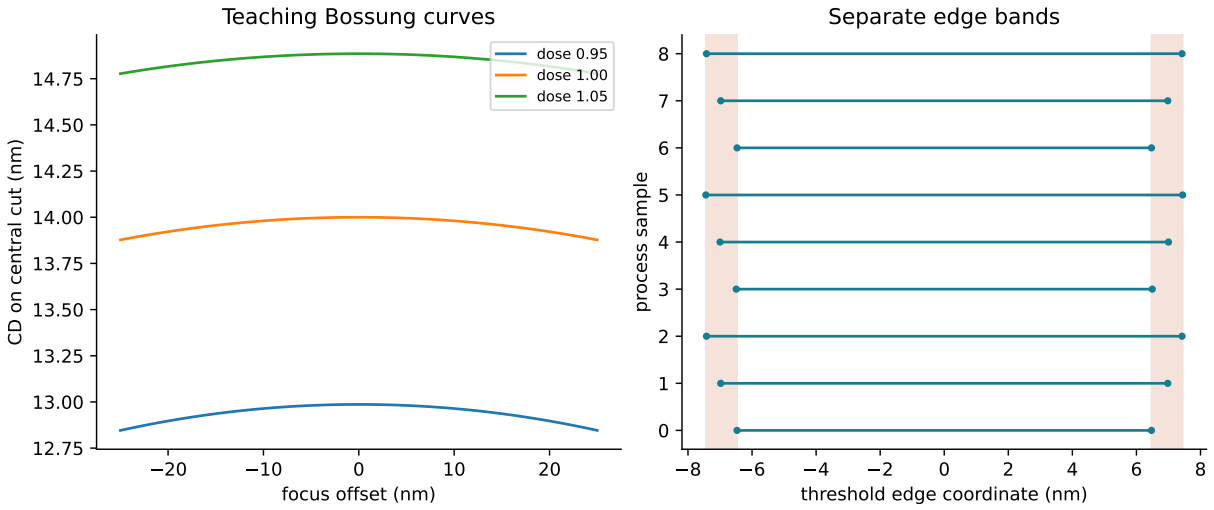


Figure 17.2: Left: teaching Bossung curves computed over a wider focus interval for illustration. Right: left and right edge positions for the nine stated focus-dose samples. Shaded regions show their individual process bands.

17.6 Independent checks built into the script

The script uses Gauss–Legendre integration in ρ^2 and uniform angular quadrature to check the Gram matrix of all 28 real modes through radial degree six. Its largest numerical departure from the identity is about 5.1×10^{-14} in the supplied run. It separately checks the balanced-quartic variance, the analytic tilt displacement, preservation of registered widths under tilt, and simple inverse-ambiguity and correlated-budget counterexamples.

It also prints the wavelength conversions, defocus RMS, 2033-row arithmetic, hypothetical full-grid size, architecture parameter count, and mask-scale consistency estimates. These checks test the calculations actually used in this book.

The source sample count is intentionally modest for transparent, quick execution. There is no claim that these pedagogical images are electromagnetically rigorous or numerically converged to a production accuracy requirement. Exact identities are checked tightly; approximate optical-model output is labelled according to its assumptions.

17.7 Experiments you can perform after reading

1. Change pitch while keeping the hole width fixed. Track how the admitted order set changes and identify which changes in the image follow from altered interference.
2. Change the sign of a pure tilt coefficient. Confirm that the shift reverses and registered widths stay fixed.
3. Add a small focus offset to astigmatism. Examine whether horizontal and vertical CDs remain equal.
4. Reduce edge slope by changing source or threshold, then apply the same dose perturbation. Compare measured edge motion with Eq. (9.6).
5. Increase source and spatial sampling. Check which plotted values stabilize and how long the calculation takes.
6. Replace a single-mode perturbation by two modes together. Compare the combined metric change with the sum of separate changes to reveal nonlinear interactions.

These exercises teach the mechanism behind the paper's dataset. They do not require pretending that the missing commercial simulation or neural-network weights are available.

Appendix A

Two derivations worth doing in full

A.1 Deriving the Zernike radial formula and orthogonality

Chapter 7 constructed low-order modes by subtracting overlaps. Here is a general construction that proves the normalization used throughout the book.

Take $m \geq 0$, $k \geq 0$, $n = m + 2k$, and $t = \rho^2$. Define

$$Q_k(t) = \frac{t^{-m}}{k!} \frac{\mathrm{d}^k}{\mathrm{d}t^k} \left[t^{m+k} (t-1)^k \right], \qquad R^m_{m+2k}(\rho) = t^{m/2} Q_k(t). \tag{A.1}$$

Although t^{-m} appears, the derivative contains a factor t^m, so Q_k is an ordinary polynomial of degree k. At $t = 1$, only the term in which all k derivatives act on $(t-1)^k$ survives. Its value is $k!$, giving $Q_k(1) = 1$ and $R^m_n(1) = 1$.

Expand the bracket:

$$t^{m+k}(t-1)^k = \sum_{s=0}^{k} (-1)^{k-s} \binom{k}{s} t^{m+k+s}.$$

Taking k derivatives gives

$$R^m_{m+2k}(\rho) = \sum_{s=0}^{k} \frac{(-1)^{k-s}(m+k+s)!}{s!(k-s)!(m+s)!} \rho^{m+2s}.$$

Set $j = k - s$, so that $m + 2s = n - 2j$ and $m + k + s = n - j$. This becomes

$$R^m_n(\rho) = \sum_{j=0}^{(n-m)/2} \frac{(-1)^j (n-j)!}{j![(n-m)/2 - j]![(n+m)/2 - j]!} \rho^{n-2j},$$

which is Eq. (7.4). The factorial formula is therefore the expanded form of a weighted polynomial orthogonalization construction.

To prove orthogonality, take $\ell < k$ and evaluate

$$\int_0^1 R^m_{m+2k} R^m_{m+2\ell}\, \rho \mathrm{d}\rho = \frac{1}{2} \int_0^1 t^m Q_k(t) Q_\ell(t) \mathrm{d}t \tag{A.2}$$

$$= \frac{1}{2k!} \int_0^1 Q_\ell(t) \frac{\mathrm{d}^k}{\mathrm{d}t^k} [t^{m+k}(t-1)^k] \mathrm{d}t. \tag{A.3}$$

Integrate by parts k times. Boundary terms vanish: the bracket has sufficient zeros at both $t = 0$ and $t = 1$. The remaining integral contains $\mathrm{d}^k Q_\ell / \mathrm{d}t^k$, which is zero because Q_ℓ has degree less than k. This proves orthogonality for different radial degrees at fixed m.

For the norm, the leading coefficient of Q_k is

$$A_k = \frac{(m+2k)!}{k!(m+k)!}.$$

Hence $\mathrm{d}^k Q_k/\mathrm{d}t^k = k!A_k$, and integration by parts gives

$$\int_0^1 [R_{m+2k}^m]^2 \rho \mathrm{d}\rho = \frac{A_k}{2} \int_0^1 t^{m+k}(1-t)^k \mathrm{d}t \tag{A.4}$$

$$= \frac{A_k}{2} \frac{(m+k)!\,k!}{(m+2k+1)!} \tag{A.5}$$

$$= \frac{1}{2(m+2k+1)} = \frac{1}{2(n+1)}. \tag{A.6}$$

The middle integral follows by repeated integration by parts: $\int_0^1 t^a(1-t)^b \mathrm{d}t = a!b!/(a+b+1)!$ for nonnegative integer a, b. Angular sine/cosine orthogonality supplies the remaining normalization factors in Eq. (7.5).

A.2 Why phase can move edges without changing total energy

For $U = \mathcal{F}^{-1}[\widetilde{t}P]$, Parseval's identity gives

$$\int |U(\boldsymbol{x})|^2 \mathrm{d}^2x = \int |\widetilde{t}(\boldsymbol{f})|^2 |P(\boldsymbol{f})|^2 \mathrm{d}^2f. \tag{A.7}$$

If only phase changes and $|P|$ stays fixed, the integrated image energy is unchanged. For oblique illumination replace $P(\boldsymbol{f})$ by $P(\boldsymbol{f}+\boldsymbol{s})$; the same conclusion holds for a fixed source point. Source averaging preserves it.

Yet local intensity contains cross terms between frequencies. Their phase differences change with W. Energy is redistributed, altering maxima, minima, slopes, and threshold edges. Phase aberration does not have to absorb light to degrade printing.

For periodic images, use energy per unit cell rather than an infinite integral over repeated cells:

$$\frac{1}{p_x p_y} \int_{\mathrm{cell}} |U|^2 \mathrm{d}x\mathrm{d}y = \sum_{m,n} |a_{mn} P_{mn}|^2.$$

Distinct orders average to zero in the integrated cross terms, but those cross terms remain essential locally. If amplitude, aperture support, or coating reflectance changes too, the total collected energy can change. The phase-only pupil model isolates one mechanism rather than describing every physical perturbation.

Appendix B

Exercises with worked solutions

Try each problem before reading its solution. Numerical examples use this book's conventions unless explicitly stated otherwise.

B.1 Phase, geometry, and diffraction

1. Convert milliwaves into length and phase

Problem. At 13.5 nm, convert $20\,\mathrm{m}\lambda$ into OPD and phase. Does this number alone specify RMS or PV?

Solution. $W = 13.5(20/1000) = 0.270\,\mathrm{nm}$ and $\phi = 2\pi(0.020) = 0.125664$ rad. The conversion gives the size of the stated OPD quantity. Whether it is a coefficient, RMS, PV, or local error must be specified separately. For an RMS-normalized coefficient, its magnitude is its RMS contribution.

2. Convert a mirror-height error

Problem. A local mirror height error is 0.10 nm at 6° incidence from the normal. Estimate OPD and phase magnitude.

Solution. $|\delta W| \simeq 2(0.10)\cos 6^\circ = 0.19890\,\mathrm{nm}$. At 13.5 nm this is about 14.734 milliwaves and 0.09257 rad. The local geometric estimate excludes coating phase and changed ray intersections.

3. Separate resolution and focus scaling

Problem. At fixed wavelength, NA increases from 0.33 to 0.55. How do λ/NA and λ/NA^2 change?

Solution. Their ratios are $0.33/0.55 = 0.60$ and $(0.33/0.55)^2 = 0.36$. The lateral diffraction scale decreases by 40 percent; the focus scale decreases by 64 percent. Process-specific constants may also change, so these are scaling comparisons.

4. Explain why one admitted order cannot draw a contact

Problem. Show that one transmitted diffraction order has uniform intensity. Why does the 38 nm example benefit from oblique light?

Solution. With $U = a\mathrm{e}^{\mathrm{i}2\pi(f_x x + f_y y)}$, $|U|^2 = |a|^2$ everywhere. At the stated wavelength and NA, first-order normalized spacing for 38 nm pitch is 1.07656, outside the unit pupil for on-axis

illumination. Oblique source points shift several orders into the pupil, allowing spatially varying interference cross terms.

5. Recover the phase-to-shift relation

Problem. Two image orders separated by frequency $1/(38\,\mathrm{nm})$ receive relative phase 0.10 rad. Find the fringe shift.

Solution. $\Delta x = -p\Delta\phi/(2\pi) = -38(0.10)/(2\pi) \simeq -0.60479\,\mathrm{nm}$. This is a differential-phase result for two orders, not a general coefficient-to-PS calibration.

6. Add amplitudes before intensities

Problem. Two coherent equal-amplitude fields each have intensity one. Find total intensity for relative phases 0, $\pi/2$, and π. What if the sources are mutually incoherent?

Solution. $I = 2 + 2\cos\Delta\phi$, giving 4, 2, and 0. For incoherent contributions the averaged cosine term vanishes and intensity is 2. Averaging intensities prematurely loses interference.

B.2 Wavefront statistics and Zernike modes

7. Normalize defocus

Problem. For $W = B\rho^2$, find its mean and piston-removed RMS. Express the nonpiston part using Z_2^0.

Solution. $\langle W\rangle = B/2$ and $\sigma_W^2 = B^2(1/3 - 1/4) = B^2/12$. The residual is $B(\rho^2 - 1/2) = [B/(2\sqrt{3})]Z_2^0$. Its unit-RMS coefficient is $B/(2\sqrt{3})$.

8. Distinguish raw and balanced spherical aberration

Problem. Why is $B\rho^4$ not pure Z_4^0? Find the best quadratic compensation in RMS.

Solution. The raw quartic overlaps piston and defocus. For $B\rho^4 + d\rho^2$, minimizing piston-removed variance gives $d = -B$. The residual after mean removal is $B(\rho^4 - \rho^2 + 1/6) = [B/(6\sqrt{5})]Z_4^0$. Its RMS is $|B|/\sqrt{180}$.

9. Count modes and total RMS

Problem. How many real disk modes exist through degree six? What RMS results if 24 orthonormal nonpiston coefficients are all 20 milliwaves?

Solution. There are $\sum_{n=0}^{6}(n+1) = 28$ modes including piston. RMS is $\sqrt{24}(20) = 97.9796$ milliwaves. Neither calculation determines the paper's unspecified single-index ordering.

10. Find a pure-tilt placement allowance

Problem. For $W = cZ_1^{-1}$ and NA 0.33, what $|c|$ corresponds to 0.4 nm vertical shift?

Solution. $Z_1^{-1} = 2v$ gives $|\Delta y| = 2|c|/\mathrm{NA}$. Thus $|c| = 0.4(0.33)/2 = 0.066\,\mathrm{nm} = 4.88889$ milliwaves at 13.5 nm. This assumes the declared basis and no later removal of the translation.

11. Decide which error a Strehl calculation omits

Problem. A report removes piston, tilt, and defocus before giving a high Strehl estimate. Does it guarantee small absolute pattern shift at a fixed wafer plane?

Solution. No. Tilt changes absolute placement while preserving registered shape; defocus compensation changes the observation plane. The residual quality excludes those freedoms. Placement and fixed-plane focus must be constrained separately.

B.3 Image edges and process variation

12. Distinguish CD change from PS change

Problem. Left and right edges move by -0.2 and $+0.3$ nm. Find width and centre changes.

Solution. $\delta\mathrm{CD} = 0.3 - (-0.2) = 0.5\,\mathrm{nm}$. $\delta\mathrm{PS} = [0.3 + (-0.2)]/2 = 0.05\,\mathrm{nm}$. Most of the change is widening, with a small net translation.

13. Use the edge-slope formula

Problem. At a bright feature's right edge, $I' = -0.1$ intensity units per nm and $\delta I = +0.02$. Find edge displacement.

Solution. $\delta x = -\delta I / I' = +0.2\,\mathrm{nm}$. The edge moves outward. If the left slope is $+0.1$ with the same perturbation, that edge moves $-0.2\,\mathrm{nm}$, so width increases 0.4 nm and the centre stays fixed.

14. Check NILS with an exactly soluble image

Problem. Let $I(x) = I_0 \exp[-x^2/(2s^2)]$ and $T = I_0/2$. Find nominal width, NILS, and width sensitivity to $\ln D$.

Solution. The positive edge is $r = s\sqrt{2\ln 2}$, width is $w = 2r$, and $|\partial \ln I/\partial x| = r/s^2$. Thus $\mathrm{NILS} = 2r^2/s^2 = 4\ln 2$. At dose D,

$$w(D) = 2s\sqrt{2\ln(2D)}, \qquad \frac{\mathrm{d}w}{\mathrm{d}\ln D} = \frac{2s}{\sqrt{2\ln(2D)}}.$$

At $D = 1$, this equals $2w/\mathrm{NILS}$, verifying the two-edge formula.

15. Show that PVB is not always CD range

Problem. A 14 nm feature translates between centres -0.3 and $+0.3$ nm over process conditions. Find edge-band widths and CD range.

Solution. Each edge translates over 0.6 nm, so $B_L = B_R = 0.6\,\mathrm{nm}$. Every width remains 14 nm, so $B_{\mathrm{CD}} = 0$. Contour variation cannot be inferred from CD range alone.

B.4 Data and neural-network reasoning

16. Audit the sample count

Problem. Why does the single-mode sweep have 984 rows but at most 961 distinct wavefronts?

Solution. $41(24) = 984$ rows, but the all-zero vector appears 24 times. Removing 23 extra copies leaves at most 961 distinct vectors. Other groups can contain further duplicates.

17. Check normalization and extrapolation

Problem. A feature trained over $[10, 30]$ nm maps to $[-1, 1]$. Find normalized values at 20 and 35 nm.

Solution. $\tilde{x} = 2(x - 10)/20 - 1$ gives 0 and 1.5. The latter input is outside the observed range. Scaling does not clip it or make extrapolation reliable.

18. Count a network's parameters

Problem. Count parameters in a dense $84 \to 64 \to 25$ network with biases.

Solution. There are $(84 + 1)64 = 5440$ in the first layer and $(64 + 1)25 = 1625$ in the second, totaling 7065. The paper's longer literal architecture has 24,485. Input entries are not themselves trainable parameters.

19. Calculate one LM step

Problem. For $e(\theta) = 2\theta - 1$, start at $\theta = 0$ with damping $\mu = 1$. Find the LM step.

Solution. $e = -1$, $J_e = 2$, and $(2^2 + 1)\Delta\theta = -2(-1)$. Hence $\Delta\theta = 0.4$ and the new residual is -0.2. With $\mu = 0$, the linear problem is solved exactly by $\Delta\theta = 0.5$.

20. Explain a valid coefficient mismatch

Problem. For $F(c_1, c_2) = c_1 + c_2$, an original pair is $(1, 0)$ and the inverse proposes $(0, 1)$. Is it necessarily wrong?

Solution. Both produce output 1. Coefficient error is nonzero, but forward consistency error is zero. An additional indicator such as $c_1 - c_2$ would distinguish them. Equivalence depends on the selected requirements.

21. Explain an invalid mean of valid inverses

Problem. For $F(c) = c^2$, targets $+1$ and -1 both yield input $y = 1$. What single prediction minimizes their average squared coefficient loss, and is it feasible?

Solution. Loss is $\hat{c}^2 + 1$, minimized at $\hat{c} = 0$. But $F(0) = 0$, not 1. Inverse ambiguity can undermine regression even when every training pair is physically valid.

B.5 Validation and budgeting

22. Compare percentage and absolute error

Problem. Reference shift is 0.005 nm and prediction is 0.015 nm. Find absolute and percentage error. Does the prediction meet a 0.4 nm magnitude limit?

Solution. Absolute error is 0.010 nm; percentage error is $100(0.010/0.005) = 200\%$. The predicted 0.015 nm magnitude is below 0.4 nm. A large percentage need not imply a physical limit violation.

23. Derive a safe two-coefficient box

Problem. A local shift change is $\delta y = 2\delta c_1 - \delta c_2$ and remaining allowance is 0.3. Assign equal independent bounds $|\delta c_j| \le a$.

Solution. Maximum absolute change is $2a + a = 3a$, so $a \le 0.1$. Testing the first coefficient alone would allow 0.15; testing the second alone would allow 0.3. Those individual limits cannot be simultaneously permitted independently.

24. Propagate correlated uncertainty

Problem. Use gradient $\boldsymbol{g} = (1, 2)^T$ and covariance $C = \begin{pmatrix} 0.01 & 0.005 \\ 0.005 & 0.04 \end{pmatrix} \mathrm{nm}^2$ to find metric variance.

Solution. $\boldsymbol{g}^T C \boldsymbol{g} = 0.01 + 4(0.04) + 4(0.005) = 0.19\,\mathrm{nm}^2$. Standard deviation is $0.43589\,\mathrm{nm}$. Ignoring covariance gives $0.17\,\mathrm{nm}^2$ and underestimates variance for this gradient.

25. Identify what remains unknown

Problem. What essential information is missing for reproducing every bar in the paper's Fig. 9 exactly?

Solution. The coefficient convention; mask layouts resolving Tables 2 and 3; source, stack, polarization, and simulator settings; metric extraction definitions; training and test arrays; filtering and scaling settings; network activations, weights, and split; and all 118 target and verification vectors. The paper explains a workflow but does not supply these artifacts. A reproduction must obtain or explicitly replace them.

Appendix C

Reading map, notation, and source audit

C.1 Verified entry points into the supplied textbooks

Printed page numbers refer to numbers on book pages. PDF page numbers are one-based viewer positions in the supplied files. The table gives verified entry points, not an assumed constant offset.

Topic	Section and printed page	PDF page
Scalar waves	Born–Wolf §1.3, p. 14	43
Reflection/refraction	Born–Wolf §1.5, p. 36	65
Layered media	Born–Wolf §1.6, p. 51	80
Characteristic matrices	Born–Wolf §1.6.2, p. 55	84
Periodic layers	Born–Wolf §1.6.5, p. 66	95
Geometrical-optics limit	Born–Wolf §3.1, p. 109	138
Stops and pupils	Born–Wolf §4.8.2, p. 186	215
Wave/ray aberrations	Born–Wolf §5.1, p. 203	232
Diffraction integral	Born–Wolf §8.3.1, p. 375	413
Circular-aperture diffraction	Born–Wolf §8.5.2, formula on p. 396	434
Microscope resolution	Born–Wolf §8.6.3, p. 419	459
Aberration diffraction	Born–Wolf §9.1.1, p. 460	559
Reference-sphere changes	Born–Wolf §9.1.2, p. 462	557
Small-error intensity	Born–Wolf §9.1.3, p. 463	556
Zernike definition	Born–Wolf §9.2.1, p. 464	555
Radial-polynomial table	Born–Wolf §9.2.1, p. 465	554
Wavefront expansion	Born–Wolf §9.2.2, p. 466	553
Aberration tolerances	Born–Wolf §9.3, p. 468	551
Small-error criterion	Born–Wolf §9.3, p. 469	550
Coherent image transfer	Born–Wolf §9.5.1, p. 483	534
Incoherent transfer	Born–Wolf §9.5.2, p. 486	531
Mutual intensity	Born–Wolf §10.4.1, p. 507	510
Extended incoherent source	Born–Wolf §10.4.2, p. 508	509
Hopkins propagation	Born–Wolf §10.4.2(b), p. 512	505
Wave-optics postulates	Saleh–Teich §2.1, p. 40	62

Topic	Section and printed page	PDF page
Complex wave amplitude	Saleh–Teich §2.2, pp. 41–42	63–64
Interference	Saleh–Teich §2.5, p. 58	80
Spatial harmonics	Saleh–Teich §4.1, p. 105	127
Angular spectrum	Saleh–Teich §4.1, pp. 106–107	128–129
Optical Fourier transform	Saleh–Teich §4.2, p. 116	138
Diffraction	Saleh–Teich §4.3, p. 121	143
Airy diffraction	Saleh–Teich §4.3, p. 123	145
Image formation	Saleh–Teich §4.4, p. 127	149
Coherent transfer function	Saleh–Teich §4.4, p. 130	152
Defocused pupil	Saleh–Teich §4.4, p. 133	155
Electromagnetic theory	Saleh–Teich §5.1, p. 152	174
Reflection/polarization	Saleh–Teich §6.2, p. 209	231

For a first pass, use Saleh–Teich for waves and Fourier imaging, then Born–Wolf for wave aberrations, Zernike theory, and partial coherence. Born–Wolf uses older notation and electromagnetic unit conventions in places. This guide uses SI for Maxwell's equations and explicitly declares its complex-phase and polynomial normalization conventions. Do not equate symbols across books by appearance alone.

The statistical-optics chapter in Saleh–Teich is absent from the provided file. Its Fourier-transform appendices are also outside the supplied excerpt. The needed definitions and derivations are included here.

C.2 Symbol dictionary

Symbol	Meaning here	Units
λ, λ_0	Vacuum wavelength; also the wavelength in vacuum image space	length
k_0	$2\pi/\lambda_0$	inverse length
$n, \widetilde{n}$	Real or complex refractive index	dimensionless
$U, \boldsymbol{U}$	Scalar or vector complex field amplitude	field scale
I	Time-averaged intensity, often normalized	power/area or relative
$\mathcal{L}$	Optical path length $\int n\mathrm{d}\ell$	length
W	Actual-minus-reference phase-equivalent OPD	length
ϕ	Phase $2\pi W/\lambda$	radians
σ_W	Piston-removed RMS, unless other removals are stated	length
ρ, θ	Normalized pupil radius and azimuth	dimensionless; rad
u, v	Normalized Cartesian pupil coordinates	dimensionless
a	Physical pupil radius; opening width in rectangle examples when stated locally	length
$\boldsymbol{f}$	Transverse spatial-frequency vector	inverse length
f_c	Image coherent cutoff NA/λ	inverse length
P	Complex coherent pupil transfer function	relative amplitude

Symbol	Meaning here	Units
$S(\boldsymbol{s})$	Source weight density	reciprocal coordinate area
Z_n^m	This guide's RMS-normalized real basis function	dimensionless
c_n^m	Coefficient multiplying that function	length
Z_i, F_i	Paper notation: coefficient and basis function	its convention
z	Image-space focus displacement	length
D	Relative dose multiplier in the contour example	dimensionless
T	Fixed exposure threshold	exposure scale
$\boldsymbol{y}$	Vector of imaging indicators	usually nm
F, G	Forward imaging map and learned inverse	by arguments
J	Imaging Jacobian $\partial \boldsymbol{y}/\partial \boldsymbol{c}$	metric / coefficient
J_e	Training-residual Jacobian versus network parameters	training scale
Θ	All trainable weights and biases	parameter-dependent
μ	LM damping; μ_{12} denotes coherence elsewhere	context-dependent
B	Physical-variable-to-wavefront influence matrix	coefficient / control

C.3 Publication issues that must remain visible

Issue	Consequence
Unspecified Zernike ordering and normalization	Named high-index modes, RMS conversion, and transfer to another simulator cannot be fixed uniquely.
Z_2–Z_{25} versus 25 outputs	There are 24 listed varied coordinates; the extra output is unexplained.
Tables 2 and 3 disagree	Plot labels cannot be converted into a unique mask prescription.
Table 3 first SQ pitch pair	$(56, 56)$ also differs from Table 2's rectangular $(56, 70)$.
Table 4 repeats PVB_X	Interpreting the second as PVB_Y is plausible but unconfirmed.
Table 4 REC horizontal CDs	Values near 27–30 nm differ strongly from the nominal 14 nm horizontal size; the cause is unresolved.
CRAO and source shaping mixed in prose	Chief-ray geometry and off-axis source distribution are distinct.
Figure 9 extrema exceed training sweep	Output extrapolation or a scale issue needs clarification; hard coefficient-box enforcement is not demonstrated.
Small-value filtering description	Retained data, threshold units, and zero treatment are not reproducible.
Training split and activations absent	Layer widths alone do not specify a trained model.
Metric extraction not fully defined	Different CD/PS/PVB conventions can give different bars.
Roadmap-to-PS allocation not derived	The 1 nm motivation is not a universal projector allowance.

Issue	Consequence
Data and network weights absent	The guide reproduces its teaching calculations, not the authors' exact dataset.

C.4 A checklist for your own future experiment

Record wavelength; image and object NA; magnification; physical and wafer-referred geometry; pupil axes and support; OPD sign and units; Zernike functions and normalization; removed modes and reference focus; source weights; polarization; mask stack and materials; image or resist model; contour measurement rules; dose-focus sampling; coefficient bounds; training design and duplicate handling; split, scaling, and filtering; network activations, loss, stopping, and random seed; and forward verification of each candidate.

That record turns an interesting calculation into something another researcher can examine, repeat, and improve.

Bibliography

[1] Z. Chen, L. Dong, H. Ding, and Y. Wei, *Aberration budget analysis of EUV lithography from the imaging performance of a contact layer in a 5 nm technology node*, Applied Optics **62**(27), 7270–7279 (2023). doi:10.1364/AO.496461. Primary research source: the supplied 10-page `Abberation Budget paper.pdf`. All three numbered equations, four tables, and twelve figures were inspected.

[2] M. Born and E. Wolf, *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light*, fourth edition, Pergamon Press. The supplied 859-page scanned file was used, including the table of contents, relevant physical-optics pages, and the reversed page run containing Zernike and partial-coherence material. Verified page locations are in Appendix C. The fourth-edition preface is dated May 1968.

[3] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, second edition, Wiley (2007). The supplied file is named `Saleh & Teich Foundations of Photonics.pdf` but the title identifies *Fundamentals*. The excerpt ends at printed page 278; references here are to content present in that excerpt.

[4] ASML, *TWINSCAN NXE:3400C*. Official description used for the NXE system designation, reflective 4:1 reduction optics, and 0.33 NA. https://www.asml.com/en/products/euv-lithography-systems/twinscan-nxe3400c. Accessed 24 September 2026. This supplementary verification does not replace the paper's simulation specification.

[5] IEEE, *International Roadmap for Devices and Systems: 2021 Update, Lithography*. https://irds.ieee.org/images/files/pdf/2021/2021IRDS_Litho.pdf. Used for the distinction between node labels and dimensions and the context of stochastic and pattern-control requirements. The paper's reference 20 instead points to the companion LER white paper, https://irds.ieee.org/images/files/pdf/2021/LER_white_paper.pdf. The 1 nm vertical-PS inference is attributed to the paper's authors; its exact allocation was not independently established from that companion.

[6] MathWorks, *mapminmax: Process matrices by mapping row minimum and maximum values to [-1 1]*. https://www.mathworks.com/help/deeplearning/ref/mapminmax.html. Official documentation for rowwise scaling and reuse of stored settings. Accessed 24 September 2026.

[7] MathWorks, *trainlm: Levenberg–Marquardt backpropagation*. https://www.mathworks.com/help/deeplearning/ref/trainlm.html. Official documentation for damping, squared-error training, and validation-failure stopping. Accessed 24 September 2026. This guide explains the paper's historical method rather than recommending a particular current MATLAB API.